PUNJAB POLICE HOUSING CORPORATION LIMITED

POLICE COMPLEX, PHASE-VII, S.A.S. NAGAR.

BID DOCUMENT

NAME OF WORK: **Supply, Installation, Testing, Commissioning, Operations & Maintenance of City Surveillance & Traffic Management System with Integrated Command & Control Centre in S.A.S. Nagar, Punjab**

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PUNJAB POLICE HOUSING CORPORATION LTD.

POLICE COMPLEX, PHASE-VII, SAS NAGAR.TEL. NO.4080111, 4080112 & FAX No.4080107

SHORT E-TENDER NOTICE NO. 22/2023-24 DATED 01.01.2024

Online tenders are invited on two bid system (Technical Bid & financial Bid) on item rate basis for the following work from reputed specialized agencies having experience of the similar work as per qualification mentioned. For detailed terms and conditions of the notice inviting tenders refer to website eproc.punjab.gov.in

Sr. No	Name of Work	i) Estimated Cost ii) Earnest Money iii) Processing Fee iv) Tender Form Cost	Time Limit(in months)
1.	Supply, Installation, Testing, Commissioning, Operations & Maintenance of City Surveillance & Traffic Management System with Command & Control Centre in S.A.S Nagar City, Punjab.	i) To be quoted by the agency ii) Rs.22.00 lacs iii) Rs.2360/- iv) Rs.5900/-	06

i)	Start Date of Downloading	02.01.2024 Time 17:00 Hrs.
ii)	Start Date of Bid submission	02.01.2024 Time 17:00 Hrs
iii)	Last date of Bid Submission	15.01.2024 Time 11:00 Hrs
iv)	Date & Time of Opening of Technical Bid	16.01.2024 Time 11:00 Hrs

CONDITIONS:

1. Bidders have to pay the Earnest Money Deposit (EMD), Tender Form Fee & Tender processing fee as mentioned above in online mode. EMD can also be paid in the shape of Bank Guarantee, in favor of "Punjab Police Housing Corporation Ltd. S.A.S. Nagar" issued by any scheduled commercial/ nationalized bank, valid for 90 days from the date of submission of bid. In case of offline Bank Guarantee, the bidder would have to deposit original Bank Guarantee 24 hours before opening of PQ/Technical bid in the office of Chief Engineer, PPHC, Phase 7, S.A.S. Nagar.
2. Interested bidders can purchase the tender documents online from website.
3. Withdrawal or modification of a Bid between the deadline for submission of bids and the expiration of the original period of bid validity specified in the DNIT or as extended shall result in the forfeiture of the Bid security /EMD.
4. Bidders/Contractors, who wish to submit online tenders can access tender documents on the website, fill it and submit the completed tender document into Electronic Tender on the website itself. Bidders/ Contractors shall attach scanned copies of all the paper, i.e., copies of the certificates as required in Eligibility criteria given in the Corrigendum/ Addendum/ Corrections/ Revised DNIT, if any will be published on the website.

5. If the date of opening of tenders happens to be a holiday then the tenders will be opened on the next working day.
6. Bid shall be valid for 90 days after the deadline date for Bid submission.
7. Bidders/ Contractors should have valid GST and PAN registration.
8. Bidders/ Contractors should have valid ESI and EPF registration. In case of unavailability of the same an undertaking should be submitted with the technical bid that ESI/EPF number will be arranged within one month from the date of allotment of work.
9. The quoted rates should be inclusive of all prevailing taxes.
10. The bid submitted by a bidder shall not be accepted, if the bidder has indulged in corrupt or fraudulent practice in the past or has made misrepresentation or has a record of poor performance.
11. MD, PPHC reserves the right to accept/ reject any or all tenders without assigning any reasons.

MANAGING DIRECTOR

SECTION-I

ELIGIBILITY, AWARD OF CONTRACT AND GENERAL TERMS & CONDITIONS

1.1) Introduction

With ever growing size of S.A.S. Nagar, Security & Traffic Management issues increased. A heightened sense of security and ever-increasing challenges require new techniques with more access to integrated systems and centralized information system for monitoring and enforcement purposes.

Punjab Police & Punjab Police Housing Corporation (PPHC) intend to establish City Surveillance & Intelligent Traffic Management Systems with Integrated Command & Control Centre in S.A.S Nagar and for procuring the systems PPHC requests proposals from eligible Firms/System Integrators with relevant experience in the domain for the project "Supply, Installation, Testing, Commissioning & Maintenance of City Surveillance & Traffic Management System with Integrated Command & Control Centre in S.A.S. Nagar, Punjab.

1.2) Project features:

This project has an objective to implement integrated Video surveillance & Traffic Management system which includes Integrated Command and Control Centre consisting of Video Management & Analysis Software, Intelligent Traffic management software, reporting dashboards & e-Challan software for SSP, S.A.S. Nagar Jurisdiction.

The Proposed system will enable the above by using Video Surveillance, analytics, ANPR SVD and RLVD systems. This system will provide

1. Surveillance of selected intersections / locations, which are identified by the officials of the concerned Police Jurisdiction.
2. Detect an event and identification of person, object and vehicles within a predefined range.
3. Automatic recognition of Number Plates of all categories of vehicles under various conditions like fog, rain, low light, twilight and darkness to detect violations and enable traffic enforcement.
4. Automatically detect speed violation at given locations under all conditions like fog, rain, dark low light, twilight and darkness to detect violations and enable traffic enforcement and capture the vehicle number plate for e-Challan.
5. Search vehicles with various input methods like vehicle registration number, time, date, location, violation, colour and any other advanced means to track the vehicles.
6. Automatically recognize hot-listed (Stolen / Wanted) to facilitate criminal investigation
7. Help in investigation for post facto analysis of incidents.

8. Automatic generation of e-challan in prescribed form by Police department either in English or Punjabi (as the case may be) with the help of Red Light Violation Detection system (RLVD) and Speed Violation Detection system.
9. Bidder must ensure the integration of the offered solution with NIC platforms like "VAAHAN & SARTHI" using SDK/API provided by the department. The solution shall also integrate with hand held e-Challan & PoS devices. The offered software should have been integrated with NIC platforms Vaahan or Saarthi for at least one project in the past.
10. All cameras and Video management software shall be ONVIF compliant for any third party integration. The same shall be verified from ONVIF Website. SDK/API to be shared by bidder for integration.

1.3) INSTRUCTIONS TO BIDDERS

a) Definitions:

Unless otherwise clearly required by the context, the following terms as used in this document shall have the respective meanings as defined below:

1. **"Agreement"** means the Agreement to be signed between the successful Bidder and PPHC, including all attachments, appendices, all documents incorporated by reference thereto together with any subsequent modifications, the DNIT, the bid offer, the acceptance and all related correspondences, clarifications, presentations.
2. "**Authorized Representative**" shall mean any person authorized by either of the parties. For bidders, the authorized representative shall be the Board Director or Power of Attorney holder from the Board.
3. **"Bid Amount"**: The cost offered in the financial bid of the DNIT inclusive of all taxesand levies.
4. **"Bidder**" means **Sole bidder / Lead member of consortium** offering the solution(s), service(s) and /or materials required in the DNIT through a sealed bid. The word Bidder when used in the pre work award period shall be synonymous with bidder when used after award of the Contract shall mean the successful Bidder or system integrator with whom PPHC signs the agreement for rendering of servicesunder this project.
5. "**Contract**" is used synonymously with Agreement signed between the successful bidder & PPHC. Lead bidder of the consortium shall be legally bound by the agreement and sign the same on behalf of the consortium.
6. **"Corrupt Practice"** means the offering, giving, receiving or soliciting of anything of value or influence the action of an official in the process of Contract execution.
7. **"DC"** Data Centre for termination of all field devices through MPLS Leased Line, Software and connectivity to ICCC equipment.
8. **"Default Notice"** shall mean the written notice of Default of the Agreement issued by one Party to the other in terms hereof.

9. **“EMD”**: Earnest Money Deposit as commitment fees
10. **“FAT” Factory acceptance Test** means the inspection of equipment proposed to be used at the project sites before delivery in the presence of respective OEM employee authorized for such inspection.
11. **“Site Acceptance Test (SAT)”** means the acceptance testing of all the deliverables deployed by Implementing Agency/ System Integrator after Go Live
12. “**Fraudulent Practice**” means a misrepresentation of facts in order to influence a procurement process or the execution of a Contract and includes collusive practice among Bidders (prior to or after Bid submission) designed to establish Bid prices at artificial non-competitive levels and to deprive PPHC of the benefits of free and open competition.
13. **“GoP”** shall stand for the Government of Punjab.
14. **“Implementation Agency / System Integrator / Vendor**” means successful Sole Bidder / Lead Bidder of consortium Sole.
15. **“Implementation Period”** shall mean the period from the date of signing of the Agreement till the completion period mentioned in this DNIT.
16. **“ICCC”** means the **Integrated Command & Control Centre** inclusive of Data Centre & Command & Control Centre.
17. **PPHC**: Punjab Police Housing Corporation Ltd as nodal agency acting for and on behalf of Punjab Police, SAS Nagar, Punjab and represented by its MD, Engineer in charge as the case may be.
18. **"Law"** shall mean any Act, notification, by law, policy, rules and regulations, directive, ordinance, order or instruction having the force of law enacted or issued by the Union Government of India and/ or the Government of Punjab or any other Government body or regulatory authority or subdivision of government agency.
19. **“LOA”** means issuing of “Letter of Acceptance” which shall constitute the intention of the Tenderer to place the Purchase Order with the successful Bidder.
20. **“OEM”** means the Original Equipment Manufacturer whose product(s) is/are offered in the solution for the DNIT requirements and shall be liable for warranty support to successful bidder / PPHC as the case may be.
21. "**Party**" means PPHC or its authorized officials and Bidder, individually and “Parties” mean PPHC and Bidder collectively.
22. “**Period of Agreement**" means period from date of issuance of Letter to proceed and three (3) year from the date of Final-Acceptance of project, extendable for another 2 years.
23. “**Proposal**” means the Eligibility, Technical and the Financial Proposal of bidder in online submission.
24. **“Detailed Notice Inviting Tender (DNIT)”,** means the detailed document seeking a set of solution(s), services(s), materials and/or any combination of them.

25. "**Service**" means provision of Contracted service viz., Supply, Installation, Testing, Commissioning, Operations & comprehensive maintenance and associated services as per the requirements of this DNIT.
26. "**Site**" shall mean the location(s) for which the Contract has been issued and where the services shall be provided as per agreement.
27. **SLA**: Service Level agreement between PPHC and the Selected Lowest cost Bidder of the Project (Implementing Agency/Service Provider).
28. "**Stakeholders**" means SSP, S.A.S. Nagar, Punjab Police/ PPHC and other departments for their respective authorities and domains.
29. "**Termination Notice**" means the written notice of termination of the Agreement issued by one Party to the other in terms hereof.
30. "**Third Party Agency"** means any agency other than the successful Bidder, appointed by PPHC for monitoring the project during commissioning and operation.
31. "**Uptime**" means the time period when specified services/network segments with specified technical and service standards as mentioned in this DNIT and agreement are available to users / stakeholders. The uptime will be calculated as follows: Total time in a quarter (in minutes) less total Service Down time (in minutes) in the quarter.
32. **"% Uptime**" means ratio of 'up time' (in minutes) in a quarter to Total time in the quarter (in minutes) multiplied by 100.
33. "**Service Down Time**" (SDT) means the time period when specified services/networksegments with specified technical and operational requirements as mentioned in this document are not available to stakeholders.
34. **"Go Live"** — Go Live is considered as operational when all facilities at Integrated Command & Control Centre and all field locations are connected live and working as per therequirements of the agreement and PPHC, providing all/ specified services as mentioned in full capacity for a period of time.
35. "**VCA**" means Video Content Analytics.
36. **"VMS"** means Video Management System.
37. **"ANPR"** means Automatic Number Plate Recognition
38. **"RLVD"** Red Light Violation Detection
39. **"SVD"** Speed violation detection

b) Local Conditions

i. Each bidder is expected to get fully acquainted with the local conditions and factors, which would have any effect on the performance of the contract and /or the cost.

ii. The bidder is expected to know all conditions and factors, which may have any effect on the execution of the contract after issue of Letter of Award as described in the bidding documents. PPHC shall not entertain any request for clarification from the bidders regarding such local conditions.

iii. It is the bidders' responsibility that such factors have properly been investigated and considered while submitting the bid proposals and no claim whatsoever including those for financial adjustment to the contract awarded under the bidding documents will be entertained by PPHC. Neither any change in the time schedule of the contract nor any financial adjustments arising thereof shall be permitted by the Tenderer on account of failure of the bidder to know the local news / conditions.

iv. The bidder is expected to visit and examine the location of offices &field sites and its surroundings and obtain all information that may be necessary for preparing the bid at their own interest and cost.

v. Bidder must take into consideration stakeholders for permissions like Municipal Corporation, Punjab Police, PSPCL and any other department / organization with governing such rights like airport, Indian defence, Gas pipeline or Telecom service providers etc. This is a suggestive list and not exhaustive.

vi. Subcontracting of project is not allowed except prior permission of PPHC.

vii. The Sole bidder/ Lead bidder of consortium or any Consortium member shall submit only one bid for this work. A company which submits more than one Bid will cause all the proposals with their participation to be disqualified.

viii. Bidders shall submit offers that comply with the requirements of the bidding documents, including the basic technical design as indicated in the drawing and specifications. Alternatives and variable bids will not be considered as rejected summarily.

1.4) BID DETAILS:

1. Scope of Bid

i. Managing Director, PPHC, Sahibzada Ajit Singh Nagar, Punjab (referred to as PPHC in these documents) invites bids (as defined in these documents and referred to as "the works").

ii. The successful bidder will be expected to complete the works by the completion date specified in the Contract data.

iii. The Bidder shall submit the bid in two parts as under:

PART – 1: PREQUALIFICATION ELIGIBILITY & TECHNICAL BID

The bidder has to submit online editable & searchable scanned copies of the following documents on web site: - (Refer Check List in Section-IV)

i)	Bid cover letter with proof of EMD, Tender Fees and processing fees submitted
ii)	Eligibility criteria documents such as per pre-qualification, Technical qualification and Financial qualification
iii)	Copies of registration with EPF, ESI, Income Tax (PAN No) and GST scanned copies.
iv)	Technical proposal and supporting documents like data sheets, technical specification compliances, MAF, System architecture etc
v)	All letters & undertakings as prescribed in the DNIT, signed by Board Director or Power of Attorney holding authorized signatory
vi)	Unpriced Bill of Quantity with make and models (complete number)
vii)	Proposed devices and Camera at mentioned Sites/ Locations

PART - 2 : FINANCIAL BID

1. The rates and prices tendered in the priced bill of quantities shall include all constructional activity, tools & plant, labour supervision, materials, comprehensive maintenance, insurance, profit, taxes and duties, together with the all general risks, liabilities and obligations set out in the contract.

2. Nothing extra will be paid due to losses/damages caused by Rains/Floods, War, epidemic, strike of the departmental officials or any other act of God or any other cause whatsoever. The same shall be got covered under appropriate insurance / coverage by bidder till making the system junction wise go-live

3. The item for which no rate or price has been entered in will not be paid for by the PPHC when executed and shall be deemed to be covered by the other rates and prices in the Bill of Quantities. Nothing extra shall be paid on account of such items.

4. Bidder shall mention rates and values of all Government Taxes & levies as applicable on the date of bidding.

5. The Tenderer shall duly fill up the item rate in the BOQ on the e-procurement website while submitting the bid in a specified format.

1.5) QUALIFICATION OF BIDDERS

a) Eligibility Criteria

For each category of pre-qualification criteria, the documentary evidence is to be produced online with digitally signed and duly certified signed in ink with seal by authorized signatory of the bidder, serially numbered and enclosed with pre-qualification/ technical bids. If the documentary proof is not enclosed for any/ all criteria, bidder's submission is liable for rejection.

The bidders shall meet the following schedule wise criteria for eligibility. The bidders must submit all supporting documents alongwith their Technical bid.

S.No.	Criteria	Documentary Evidence
1.	The bid shall be submitted by an individual company as a Sole Bidder or as a Consortium of 2 companies for this project. The Sole Bidder / Lead bidder & Consortium members should be registered under Indian Companies Act, 1956/ 2013/ Sole proprietorship. The Sole Bidder or the Lead Bidder, in case of a Consortium, should have completed atleast 5 years of Operations in India, as on bid submission date	Lead bidder / Consortium agreement & undertaking with roles and responsibilities of scope of work in this project Copy of Certification of Incorporation/ Registration Certificate Memorandum of Association and Articles of Association Work order/ Purchase Order/ Contract Agreement to certify that the Bidder has been in Operations for atleast 5 years in India
2.	Average Annual Turnover during the last three financial years (FY 2020-21, 2021-22 & 2022-23) should be minimum INR 10 Crores. i) In case of sole bidder: INR 10 Crores. ii) In case of a Consortium a) Lead Bidder - Min. INR 6 crore b) Consortium member - Min. INR 4 crore.	Audited financial statements for the last three Financial Years i.e. 2020-21, 2021-22 and 2022-23. And Certificate from the Statutory Auditor/ Chartered Accountant on turnover details for the last three FYs. **Note:** In case the financial statements for FY 2022-23 are not audited then Provisional Certified Financial statement for FY 2022-23 by CA can be provided.
3.	The Net Worth of the Sole/ Lead bidder and consortium partner shall be positive for each of the last 3 audited financial years.	Certificate from the Statutory Auditor/ Chartered Accountant on positive net-worth for the last three (3) financial years.
4	The bidder must have completed/ executed similar nature of works in India/ abroad during the last 10 years as on bid submission date, as mentioned below:- 01 Project of min. 7 Crores, having minimum 300 cameras or 02 Projects, each of min. 5 Crores, having minimum 200 cameras or 03 Projects, each of min. 3.5 Crores, having minimum 150 cameras **Similar Project-** Supply, Installation, Testing, Commissioning in any City Surveillance project consisting of City-wide video surveillance (with Video analytics/ VMS)/ Intelligent Traffic Management Systems/ ATMS on highways/ ITS on highways with e-Challan system connected with vehicle data bases and Integrated Command and Control Centre. The bidder should also have experience of Operation & Maintenance of the similar project for at least one year.	Work order/ purchase order/ agreement and Satisfactory completion Certificate from the concerned authority, showing nature of work, value and number of cameras installed.
5.	The Bidder should have a valid GST, PAN, ESI & EPF registration with latest updated return filed.	Copy of GST, PAN, ESI & EPF Registration certificate with latest compliances. In case of unavailability of EPF/ESI number, an Undertaking should be submitted with the technical bid that

		ESI/EPF number will be arranged within one month from the date of allotment of work.
6.	The Sole Bidder or any of Consortium members, should jointly possess any Two (2) of the below certifications, which are valid at the time of bidding: – ISO 9001:2015 for Quality Management System – ISO 27001:2013 for Information Security Management System. – ISO 20000:2018 for IT Service Management.	Copies of valid certificates
7.	The Bidder should have direct authorization from the Original Equipment Manufacturer (OEM) for the equipment's i.e. Cameras, Software, Active Network equipment, Bandwidth service provider, Displays, Servers, Desktop and Storage offered.	Manufacturer's Authorization Form to be submitted as per the Format terms.
8.	The Sole Bidder and each of Consortium members and its Partner/ Directors, at the time of bid submission, shall not have been blacklisted/ debarred by any Central Government/ State Government/ Semi Government/ Corporation/ PSU or any other Government Institution/ Regional Authority in India.	The Sole Bidder and each of Consortium members shall furnish an undertaking signed by the Authorized signatory of the company to be provided on Non-Judicial Stamp Paper duly attested by notary public.
9.	The Bidder should have an office with project / service team in SAS Nagar/ Chandigarh UT/ Panchkula. Note: In case Bidder does not have a local office at the time of bid submission, Bidder shall furnish an undertaking that an office shall be opened within ten days of issue of Letter of Acceptance.	Notarized copy of ownership document/ Rent agreement. or Undertaking signed by authorized signatory of bidder stating the local office will be established within ten (10) days) days from the issue of LOA date.

b) Mandatory qualifications –

In addition, each bidder shall submit the following information for his qualification

1) Copy of Permanent Account Number (PAN) issued by Income Tax Department andthe latest ITR filed.
2) Affidavit/undertaking of not having been black-listed by any Govt. /Semi Govt. Organization/ Corporation at any stage and/or debarred by PPHC or any other Regional Development Authority.
3) Affidavit/undertaking that information being submitted is correct and true, and that any false information shall lead to disqualification at any stage.
4) GST Registration Number (GST) issued by concerned department with copy of latest return filed.
5) EPF registration certificate from Provident Commissioner.
6) ESI registration certificate.
7) The bidder should comply with Orders/ Instructions regarding Restrictions under rule144(xi) of the General Financial Rule (GFRs) 2017 issued by Govt. of India, Ministry of Finance Department. Declaration should be submitted on letter head.

8) Any other qualification information specified in the Conditions of Particular Application.

c) <u>Mandatory Requirement of OEM:</u>

i) The Camera OEM shall be in IP CCTV manufacturing for 10 years. The documentary proof shall be submitted

ii) OEM of CCTV should be registered in India for Last 7 years. Proof of the same should be attached with the technical bid.

iii) The OEM of all types of cameras should have at least 1 project reference for supplies in India for SITC of City Surveillance/ ITMS project completed with minimum deployment of 300 cameras, each for a Central Govt./ State Govt. department.

Or

2 projects reference for supplies in India for SITC of City Surveillance/ ITMS project completed with minimum deployment of 200 cameras, each for a Central Govt./ State Govt. department.

Or

3 projects reference for supplies in India for SITC of City Surveillance/ ITMS project completed with minimum deployment of 150 cameras, each for a Central Govt./ State Govt. department.

iv) The OEM of ITMS should have at least 1 project reference for supplies in India for SITC of City Surveillance/ ITMS project completed for Central Govt./ State Govt. department with atleast 300 cumulative licenses for RLVD, ANPR, SVD, No Helmet, Triple riding features in the project.

Or

2 projects reference for supplies in India for SITC of City Surveillance/ ITMS project completed for Central Govt./ State Govt. department with atleast 200 cumulative licenses, each for RLVD, ANPR, SVD, No Helmet, Triple riding features in each project.

Or

3 projects reference for supplies in India for SITC of City Surveillance/ ITMS project completed for Central Govt./ State Govt. department with atleast 150 cumulative licenses, each for RLVD, ANPR, SVD, No Helmet, Triple riding features in each project.

v) OEM of camera should have membership of ONVIF organization.

vi) The Camera and software OEM should not have been debarred by ONVIF.

vii) Quoted Camera should have MAC IP Address on the same name of Camera OEM

viii) The camera to be provided by the bidder, all components / parts / assembly / software used in the hardware and software should not be complying with GB28181, GB/T 28181-2011 standards.

ix) MTBF of minimum 100000 hours for cameras.

x) The network cameras supplied must be compliant with RoHS certification.

xi) The quoted Model should be not any country specific. It should be a global model.

xii) The Cameras to be provided by the bidder should not have Hisilicon chipset/Processor.

xiii) Chipset of the network cameras should not be from the countries sharing land border with India.

xiv) The OEM of all types of Softwares, Active Network equipment, IT (Servers, Desktop) and Storage should have been registered entities for the last 5 financial years in India and should have customer support center and spares in India. The OEM should affirm this in MAF certificate.

xv) The quoted products / Systems (Cameras, Software, Active Network equipment, IT (Servers, Desktop) and Storage) for this project should not be declared End of Life (EOL) for 3 years and availability of spares for 5 Years from the date of bid submission. Undertaking of OEMs is required with technical bid.

xvi) Each of the proposed OEMs (Cameras, Software, Active Network equipment, IT (Servers, Desktop) and Storage) should provide 24x7 technical support in India, or should provide an undertaking that they would establish the requisite infrastructure and capability to provide 24x7 technical support, on emerging a winner in this bidding process

xvii) At all times during the performance of the Services, the Bidder shall abide by all applicable security rules, policies, standards, guidelines and procedures set forth by Govt. of India and Govt. of Punjab.

xviii) The SVD systems and its components shall comply with the Government norms defined for such systems along with Government recognized certification for the system and its acceptability as evidence in the court of law & submit such certification.

xix) Internet Service Provider (ISP) must be a Class A license holder from Department of Telecommunication (DOT) for providing bandwidth services and must have sufficient experience for providing Internet services through their own network.

xx) The OEM of CCTV Camera should have ISO 27001:2013 or latest, for Information Security Management System.

xxi) The OEM of ITMS & VMS should have any three of the following certifications:-

- ISO 9001
- ISO 27001
- ISO 14001
- Cmmi ML3

xxii) Housing of cameras, if required, must be from the same OEM/ as per recommendations of OEM of Cameras,

1.1) CLARIFICATION OF DNIT

a) To assist in the examination, evaluation, comparison and qualification of the Bids, the bid evaluation committee of the PPHC may, at its discretion, may ask any bidder for a clarification regarding its Bid. The committee's request for clarification and the response of the bidder shall be through the e-mail.

b) Any clarification submitted by a bidder with regard to its Bid that is not in response to a request by the committee shall not be considered.

c) No change in the prices or substance of the Bid shall be sought, offered, or permitted, except to confirm the correction of arithmetic errors discovered by the committee in the evaluation of the financial Bids.

d) No substantive change to qualification information or to a submission, including changes aimed at making an unqualified bidder, qualified or an unresponsive submission, responsive shall be sought, offered or permitted.

e) PPHC reserves the right to accept or reject any variation, deviation, or alternative offer. Variations, deviations, and alternative offers and other factors which are in excess of the requirements of bidding documents or otherwise result in unsolicited benefits for PPHC shall not be taken into account in Bid evaluation.

1.2) AMENDMENT OF BIDDING DOCUMENTS

a) PPHC may modify the bidding documents by issuing addenda as deemed fit.

b) Any addendum / corrigendum & communication shall be part of the DNIT document.

1.3) CURRENCY OF BID AND PAYMENT

The unit rates and the prices shown in BOQ shall be in Indian Rupees only.

1.4) BID VALIDITY

a) Bids shall remain valid for a period not less than ninety (90) days from the last date for bid submission.

b) In exceptional circumstances, prior to expiry of the original time limit, PPHC may request that the bidders may extend the period of validity for a specified additional period. The request and the bidders' responses shall be made by email. A bidder may refuse the request without forfeiting the bid security. A bidder agreeing to the request will not be required or permitted to modify the bid submission but will be required to extend the validity of the bid security for a period of the extension.

1.5) BID SECURITY/EARNEST MONEY DEPOSIT.

i. Bidders have to pay the Earnest Money Deposit (EMD) through online mode. EMD can also be paid in form of Bank Guarantee in favor of “Punjab Police Housing Corporation Ltd. S.A.S. Nagar” issued by any scheduled / nationalized bank, valid for 90 days from the last date of bid submission. The bidder has to deposit original BG 24 hours before opening of PQ/Technical bid. EMD deposited by the bidder shall not bear any interest.

ii. EMD of unsuccessful bidders shall be refunded online through e-procurement portal after issuance of LoA to the L1 bidder (By default)

iii. The EMD received in the shape of BG in physical mode shall be refunded after issuance of LoA to the L1 bidder.

iv. The Tender Form Fee & EMD in any shape other than described above shall not be accepted.

v. Any bid not accompanied with an acceptable Earnest Money deposit (EMD) and not secured as indicated in RFP, shall be rejected by the PPHC under non-responsive clause.

vi. The Bid Security of the successful Bidders shall be returned after the receipt of Performance Security and Agreement signing for the project.

vii. The Bid Security may be forfeited:-

If the Bidder with draws the Bid after Bid opening during the period of Bid validity If the Bidder does not accept the correction of the Bid Price, as per the RFP; or

In the case of a successful Bidder, if the Bidder fails within the specified time limit to:-

a. Sign the Agreement; or

b. Furnish the required Performance Security.

viii. In addition to submission of Performance Security, 5% amount from each bill will also be retained as additional security.

1.6) MODIFICATION AND WITHDRAWAL OF BIDS

a) Bidders may modify or withdraw their bids as per online e-procurement portal process.

b) No bid shall be modified after the deadline for submission of Bids.

c) Withdrawal or modification of a Bid between the deadline for submission of bids and the expiration of the original period of bid validity specified in the DNIT or as extended shall result in the forfeiture of the Bid security /EMD

1.7) <u>BID OPENING, EVALUATION & AWARD</u>

PPHC shall open the bids online and the same shall be evaluated by Technical Committee. The Bidders or their representative may attend the opening of Technical/ financial bids by online or physical attendance.

1. Opening of Technical bids:

Technical Bid shall be opened on bid opening date and evaluation of the same with respect to Qualification criteria & Technical Evaluation required as per RFP shall be taken up.

2. Clarification of Bids:

To assist in the examination, evaluation, and comparison of Bids, the PPHC may, at its discretion, ask any Bidder for clarification of the Bid, including breakup of the unit rates.

The request for clarification and the response shall be in writing or online, but no change in the price or substance of the Bid shall be sought, offered, or permitted in the evaluation of the Bids. The total amount of financial bid as mentioned in words shall be considered as final bid in case of any error in financial bid offered in figures is found. The lower amount as arithmetically correct shall be considered as final.

PPHC reserves the right to accept or reject any variation, deviation, or alternative offer. Variations, deviations, and alternative offers and other factors which are in excess of the requirements of bidding documents or otherwise result in unsolicited benefits for PPHC shall not be taken into account in Bid evaluation.

3. Opening of Financial bids:

After evaluation of Prequalification, the bidders found qualified in pre-qualification criteria will be shortlisted for Demonstration and PoC. After Demonstration and PoC, Financial bids of only qualified bidders in technical evaluation process shall be opened.

4. Method of Selection:-.

i. The Bidder shall be selected on the basis of lowest quoted rates (L1).

Note: For Demonstration

Actual equipment set up shall be created at the PPHC designated location and makes and models offered shall be made active as offered in technical bid in compliance to the bid requirements.

All parameters of minimum specifications read in conjunction with the requirements and scope of work shall also be evaluated in real time through demonstration of the said equipment through the system design.

1.8) COST OF BIDDING & SUFFICIENCY OF BIDDING

a) The bidder shall bear all costs associated with the preparation and submission of his Bid, and the PPHC will in no case be responsible and liable for those costs.

b) The bidder shall be deemed to have satisfied himself before bidding as to the correctness and sufficiency of his bid information for the works and of the rates and prices given in the relevant Schedule or quoted by him, which rates and prices shall, except as otherwise provided, cover all his obligations under the Contract and all matters and things necessary for the proper completion of the works.

1.9) SITE VISIT

a) The bidders shall be expected to visit and examine the Sites of the Project, at their own responsibility and expenses and obtain all information (including that on the risks, contingencies and other circumstances which may affect or influence the bid) that may be necessary for preparing the Bid and entering into a contract.

b) No extra charges consequent on any misunderstanding or otherwise shall be allowed. The bidder may reassure themselves at own cost about the soil properties at the site. PPHC shall not be responsible for any variation in local site conditions from that given in this document.

1.10) NON-DISCLOSURE & CONFIDENTIALITY

(a) Information relating to the examination, clarification, evaluation, and comparison of Bids shall not be disclosed to Bidders or any other persons not officially concerned with such process until the award to the successful Bidder has been announced. Any effort by a Bidder to influence PPHC's processing of Bids or award decisions may result in the rejection of their Bid.

(b) Any effort by the Bidder to influence PPHC in the bid evaluation, bid comparison or contract award decisions may result in the rejection of the Bidders' bid.

(c) The Bidder should note that before any of its employees or assignees is given access to the

Confidential Information, each such employee and assignees shall agree to be bound by the term of this tender and such rules, policies, standards, guidelines and procedures by its employees or agents.

1.11) CORRUPT OR FRAUDULENT PRACTICES

(a) PPHC will reject a proposal for award if it determines that the Bidder recommended for award has engaged in corrupt or fraudulent practices in competing for the contract in question and will declare a firm ineligible, either indefinitely or for a stated period of time, to be awarded a contract with PPHC and any other agencies, if it at any time determines that the firm has engaged in corrupt or fraudulent practices in competing for the contract, or in execution.

(b) For the purpose of this clause, "Corrupt Practice" means the offering, giving, receiving or soliciting of anything of value to influence the execution. "Fraudulent Practice" means a misrepresentation of facts in order to influence a procurement process or the execution of a contract to the detriment of the Borrower, and includes collusive practice among Bidders (prior to or after bid submission) designed to establish bid prices at artificial non-competitive levels and to deprive the Borrower of the benefits of free and open competition".

(c) As used herein, the term "Confidential Information" means any information, including information created by or for the other party, whether written or oral, which relates to internal controls, computer or data processing programs, algorithms, electronic data processing applications, routines, subroutines, techniques or systems, or information concerning the business or financial affairs and methods of operation or proposed methods of operation, accounts, transactions, proposed transactions or security procedures of either party or any of its affiliates, or any client of either party, except such information which is in the public domain at the time of its disclosure or thereafter enters the public domain other than as a result of a breach of duty on the part of the party receiving such information. It is the express intent of the parties that all the business process and methods used by the Bidder in rendering the services hereunder are the Confidential Information of the Bidder. A Non- disclosure agreement (NDA) shall be signed by the bidder in favour of PPHC as per format provided to successful bidder.

(d) The Bidder shall keep confidential any information related to this tender with the same degree of care as it would treat its own confidential information. The Bidder shall note that the confidential information will be used only for the purposes of this tender and shall not be disclosed to any third party for any reason whatsoever.

(e) The Bidder should not disclose to any other party and keep confidential the terms and conditions of this Contract agreement, any amendment hereof, and any Attachment or Annexure hereof.

(f) The obligations of confidentiality under this section shall survive rejection of the contract.

1.12) PUBLICITY

i. Any publicity by the Bidder in which the name of the said project is to be used, should be done only with the explicit written permission from PPHC.

ii. Right of Selection/Rejection– Waiver of Informalities or Irregularities:

1.13) SCOPE OF WORK:

Setting up of "City Surveillance & Traffic Management System" in Jurisdiction of Senior Superintendent of Police, S.A.S. Nagar, Punjab. The broad scope of work for this assignment is given under various relevant sections of this DNIT

i) General requirements

a) The selected bidder shall perform detailed site surveys for the purpose of installation & commissioning of the field equipment at all locations under the scope of the Project. The Supply Installation Testing and Commissioning (SITC) shall only commence after the detailed survey report is duly approved by PPHC.

b) The selected bidder shall identify and obtain necessary No Objection Certificate (NOC), statutory clearances, approvals for erecting the poles, carry out civil work and installing cameras etc. including provisioning of the required power and network bandwidth.

c) The selected bidder shall manage all necessary paperwork to pursue permission from respective authorities to initiate & complete the scope of work. The agency will get approval (RoW) from the local government bodies/ Municipal Corporation etc on behalf of PPHC for any clearance required to execute the work. PPHC shall provide necessary assistance in application process as requested by successful bidder. Any non-refundable fees / charges payable for this account within radius of 50 mtrs from the centre of the junction / location shall be borne by PPHC on production of valid demand note by concerned authority. This permission shall apply to works executed by agency under the scope of the said project from the junction box to respective poles only. No RoW charges or applicationsshall be made for bandwidth service provider scope of work.

d) The Contractor / vendor of the project shall enter into tripartite agreement with the bandwidth service provider and PPHC for enabling direct billing to PPHC by the license holder bandwidth service provider as per the amounts quoted by bidder in Part II of the bid i.e., Financial bid. The bidder shall submit a Memorandum of Understanding with the bandwidth service provider to this effect in the technical bid submission along with an authorization to bid.

e) Payment on account of MPLS connectivity will be made for the period of date of Go-Live to expiry of Warranty period only. The SI will have to arrange MPLS connectivity at his own cost, as required for testing purpose, before the date of Go-live.

f) The scope shall not be restricted to the items listed in the schedule or requirement and specifications only. Whatever additional items, components and equipment that are needed to make the complete system functional, shall be included in the BOM when bidding.

g) Contractor to get power supply meters installed from PSPCL in the name of PPHC/ Police department. Contractor to pay electricity connection charges and consumption charges for the project duration and after Go Live and get reimbursement from PPHC as per actual bills. The

power load calculations in terms of load 22 and expected units / month, for each location shall be provided by bidder in technical bid for budgeting the power load and billing.

ii) Sites/ Locations:

Sr. No	Location	Location Type
1.	New Sunny Enclave gate	ITMS
2.	Cheema Boiler Chowk	ITMS
3.	Radha Swami Chowk	ITMS
4.	T-point Ajit Smachar Sec 90/Phase 8B Dividing	ITMS
5.	Landra Location (2 Chowks)	ITMS
6.	Purab Apartments Crossing Sec 88/89	ITMS
7.	Sec. 105/106 Dividing	Entry/ Exit
8.	Gurudwara Singh Shaheedan Chowk	ITMS
9.	IISER Chowk	ITMS
10.	Airport Chowk (Bakarpur)	ITMS
11.	Dairi T-Point (Landran Banur Road)	Entry/ Exit
12.	PCA Stadium Crossing (Ph 9/10)	ITMS
13.	Phase 7 Crossing	ITMS
14.	Chawla Chowk Crossing	ITMS
15.	Phase 3/5 crossing	ITMS
16.	Madanpur chowk	ITMS
17.	Micro Tower, Phase 2/3A crossing (2 locations)	ITMS/ Entry/ Exit
18.	Max Hospital	ITMS/ Entry/ Exit
19.	Near TDI/ Gilco gate (Airport Road)	SVD/ Entry/ Exit
20.	Airport Chowk (Bakarpur) to Zirakpur Road	SVD/ Entry/ Exit

iii) City Surveillance System

a) Cameras with suitable environmental and power protections to be provided. All cameras shall stream in zero lux light.

b) Video Management platform for stream processing, management and recording. The system shall have easy data retrieval function.

c) Video Analytics as per detection parameters mentioned for need based alerts anddata analysis.

d) The live video feed shall be streamed in real-time to Data Centre and Integrated Command & Control Centre located at Police Station Sohana, Sector 79, SAS Nagar, Punjab.

iv) Intelligent Traffic Management System (ITMS)

a) Red-Light Violation Detection (RLVD), Speed Violation Detection System (SVDS) shall be the part of the ITMS component.

b) Automatic Number Plate Recognition (ANPR) System shall have capability to capture the number plate of the vehicles associated with violating any of the above traffic rules. The system shall also provide data of all vehicles passing the said location.

c) The ANPR system will be evaluated for capturing High Security Registration Plates as well as shall be able to detect nonstandard number plates also.

d) It is further mandatory that the e-Challan platform shall have atleast one integration with NIC based

databases like Vaahan / Saarthi for integrity test. The solution shall alsointegrate with hand held e-Challan & PoS devices

v) Communication Network

1. Bidder shall arrange MPLS based Network with required bandwidth at each location, which can support all the planned initiatives i.e. City Surveillance, Traffic Management and connect to the data centre at Integrated Command & Control Centre at PS Sohana, Sector 79, SAS Nagar and scalable to accommodate future IT requirements of the city. The payment for this service shall be made by PPHC / Punjab Police to the service provider as quoted by bidder & agreement with bandwidth provider and enclosed in financial bid. The agreement between bandwidth Service provider & bidder (without financials) shall be annexed with technical bid and the agreement between successful bidder and proposed bandwidth service provider for such rates as quoted in the financial bid for a period of three years from start of the bandwidth service, shall be annexed withthe financial bid, in support of this cost.

2. The vendor / service provider & Bandwidth network provider shall enter into tripartite agreement with the successful bidder / vendor of the project and PPHC for enabling direct billing to PPHC by the license holder bandwidth service provider. Clause 1.58 will also be applicable on bandwidth service provider.

Note: Bidder shall arrange MPLS based end to end underground Fibre network. Max average utilization should not be more than 80% in a month, at each location. ISP will be liable to give access to the department to MRTG software/ tool for monitoring of daily utilization.

In case, average utilization goes beyond 80% of the offered bandwidth. The SI will be liable to enhance the same within the quoted cost. PPHC will not be liable for any increase the payments. If any additional demand made by the ISP, the same shall be deducted from the bill of SI.

vi) Date Centre & Integrated Command & Control Centre (ICCC)

a) The Bidder will be provided Data Centre & Integrated Command & Control Centre space at Police Station Sohana, Sec. 79, S.A.S. Nagar, Punjab.

b) The Bidder is responsible for SITC & O&M of systems under BOQ or any additional items to make the Data Centre and Command & Control Centre functional as per satisfaction of Engineer in Charge.

vii) <u>Operations & Comprehensive Maintenance</u>

(1) Equipment

a) The equipment, its installation and configurations shall remain under Warranty/ defect liability period for three years (1+2 years) from the date of Final Acceptance after Go Live. Any upgrades & updates in software shall be provided free of cost to keep the system updated and healthy during this period.

b) **Preventive Maintenance —** Vendor shall conduct periodic preventive checks as per the Standard

manuals by OEM and approved maintenance schedule to maintain the required uptime. Vendor shall plan system maintenance schedule with the approval of concerned engineer / officers where shut down shall be required.

c) **Diagnostic Maintenance –** Vendor shall maintain sufficient spares of equipment in order to provide the required uptime or better. Under no conditions shall the system be under downtime beyond.

d) Agency will be liable for periodic comprehensive maintenance/ Servicing of DG sets during warranty period, including supply of all consumable items but except diesel. Servicing/ repair is to be done from the authorized service agent of concerned OEM.

(2) Manpower

a) Project Manager: (01 no.) The Bidder shall be responsible to deploy dedicated Project Manager(s) with minimum 10 years' experience in similar field including City surveillance, Traffic Management and Command & Control Centre till the completion of the project.

The qualification of the project Manager should be qualified B.E./ B.Tech Electronics/ Electrical/ IT/ Computer Science and experience of 10 years in similar field.

CV's for Project Manager (s) shall be provided with the bid and evaluated during presentation scoring

b) Field Engineer: (Minimum 4 nos.) B. Tech / Diploma with 3 years' experience in similar projects as per requirements till the completion of the project and O&M period.

c) Any additional manpower, if required will be provided by the SI at his own cost.

1.14) DATA RETRIEVAL:

SI shall be liable to put in best efforts for retrieval of data without any extra cost, in case any data loss, HDD Crash due to any reason to the entire satisfaction of Engineer-in- Charge and Police Department. In case OEM is not providing data recovery service to SI, the SI may get data retrieved fromany third party, lab/ organization. Data integrity and Privacy need to be maintained for the retrieval process either by Bidder/ OEM or by third party.

The SI is exempted if data loss happens because of below reasons

1. Natural calamity/Act of God
2. Physical Damage by third party
3. Fire or any other unforeseen reason

Note: If the Servers & Hard disks are required to be carried away from the Command & Control Centre to any laboratory for data retrieval, the SI will have to make standby arrangement at his own cost till restoration of the same. SI will submit Undertaking for the same (Annex. M) along with the Technical bid

1.15) PAYMENT SCHEDULE

S.NO	PAYMENT TERMS	PAYMENT
1	Delivery and Inspection of Hardware/ Equipment at Site/ PPHC Stores	Up-to 60% of Project value (CAPEX) on prorata basis.
2	Installation, testing & commissioning	Up-to 10% of Project value (CAPEX) on prorata basis.
3	Final acceptance test by the deptt. and Go Live of the Project as per the definition mentioned in RFP.	Up-to 20% of Project value (CAPEX) on prorata basis.
4	Free AMC/ Warranty Period:	i) 10% of CAPEX will be paid in 4 Quarterly Installments on prorata basis after the end of each quarter, during free AMC period, after verification of SLA's as per the satisfaction of PPHC. ii) Payment against MPLS connectivity will be released at the end of each quarter.
5	Additional comprehensive AMC Period:	Payment of OPEX and MPLS connectivity will be released on monthly basis, on production of bill and EPF and ESI payment receipt.

1.16) PROJECT TIMELINES

Sr. No	Milestone	Deliverables	Timeline
1.	Date of Agreement signing	(within 15 days from LoA along with submission of Performance Bank Guarantee)	T
2.	**Project Planning**	Detailed Project Plan Survey and Detailed Design of all the solutions components Design Approvals Required Infrastructure Plan & Approval	T + 10 days
3.	**Delivery Of BOM**	All the BOM mentioned hardware and softwares shall be procured and delivered Stock entry of Hardware and Verification of Hardware by PPHC/Representative of PPHC	T + 90 days
4.	**Installation, Commissioning of Hardware**	Weekly and Monthly Progress Reports Hardware Installation and configuration Necessary Software Configuration Testing-Performance, Scalability, Systems Integration, Security Testing, Systems Acceptance Test, etc. Develop Training Materials	T + 150 days
5.	**Testing & Go-Live**	Final Acceptance Testing Training & Change management User Training Operational System Acceptance Go-Live of Project	T + 165 days
		Go Live Certification (T1)	T + 180 days

7.	**Operation & Comprehensive Maintenance:-** System Integrator (SI) has to follow the SLA's defined during the maintenance phase. System Integrator (SI) will be solely responsible for the deliverables. SLA Compliance Reports, Audits		T1+36 months from the date of Go-Live of whole project

NOTE:-

- *T is the date of signing of contract with the System Integrator (SI).*
- *T1 is the date of Go-Live of the contract.*

1.17) LANGUAGE AND LAW

The language of the Contract shall be English and the law governing the Contract are stated in the Contract Data.

1.18) ENGINEER'S DECISIONS

Except where otherwise specifically stated, the Engineer shall decide contractual matters between the PPHC and the Contractor in the role representing the PPHC. Any clarifications on the Contract Data shall be given by the Engineer in charge.

1.19) COMMUNICATIONS

Communications between parties which are referred to in the conditions are effective only when in writing. A notice shall be effective only when it is delivered under email / Registered post / Speed post / Facsimile (fax) or other electronic means. The notice sent by Registered post or Speed post shall be effective on delivery or at the expiry of the normal delivery period as undertaken by the postal service. The notice sent by Facsimile (fax) or other electronic means shall be effective on confirmation of the transmission

1.20) PERSONNEL

The Contractor shall employ the key personnel named in the Schedule of Key Personnel as referred to in the Contract Data to carry out the functions stated in the Schedule or other personnel approved by the Engineer. The Engineer will approve any proposed replacement of key personnel only if their qualifications, abilities, and relevant experience are substantially equal to or better than those of the personnel listed in the Schedule.

If the Engineer asks the Contractor to remove a person who is a member of the Contractor's staff or his work force stating the reasons the Contractor shall ensure that the person leaves the Site within seven days and has no further connection with the work in the Contract.

1.21) PPHC'S RISKS

PPHC carries the risks which this Contract states are PPHC's risks, and the Contractor carries the risks which this Contract states are Contractor's risks.

PPHC is responsible for the accepted risks which are (a) in so far as they directly affectthe execution of the Works within the country, the risks of war, hostilities, invasion, act of foreign enemies, rebellion, revolution, insurrection or military or usurped power, civil war, riotcommotion or disorder (unless restricted to the Contractor's employees), and contaminationfrom any biological, nuclear fuel or nuclear waste or radioactive toxic explosive, or (b) a cause due solely to the design of the Works, other than the Contractor's design.

1.22) CONTRACTOR'S RISKS

All risks of loss of or damage to physical property and of personal injury and death which arise during and in consequence of the performance of the Contract other than the excepted risks are the responsibility of the Contractor.

1.23) INSURANCE

a) The Contractor shall provide, in the joint names of PPHC and the Contractor, insurance cover from the Start Date to the end of the Defects Liability Period, in the amounts and deductibles stated in the Contract Data for the following events which are due to the Contractor's risks:

i) loss of or damage to the Works, Plant and Materials;

ii) loss of or damage to Equipment; loss of or damage of property (except the Works, Plant, Materials and Equipment) inconnection with the Contract; and personal injury or death.

b) Policies and certificates for insurance shall be delivered by the Contractor to the Engineer –in-charge for the approval before the Start Date. All such insurance shall provide for compensation to be payable in the types and proportions of currencies required to rectify the loss or damage incurred.

c) If the Contractor does not provide any of the policies and certificates required, PPHC may affect the insurance which the Contractor should have provided and recover the premiums the PPHC has paid from payments otherwise due to the Contractor or, if no payment is due, the payment of the premiums shall be a debt due.

d) Alterations to the terms of an insurance shall not be made without the approval of the Engineer-in-charge.

e) Both parties shall comply with any conditions of the insurance policies.

1.24) CONTRACTOR TO CONSTRUCT THE WORKS

a) The Contractor shall construct and install the works in accordance with the Specification and Drawings, and as per instructions.

b) List of materials to be used in work is annexed with bid documents. Engineer-In chargereserves the right to change the brand of material to be used according to its availability in the market in case of emergency.

c) The Contractor shall commence execution of the Works on the Start Date and shall carry out the Works in accordance with the program submitted by the Contractor, as updated with the approval of the Engineer-in-charge and complete them by the Intended Completion Date.

1.25) APPROVAL BY THE ENGINEER

The Contractor shall submit Specifications and Drawings showing the proposed Works to PPHC to approve them if they comply with the requirements.

1.26) SAFETY

The Contractor shall be responsible for the safety of all activities & personnel at Site(s).

1.27) ACCESS TO THE SITE

a) The Contractor shall allow the Engineer and any person authorized by the Engineer access to the Site, to any place where work in connection with the Contract is being carried out or is intended to be carried out and to any place where materials or plant are being manufactured / fabricated / assembled for the works.

b) All the work under or in course of execution or executed in pursuance of the contract shall at all times be open to the inspection and supervision of the Engineer and his senior/subordinate and the Contractor shall at all times during the usual working hours and at all other times at which reasonable notice of the intention of the Engineer or his senior/subordinate to visit the works shall have been given to the Contractor, either himself to be present to receive orders and instructions or have a responsible agent, duly accredited in writing, present for that purpose. Orders given to Contractor's agent shall be considered to have the same force as if they had been given to the Contractor himself.

1.28) DISPUTE REDRESSAL MECHANISM

If any dispute or differences of any kind what-so-ever arise between the PPHC/ its authorized representatives and the System Integrator (SI), in connection with or arising out of this contract or the execution of work, these shall be resolved as under:-

Internal Escalation:

Whether before its commencement or during the progress of Project or after the termination, abandonment or breach of the contract, the dispute shall in the first instance supported with complete documents and further documents, if any, required by him, be referred for settlement to the Executive Engineer (PPHC) of the work and he shall, within a period of sixty (60) days after being requested in writing by the System Integrator (SI) to do so, convey his decision to the System Integrator (SI).

If the System Integrator (SI) is not satisfied with the decision of Executive Engineer (PPHC) or he does not convey his decision within a period of sixty (60) days then he shall make an appeal to the Chief Engineer, PPHC whose decision shall be final and binding upon the System Integrator (SI).

However, in case the work is already in process, the System Integrator (SI) shall proceed with the execution of the works, including maintenance thereof, pending receipt of the decision of the Executive Engineer/ Chief Engineer, as the case may be as aforesaid, with all due diligence.

Arbitration:

In case dispute between the PPHC and the SI, which has not been settled amicably, the SI can request the Managing Director, PPHC to refer the dispute for Arbitration under Arbitration and Conciliation Act, 1996. Such disputes shall be referred to the Arbitrator nominated by the PPHC. However, the System Integrator may request for changing the arbitrator, which shall be appointed by Hon'ble Punjab and Haryana High Court. The Indian Arbitration and Conciliation Act, 1996 and any statutory modification or reenactment thereof, shall apply to these arbitration proceedings. Arbitration proceedings will be held at Chandigarh/ S.A.S. Nagar, Punjab. The decision of the arbitrator shall be final and binding upon both the parties. All arbitration awards shall be in writing and shall state the reasons forthe award**.**

1.29) TIME CONTROL

a) Within the time stated in the Contract Data the Contractor shall submit to the Engineer for approval a Program showing the general methods, arrangements, order, and timing for all the activities in the Works along with monthly cash flow forecast.

b) An update of the Program shall be a program showing the actual progress achieved on each activity and the effect of the progress achieved on the timing of the remaining work including any changes to the sequence of the activities.

c) The Contractor shall submit to the Engineer, for approval, an updated Program at intervals no longer than the period stated in the Contract Data. If the Contractor does not submit an updated Program within this period, the Engineer may withhold the amount stated in the Contract Data from the next payment certificate and continue to withhold this amount until the next payment after the date on which the overdue Program has been submitted.

d) The Engineer's approval of the Program shall not alter the Contractor's obligations. TheContractor may revise the Program and submit it to the Engineer again at any time. A revised Program is to show the effect of Variations and Compensation Events.

1.30) EXTENSION OF THE INTENDED COMPLETION DATE

a) If the contractor shall desire an extension of the time for completion of the work on the ground of his having been unavoidably hindered in its execution or any other ground, heshall apply in writing to the Engineer (with corresponding time extension in Performance Security) within seven (7) days of the date of hindrance (but 30 days before the expiry of time limit) on account of which he desires such extent in as aforesaid" and the competent authority shall if in his opinion (which shall be final) reasonable grounds be shown therefore, authorize such extension of time as may in his

opinion be necessary or proper. No application for extension of time received late or addressed to the Sub-Divisional Engineer or any officer other than the Engineer shall be considered valid. If the contractor fails to apply for extension as aforesaid and the work is not completed within the time limit, the contract shall be determined absolutely after action under relevant clauses. Hinderance register shall be maintained by the contractor and reviewed by the PPHC Engineer-in- charge.

b) The Engineer shall decide whether and by how much to extend the Intended Completion Date within 21 days of the Contractor asking the Engineer for a decision upon the effect of a Compensation Event or Variation and submitting full supporting information. If the Contractor has failed to give early warning of a delay or has failed to cooperate in dealing with a delay, the delay by this failure shall not be considered in assessing the new Intended Completion Date.

c) In case if there is any delay, attributable either to the contractor or the PPHC or to both, and the contractor seeks and obtains extension of time for execution on that account, he will not be entitled to claim compensation of any nature, on the ground of such delay, in addition to the extension of time obtained by him.

1.31) DELAYS ORDERED BY THE ENGINEER

The Engineer may instruct the Contractor to delay the start or progress of any activity within the Works.

1.32) MANAGEMENT MEETINGS

a) Either the Engineer or the Contractor may require the other to conduct a management meeting. The business of a management meeting shall be to review the plans for remaining work and to deal with matters raised in accordance with the early warning procedure.

b) The Engineer shall record the business of management meetings and is to provide copies of his record to those attending the meeting and to the PPHC. The responsibility of the parties for actions to be taken is to be decided by the Engineer either at the management meeting or after the management meeting and stated in writing to all who attended the meeting.

1.33) EARLY WARNING (RISK & MITIGATION PLAN)

a) The Contractor is to warn the Engineer at the earliest opportunity of specific likely future events or circumstances that may adversely affect the quality of the work, increase the Contract Price or delay the execution of works. The Engineer may require the Contractor to provide an estimate of the expected effect of the future event or circumstance on the Contract Price and Completion Date. The estimate is to be provided by the Contractor as soon as reasonably possible.

b) The Contractor shall cooperate with the Engineer in making and considering proposals for how the effect of such an event or circumstance can be avoided or reduced by anyone involved in the work and in carrying out any resulting instruction of the Engineer.

1.34) QUALITY CONTROL

a) Contractor will enable material testing of work items if required by PPHC and engage Govt / NABL certified test laboratory reports. The Engineer shall check the Contractor's work and notify the Contractor of any Defects that are found. Such checking shall not affect the Contractor's responsibilities. The Engineer may instruct the Contractor to search for a Defectand to uncover and test any work that the Engineer considers may have a Defect.

b) The contractor shall permit the PPHC's Technical auditor to check the contractor's work and notify the Engineer and Contractor of any defects that are found. Such a check shall not affect the Contractor's or the Engineer's responsibility as defined in the Contract Agreement.

c) If the Engineer instructs the Contractor to carry out a test not specified in the Specification to check whether any work has a Defect and the test shows that it does, the Contractor shall pay for the test and any samples. If there is no Defect the test shall be a Compensation Event.

1.35) CORRECTION OF DEFECTS

a) The Engineer shall give notice to the Contractor of any Defects at any time during the construction or before the end of the Defects Liability Period,(which begins at Completion and is defined in the Contract Data). The Defects Liability Period shall be extended for as long as Defects remain to be corrected.

b) Every time notice of a Defect is given, the Contractor shall correct the notified Defect within the length of time specified by the Engineer's notice.

c) If the Contractor has not corrected a Defect within the time specified in the Engineer's notice, the Engineer will assess the cost of having the Defect corrected and recover twice of this amount from the payments due to the Contractor or retention money.

Note: Where in certain cases, the technical specifications provide for acceptance of works within specified tolerance limits at reduced rates, Engineer will certify payments to Contractor accordingly and the Contractor shall be bound by the same.

1.36) COST CONTROL

a) Bill of Quantities

i. The Bill of Quantities shall contain items for the Supply, installation, testing, commissioning and comprehensive maintenance work to be done by the contractor as per DNIT requirements and specifications.

ii. The Bill of Quantities is used to calculate the Contract Price. The Contractor is paid for the quantity of the work done at the rate in the Bill of Quantities for each item.

b) Changes in the Quantities

i. PPHC shall have the right to make any variations, alterations to the work or substitutions for the original quantities, specifications, drawings, designs and instructions that may appear to be

necessary or advisable during the progress of the work, and the Contractor shall be bound to carry out the work in accordance with any instructions which may be given to him in writing, signed by the Engineer. Such alterations/ additions/substitutions shall not invalidate the contract and shall be carried out by the Contractor on the same conditions in all respect on which he agreed to do the main work. The time of completion of the work shall be extended in the proportion that the altered, additional or substituted works bears to the original contract work and the certificate of the Engineer shall be conclusive as to such proportion.

ii. In case quantities of DNIT/ exceed beyond 10% then the payment shall be made at the rate decided by the MD, PPHC but it would not be above quoted rate.

iii. NS extra item as approved by the Managing Director, PPHC based on prevailing market rates at the time of execution.

iv. If requested by the Engineer, the Contractor shall provide the Engineer with a detailed cost breakdown of any composite rate in the Bill of Quantities.

c) Variations

i. All Variations shall be included in updated Programs produced by the Contractor.

ii. Items of bill of quantity may vary by +/- 25% at the same rates and terms & conditions of the agreement**.** Completion schedule shall vary accordingly as per PPHC.

d) Payments for Variations

i. The Contractor shall provide the Engineer with a quotation (with breakdown of unit rates) for carrying out the Variation when requested to do so by the Engineer. The Engineershall assess the quotation, which shall be given within seven days of the request or within any longer period stated by the Engineer and before the Variation is ordered.

ii. If the work in the Variation corresponds with an item description in the Bill of Quantities and if, in the opinion of the Engineer, the quantity of work above the limit stated in Sub Clause 45 (b) or the timing of its execution do not cause the cost per unit of quantity to change, the rate in the bill of Quantities shall be used to calculate the value of the Variation. If the cost per unit of quantity changes, or if the nature or timing of the work in the Variation does not correspond with items in the Bill of Quantities, the quotation by the Contractor shall be in form of new rates for the relevant items of work.

iii. If the Contractor's quotation is unreasonable, the Engineer may order the Variation and make a change to the Contract Price which shall be based on Engineer’s own forecast of the effects of the Variation on the Contractor's cost.

iv. If the Engineer decides that the urgency of varying the work would prevent a quotationbeing given and considered without delaying the work, no quotation shall be given and the Variation shall be treated as a Compensation Event.

v. The Contractor shall not be entitled to additional payment for costs that could have been avoided by giving early warning.

e) Payment Certificates

i. The Contractor shall submit to the Engineer monthly statements/bills of the estimated value of the work completed less the cumulative amount certified previously, along with copies of the following documents:-

a) Measurements and quantities of items of workdone since last bill.

b) Copies of quality control tests in specified format covering the work done since last bill.

c) Copies of instructions recorded in the instruction book containing the instructions and compliance made thereof, covering the work done since last bill. The Contractor shall submit all bills on proper printed invoice format as regulated by government or printed formsto be had on application from the office of engineer.

ii. In case the contractor is unable to prepare the bill, he shall request in writing to the Engineer-in-charge for preparation of his bill and the measurements made by the Junior Engineer/Sub Divisional Engineer shall be final and binding upon the contractor.

iii. The Engineer shall check the Contractor's monthly statements within 30 days and certify the amount to be paid to the Contractor after taking into account any deductions for retention, other recoveries in terms of the contract and taxes at source, as applicable underthe law, and the credit or debit for the month in question in respect of materials for the works in the relevant amounts. The Contractor shall be required to sign the corrections made, if any, in token of acceptance of the same, before release of payment.

iv. The value of work executed shall be determined by the Engineer.

v. The value of work executed shall include the valuation of Variations and Compensation Events. The Contractor shall be deemed to have waived off all claims not included in such bills/statements and will have no right to enforce any claim not so included, whatsoever be the circumstances.

vi. The Engineer may exclude any item certified in a previous certificate or reduce the proportion of any item previously certified in any certificate in light of later information

d) Payments

i. PPHC shall pay the Contractor the amounts as certified by the Engineer in Charge.

ii. Any excess payment made to the Contractor inadvertently or otherwise under this contract or on any account whatsoever and any other sum found to be due to the Government by the Contractor in respect of this contract or any other contract or work-order or any account whatsoever may be deducted from any sum payable by the Government to the Contractor either in respect of this contract or any work order or contract or any other account by any other department of the

Government.

iii. The Engineer may refuse or suspend payments on account of a work when executedby a firm or by a Contractor described in his bid as a firm, unless receipts are signed by all the partners, or one of the partners or an authorized representative of the Contractor who produces a valid authority in writing enabling him to give effectual receipts on behalf of the firm.

iv. All the intermediate payments shall be regarded as payments by way of advance against the final payment only and not as payments for work actually done and completed ; and shall not preclude the requiring of bad, unsound , imperfect or unskilled work to be removed and taken away and reconstructed or re-erected or be considered as an admissionof the performance of the contract or any part thereof in any respect of the occurring of any claim , nor shall it conclude , determine or effect in any way the powers of the Engineer under these conditions or any of them as to the final settlement and adjustment of the accounts or otherwise or in any other way , vary or affect the contract.

v. The receipt of the Accountant or Clerk for any money paid by the Contractor to the department will not be considered as a receipt of payment to the Engineer and the Contractor shall be responsible for ensuring that he procures a receipt duly signed by the Engineer or his Sub Divisional Engineer.

vi. Whenever any claim for the payment of a sum of money arises out of or under the contract against the contractor, the PPHC shall be entitled to recover such sum by appropriating in whole or in part, the security, if any, deposited by the contractor, and for thepurpose aforesaid, shall be entitled to sell and/or realize securities forming the whole or partof any such security deposit. In the event of the security being insufficient, the balance and ifno security has been taken from the contractor, the entire sum recoverable shall be recovered by appropriating any sum then due or which at any time thereafter may become due to the contractor under the contract or any other contract with the PPHC. If such sum even be not sufficient to cover the full amount recoverable, the contractor shall on demand pay to the purchaser the balance remaining due

1.37) COMPENSATION EVENTS

The following are Compensation Events unless they are caused by the Contractor:-

a) The Engineer does not give access to a part of the Site by the Site Possession Datestated in the Contract Data.
b) The PPHC modifies the schedule of other Contractors in a way which affects thework of the Contractor under the contract.
c) The Engineer orders a delay or does not issue drawings, specifications and /or instructions required for execution of works on time.
d) The Engineer gives an instruction for dealing with an unforeseen condition, causedby the PPHC, or other reasons.
e) The effect on the Contractor of any of the PPHC's Risks.
f) The Engineer unreasonably delays issuing a Certificate of Completion.

g) Other Compensation Events listed in the Contract Data or mentioned in the Contract**.**

h) If a Compensation Event would prevent the work being completed before the Intended Completion Date, the Intended Completion Date would be extended. The Engineer shall decide whether by how much the Intended Completion Date shall be extended.

1.38) TAXES

The rates quoted by the Contractor shall be deemed to be inclusive of all the taxes, levies, etc. Bidder shall mention rates of the tax slabs and the amount of tax for each item. Variations as notified by the concerned authority from time to time and also all the new taxes and levies that may be imposed shall be considered as payable for the performance of this Contract. Taxes shall be payable as per actual rates prevailing at the time of invoicing.

The Contractor shall comply with the proper bye-laws and legal orders of the local body or public authority under the jurisdiction of which the work is executed and pay all fees and charges for which he may be liable. Nothing extra shall be payable on this account

1.39) PERFORMANCE SECURITY

a) PPHC shall take Performance Bank Guarantee (PBG) amounting to 5% of the entire contract value and MPLS connectivity (including taxes & levies) as retention security by Contractor and ISP for Project cost and MPLS cost respectively, until 3 months beyond the warranty/ defect liability period that shall cover removal of defects, imperfections, or taking remedial measures in the work executed. This retention shall be inclusive of the initial bid security deposited by the bidder..

b) 50% of the PBG amount shall be released to the Contractor after three months of completion of the implementation of the Project / Work (Go-Live) as certified satisfactory by the Engineer with respect to removal of all defects, imperfections, short comings and taking remedial measures, that may be necessary and after recording of final measurements of work done, for which the certificate of the Engineer would be conclusive. A fresh PBG of 50% of Contract value shall be submitted by contractor prior to release of this amount, which shall be released after three months of expiry of Warranty/ Defects Liability Period (Operations & comprehensive Maintenance) and after the Engineer has certified that all Defects notified by the Engineer to the Contractor before the end of this period have been corrected.

1.40) REFUND OF SECURITY

50% of the Security deducted from the bills shall be refunded to the Contractor after three months of completion of the implementation of the Project / Work (Go-Live) and balance 50% of the same will be refunded after three months of expiry of Warranty/ Defects Liability Period (Operations & Comprehensive Maintenance) and after the Engineer has certified that all Defects notified by the Engineer to the Contractor before the end of this period have been corrected.

1.41) LIQUIDATED DAMAGES

i. The Contractor shall ensure due diligence to achieve progress of work as per the milestones in the DNIT.

ii. In case of delay, the Contractor shall, notwithstanding issuance of a prior notice in this regard, pay prospectively as liquidated damages that PPHC may levy, for every week of the period for which the work remains un-commenced after 10 days of the issue of acceptance letter; or the minimum progress of work stated above is not achieved; or the work remains unfinished after the completion date.

iii. In case of continued default or shortfall in progress, PPHC may go on enhancing the levy of liquidated damages, each time limited to 1% of the amount of contract per week of further default subject to maximum limit of 10% the Contract value.

iv. The Chief Engineer, PPHC on representation from Contractor, may reduce the amount of liquidated damages after passing speaking orders and his decision in writing shall be final and binding.

v. If the Intended Completion Date is extended after liquidated damages have been paid, PPHC shall correct any overpayment of liquidated damages by the Contractor by adjusting the next payment certificate.

1.42) COST OF REPAIRS

Loss or damage to the Works or Materials to be incorporated in the Works between the Start Date and the end of the Defects Correction periods shall be remedied by the Contractor at the Contractor's cost if the loss or damage arises from the Contractor's acts or omissions.

1.43) FINAL ACCOUNT

The Contractor shall supply to the Engineer a detailed account of the total amount that the Contractor considers payable under the Contract before the end of the Defects Liability Period. The Engineer shall issue a Defect Liability Certificate and certify any final payment that is due to the Contractor within 60 days of receiving the Contractor's account if it is correct and complete. If it is not, the Engineer shall issue within 60 days a schedule that states the scope of the corrections or additions that are necessary. If the Final Account is still unsatisfactory after it has been resubmitted, the Engineer shall decide on the amount payable to the Contractor and issue a payment certificate, within 60 days of receiving the Contractor’s revised account.

1.44) OPERATING AND MAINTENANCE MANUALS

a) “As built” drawings and operating and maintenance manuals shall be submitted by the Contractor on commissioning milestone.

b) If the Contractor does not supply the Drawings and/or manuals by the dates stated in the Contract Data, or they do not receive the Engineer's approval, the Engineer shall withhold the Final bill amount from payments due to the Contractor.

1.45) TERMINATION

a) The Engineer on behalf of the PPHC or the Contractor may terminate the Contract if the other party causes a fundamental breach of the Contract. For this purpose, 14 days' notice in writing shall be served by either party on the other party clearly mentioning the particular grounds of Breach of Contract with a copy to the PPHC

b) Fundamental breaches of Contract include, but shall not be limited to the following:

i. The Contractor stops work for 15 days when no stoppage of work is shown on the current programme and the stoppage has not been authorized by the Engineer;
ii. The PPHC or the Contractor is declared bankrupt or goes into liquidation other than for a reconstruction or amalgamation;
iii. The Engineer gives Notice that failure to correct a particular defect is a fundamental breach of Contract and the Contractor fails to correct it within a reasonable period of time determined by the Engineer;
iv. The Contractor does not maintain a security which is required;
v. The Contractor has delayed the completion of works by the number of days for which the maximum amount of liquidated damages can be paid as defined in the Contract Data;
vi. If the Contractor, in the judgment of the Engineer or the PPHC has engaged in corrupt or fraudulent practices in competing for or in executing the Contract.
vii. If the Contractor who has been given a notice in writing by the Engineer, fails to rectify, reconstruct or replace any defective work or continues the execution of work in an inefficient, improper, un workman — like manner or not in accordance with sound Engineering practices or without complying with the directions and requirements within a period of 10 days of the issue of said notice.
viii. If the Contractor being a company shall pass a resolution or a court shall make an order of the effect that the company shall be wound up or if a receiver or a manager on behalf of the credit or shall be appointed or if circumstances shall arise which entitle the court of creditor to appoint a receiver or manager or to make a winding up order.
ix. If the contractor who has been given notice in writing, fails to speed the work to complete it within time frame and the Engineer opines that work will not be completed in time.
x. Notwithstanding the above, the Engineer on behalf of the PPHC may terminate the Contract for convenience
xi. If the Contract is terminated the Contractor shall stop work immediately, make the Site safe and secure and leave the Site as soon as reasonably possible.
xii. After the termination of the contract under this clause, the PPHC shall be at liberty to get the

balance work executed through some other contractual agency or through departmental means or to abandon the balance work altogether or to modify the design and scope of the work in any manner. The Contractor shall have no claim against the PPHC in this regard.

1.46) PAYMENT UPON TERMINATION

a) If the Contract is terminated because of a fundamental breach of Contract by the Contractor, the Engineer shall issue a certificate for the value of the work done less advance payments received upto the date of the issue of the certificate, less other recoveries due in terms of the contract, less taxes due to be deducted at source as per applicable law and less 5% of the contract value towards compensation for the breach of contract. The total amount of liquidated - damages and compensation for breach of contract shall, however, be limited to 10% of the contract value or the amount available with the Engineer (in the shape of retention money, performance security and due amount of work done if any), whichever is less. The requisite amount for which the Contractor may become liable shall be realized by encasing the performance security furnished by the Contractor and/ or from other amounts due to the Contractor in respect of this work.

b) If the Contract is terminated at the PPHC's convenience or because of a fundamental breach of Contract by the PPHC, the Engineer shall issue a certificate for the value of the work done. This work value shall take into account the cost of balance material brought by the Contractor and available at site, the reasonable cost of removal of Equipment, repatriation of the Contractor's personnel employed solely on the Works, and the Contractor's costs of protecting and securing the works. The advance payment received upto to the date of the certificate, other recoveries due in terms of the contract and the taxes due to be deducted at source as per applicable law, shall be detected from the work value.

c) No Compensation for Alteration in or Restriction in Works

If at any time after the commencement of the work the Government, for any reason whatsoever, does not require the whole Project/Work or part thereof to be carried out, the Engineer shall give notice in writing of the fact to the Contractor, who shall have no claim to any payment or compensation whatsoever on account of any profit or advantage, which he might have derived from the execution of the work in full, but which he did not derive in consequence of the full amount of work not having been carried out, neither shall he have any claim for compensation by reasons of any alteration having been made in the original specifications, drawings, designs and instructions, which shall involve any curtailment of the work originally contemplated.

1.47) PROPERTY RIGHTS

All materials on the Site, Plant, Equipment, Temporary Works and Works are deemed to be the property of the PPHC, if the Contract is terminated because of a Contractor's default.

1.48) SERVICE LEVEL STANDARDS & AGREEMENT:

Service Level Agreement (SLA) is to be signed between PPHC and the selected Bidder for the project. SLA defines the responsibility of the vendor in ensuring the project performance based on the agreed Indicators as detailed in the Agreement. This section defines various Service Level Indicators for the Project. Sufficient design considerations and requisite spare & manpower availability should be ensured to achieve the uptime standards.

PPHC would monitor the compliance of SLA achievements at regulated intervals and release payment for Operations & Maintenance accordingly.

The table below summarizes the indicative Performance Indicators for the services to be offered by the selected bidder. The equipment and services availability should be recorded electronically (through NMS including uptime %) and physically by the SI.

S. No.	SLA Parameter	SLA Target
1	Availability of each component in each quarter of year– • Surveillance Cameras • Traffic Management Cameras/ devices in field • Outdoor UPS Power • Outdoor Switches and all field equipment.	97%
2	Availability of each component in each quarter of year • Control Room Switches and other Active/ Passive components • NMS, VMS, ITMS • Video Wall • Network connectivity	99.5%
3	Availability of "Data Centre & Command and Control Centre (CCC) equipment, Storage and software	99.5%
4	Availability of O&M Personnel for call logging, troubleshooting, & rectification	99%

1.49) SEVERITY LEVEL DEFINITION

S. No	Severity	Issue Resolution Time
1	Level 1	Within 2 hours of occurrence
2	Level 2	Within 8 hours of occurrence
3	Level 3	Within 24 hours of occurrence

Level 1:	The service / network outage, security or performance relatedissues impacting the services availability/ performance and leading to unavailability of whole system**.**
Level 2:	The service / network outage, security or performance related issues impacting the services availability/performance and leading to unavailability of the services at one or more locations
Level 3:	The network outage, security or performance related issues impacting the network availability/performance as per SLA penalty chart below in clause 1.58

1.50) MEASUREMENT OF SLA

The monitoring & measurement of SLA and audit shall be performed by the actual status, enterprise management system logs, Service Provider network logs and decision of PPHC technical committee shall be final and binding in this regard.

The successful Bidder shall establish Enterprise & Network Management System (EMS & NMS) for monitoring and measurement of the SLA parameters identified for the project. The enterprise management system shall be able to configure IT policies on all devices. The systems implemented for the project shall conform to the open network management standards such as Simple Network Management Protocol (SNMP) and Remote Monitoring (RMON) features.

The successful Bidder shall provide the logs from time to time for monitoring and audit of the SLA parameters identified for the project operations. PPHC technical committee reserves the right to periodically change the measurement points and methodologies it uses without notice to the Service provider of the project.

1.51) NETWORK OPERATIONS MANAGEMENT:

The Contractor is required to provide an escalation matrix with contact details servicesrelated issues reported by the staff or any other related stakeholders of this project.

1.52) PENALTIES:

The successful Bidder shall be penalized on service disruption beyond permissible limits as mentioned in the Service level agreement. The upper limit of penalty would be capped at 10% of the OPEX value for each quarter. In case the calculated penalty crosses 10% penalty of the OPEX value in 2 subsequent quarters, GSCDCL reserves the right to unconditionally invoke the termination clause.

Description	Availability	Penalty Value
Availability of each component in each quarter of year:– Surveillance Cameras Traffic Management Cameras/ devices in field Outdoor UPS Power Outdoor Switches and all field equipment.	Within SLA	Nil
	Between 96% to <97%	0.1% on the OPEX payable for respective quarter
	For Every 0.5% drop from <96%	Additional 0.2% on the OPEX payable for respective quarter
Availability of each component in each quarter of year:-	Within SLA	Nil
	Between 98.5% to <99.5%	0.1% on the OPEX payable

Control Room Switches and other Active/ Passive components NMS, VMS, ITMS Video Wall Network connectivity	For Every 0.5% drop from <98.5%	Additional 0.2% on the OPEX payable for respective quarter
Availability of "Data Centre & Command and Control Centre (CCC) equipment, Storage and software	Within SLA	Nil
	Between 98.5% to <99.5%	0.1% on the OPEX payable for respective quarter
	For Every 0.5% drop from <98.5%	Additional 0.2% on the OPEX payable for respective quarter
Availability of personnel for logging calls and problem rectification	For <= 1% of the calls not getting responded in less than or equal to 30 minutes per quarter	Nil
	For > 1% of the calls not getting responded in less than or equal to 30 minutes per quarter	0.1% of the OPEX payable for respective quarter
Availability of Project Manager and Field Engineers having experience as mentioned in the Technical Bid of the agency	Within SLA	Nil
	Absence of for 01 day (upto 07 days)	Rs. 3000/- per day (For Project Manager) Rs. 1000/- per day, per person (For Field Engineer)
	Absence for more than 07 days (upto 14 days)	Rs. 5000/- per day (For Project Manager) Rs. 2000/- per day, per person (For Field Engineer)
	Absence for more than 14 days in a month	No payment for the month

Note: Penalties on account of service disruption in Network connectivity beyond permissible limits will be deducted from the payment of Bandwidth network provider.

1.53) OPERATIONAL PENALTIES

In the event the Service provider is unable to meet any one of the SLA parameters defined in this DNIT during two quarters in a year or four quarters during the three years of the contract, PPHC reserves the right to terminate the contract. **Mention of penalties served shall be reflected in any Experience/ Completion Certificate issued in lieu of this project.**

Note:

- If the original equipment is replaced by temporary substitute equipment within resolution period, the penalty will not be charged for one week from the date of substitution /replacement.

- If the same is not repaired/ replaced within one week, the penalty at the above rate will be charged thereafter for one week and the same shall be 2 (two) times thereafter.
- If the total penalty reaches an amount equal to or more than 10% of the order value, the same shall invoke from the PBG and the process of termination of service as defined in the DNIT may be initiated by PPHC.

1.54) PENALTIES FOR MISUSE

In case of misuse of data / network / equipment at the instance of the selected Bidder, the penalty imposed on the Bidder, without prejudice by PPHC/ GoP, other remedies under the Agreement shall be 200% of the final bid amount under the Agreement and also PPHC/ GoP may terminate the Agreement.

1.55) SPECIAL CONDITIONS OF CONTRACT:

a. Force Majeure

Neither party shall be responsible to the other for any delay or failure in performance of its obligations due to any occurrence commonly known as Force Majeure which is beyond the control of any of the parties, including, but without limited to, fire, flood, explosion, acts of God or any Governmental body, public disorder, riots, embargoes, or strikes, acts of military authority, epidemics, strikes, lockouts or other labour disputes, insurrections, civil commotion, war, enemy actions. If a Force Majeure arises, the Bidder shall promptly notify Tenderer in writing of such condition and the cause thereof. Unless otherwise directed by Tenderer, the successful Bidder shall continue to perform his obligations under the contract as far as is reasonably practical, and shall seek all reasonable alternative means for performance not prevented by the Force Majeure event. The successful Bidder shall be excused from performance of their obligations in whole or part as long as such causes, circumstances or events shall continue to prevent or delay such performance.

b. Intellectual Property Rights

All intellectual property rights for the work performed under this DNIT as far as data is concerned shall lie with PPHCs. This clause is applicable to all data in any form or format designed and developed for PPHC under this DNIT by the Bidder. The Bidder shall not use such data for any other purpose during and after the term of contract.

c. Indemnity

Successful Bidder/SI shall indemnify, protect and save PPHC against all claims, proceeding, liabilities, losses, costs (including legal costs), damages, expenses and action suits, resulting from infringement of any patent, trademarks, copyrights, any other statutory infringements in respect of all the hardware / software supplied by them or expenses whatsoever arising out of or resulting from any loss or damage to the property or personnel of PPHC, relating to the

performance of the Project whether or not such loss or damage is caused or contributed to by negligence or other default of SI

/ Vendors or their Authorized User.

d. SUB-CONTRACTING

The contractor shall be required to obtain consent from the PPHC for:

a) the sub-contracting of any part of the Works for which the Sub-contractor is named in the contract;
b) the provision of labour; and
c) the purchase of materials which are in accordance with the standards specified in the Contract.

Beyond this if the contractor proposes sub-contracting any part of the work during execution of works, because of some unforeseen circumstances to enable him to complete the work as per terms of the contract, the Engineer will consider the followingbefore according approval:

- The contractor shall not sub-contract the whole of the Works.
- The contractor shall not sub-contract any part of the Work without prior consent of the Engineer. Any such consent shall not relieve the contractor from any liability or obligations under the contract and he shall be responsible for the acts, defaults and neglects of any sub-contractor, his agents or workmen as fully as if they were the acts, defaults or neglects of the contractor, his agents or workmen.
- The Engineer should satisfy whether (a) the circumstances warrant such sub-contracting; and (b) the sub-contractors so proposed for the Work possess the experience, qualifications and equipment necessary for the job proposed to be entrusted to them in proportion to the quantum of work to be sub-contracted.

If payments are proposed to be made directly to that sub-contractor, this should be subject to specific authorization by the prime contractor so that this arrangement does not alter the contractor's liability or obligations under the contract.

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e. Validity of Licenses

Bidders should guarantee that any software supplied to PPHC and to any other location as part of the contract mentioned in this DNIT are licensed and legally obtained. Licenses supplied for all components including all hardware and software should be “Permanent, Perpetual and Lifetime Licenses”, which should enable the department to use the same without any restrictions during the project as well as after the contract with the selected Bidder is over

f. Transfer of Ownership

i. System Integrator (SI) must transfer all titles to the assets and goods procured for the purpose of the project to the PPHC at the time of Acceptance of System. This includes all

licenses, titles, source code, certificates, hardware, devices, equipment's etc. related to the system designed, developed, installed and maintained by System Integrator (SI). System Integrator (SI) is expected to provide source code, transfer IPR and ownership right of only those solutions which would be customized by System Integrator (SI) for the use of PPHC. For any pre-existing work, System Integrator (SI) and PPHC shall be held jointly responsible and its use in any other project by System Integrator (SI) shall be decided on mutual consent.

ii. Forthwith upon expiry or earlier termination of the Contract and at any other time on demand by the PPHC, System Integrator (SI) shall deliver to the PPHC all Documents provided by or originating from the PPHC and all Documents produced by or from or for System Integrator (SI) in the course of performing the Services, unless otherwise directed in writing by the PPHC at no additional cost. System Integrator (SI) shall not, without the prior written consent of the PPHC store, copy, distribute or retain any such Documents.

g. PROTECTION OF ENVIRONMENT:

The contractor shall take all reasonable steps to protect the environment on and off the Site and to avoid damage or nuisance to persons or to property of the public or others resulting from pollution, noise or other causes arising as a consequence of his methods of operation.

During continuance of the contract, the contractor and his sub-contractors shall abide at all times by all existing enactments on environmental protection and rules made thereunder, regulations, notifications and bye-laws of the State or Central Government, or local authorities and any other law, bye-law, regulations that may be passed or notification that may be issued in this respect in future by the State or Central Government or the local authority.

Salient features of some of the major laws that are applicable are given below:

The Water (Prevention and Control of Pollution) Act, 1974, this provides for the prevention and control of water pollution and the maintaining and restoring of wholesomeness of water. 'Pollution' means such contamination of water or such alteration of the physical, chemical or biological properties of water or such discharge of any sewage or trade effluent or of any other liquid, gaseous or solid substance into

water (whether directly or indirectly) as may, or is likely to, create a nuisance or render such water harmful or injurious to public health or safety, or to domestic, commercial, industrial, agricultural or other legitimate uses, or to the life and health of animals or plants or of aquatic organisms.

The Air (Prevention and Control of Pollution) Act, 1981, This provides for prevention, control and abatement of air pollution. 'Air Pollution' means the presence in the atmosphere of any 'air pollutant', which means any solid, liquid or gaseous substance (including noise) present

in the atmosphere in such concentration as may be or tend tobe injurious to human beings or other living creatures or plants or property or environment.

The Environment (Protection) Act, 1986, This provides for the protection and improvement of environment and for matters connected therewith, &the prevention of hazards to human beings, other living creatures, plants and property. 'Environment' includes water, air & land the inter-relationship which exists among and between water,air and land, and human beings, other living creatures, plants, micro-organism and property.

The Public Liability Insurance Act, 1991, This provides for public liability insurance for the purpose of providing immediate relief to the persons affected by accident occurring while handling hazardous substances and for matters connected herewith or incidental thereto. Hazardous substance means any substance or preparation, which is defined as hazardous substance under the Environment (Protection) Act 1986, and exceeding such quantity as may be specified by notification by the Central Government.

1.56) LABOUR LAWS

1. Labour:

The Contractor shall, unless otherwise provided in the Contract, make his own arrangements for the engagement of all staff and labour, local or other, and for their payment, housing, feeding and transport.

The Contractor shall, if required by the Engineer, deliver to the Engineer a return in detail, in such form and at such intervals as the Engineer may prescribe, showing the staff and the numbers of the several classes of labour from time to time employed by the Contractor on the Site and such other information as the Engineer may require.

2. Compliance with labour regulations:

During continuance of the contract, the Contractor and his sub-contractors shall abide at all times by all existing labour enactments and rules made thereunder, regulations, notifications and bye laws of the State or Central Government or local authority and any other labour law (including rules), regulations, bye laws that may be passed or notification that may be issued under any labour law in future either by the State or the Central Government or the local authority. Salient features of some of the major labour laws that are applicable to construction industry are given below. The Contractor shall keep the PPHC indemnified in case any action is taken against the PPHC by the competent authority on account of contravention of any of the provisions of any Act or rules made thereunder, regulations or notifications including amendments. If the PPHC is caused to pay or reimburse, such amounts as may be necessary to cause or observe, or for non-observance of the provisions stipulated in the notifications/bye laws/Acts/Rules/regulations including amendments, if any, on the part of the Contractor, the Engineer/PPHC shall have the right to deduct any money due to the Contractor including his amount of performance security. The

PPHC/Engineer shall also have right to recover from the Contractor any sum required or estimated to be required for making good the loss or damage suffered by the PPHC.

The employees of the Contractor and the Sub-Contractor in no case shall be treated asthe employees of the PPHC at any point of time.

3. SALIENT FEATURES OF SOME MAJOR LABOUR LAWS.

a) **Workmen Compensation Act 1923:** The Act provides for compensation in case of injury by accident arising out of and during the course of employment.

b) **Payment of Gratuity Act 1972:** Gratuity is payable to an employee under the Act on satisfaction of certain conditions on separation if an employee has completed 5 years' service or more or on death the rate of 15 days wages for every completed year of service. The Act is applicable to all establishments employing 10 or more employees.

c) **Employees P.F. and Miscellaneous Provision Act 1952:** The Act Provides for monthly contributions by PPHC plus workers @ 10% or 8.33%. The benefits payable under the Act are:

(i) Pension or family pension on retirement or death, as the case may be.

(ii) Deposit linked insurance on the death in harness of the worker.

(iii) Payment of P.F. accumulation on retirement/death etc.

d) **Maternity Benefit Act 1951:** The Act provides for leave and some other benefits to women employees in case of confinement or miscarriage etc.

e) **Contract Labour (Regulation & Abolition) Act 1970:** The Act provides for certain welfare measures to be provided by the Contractor to contract labour and in case the Contractor fails to provide, the same are required to be provided, by the Principal PPHC by Law. The Principal PPHC is required to take Certificate of Registration and the Contractor is required to take license from the designated Officer. The Act is applicable to the establishments or Contractor of Principal PPHC if they employ 20 or more contract labour.

f) **Minimum Wages Act 1948:** The PPHC is supposed to pay not less than the MinimumWages fixed by appropriate Government as per provisions of the Act if the employmentis a scheduled employment. Construction of Buildings, Roads and Runways are scheduled employments.

g) **Payment of Wages Act 1936:** It lays down as to by what date the wages are to be paid, when it will be paid and what deductions can be made from the wages of the workers.

h) **Equal Remuneration Act 1979:** The Act provides for payment of equal wages for work of equal nature to Male and Female workers and for not making discrimination against Female employees in the matters of transfers, training and promotions etc.

i) **Payment of Bonus Act 1965:** The Act is applicable to all establishments employing 20 or more employees. The Act provides for payments of annual bonus subject to a minimum of 8.33% of wages and maximum of 20% of wages to employees drawing Rs.21,000/-per month or less. The bonus to be paid to employees getting Rs.7000/- per month or above

upto Rs.21,000/- per month shall be worked out by taking wages as Rs.7000/-per month only. The Act does not apply to certain establishments. The newly set-up establishments are exempted for five years in certain circumstances. Some of the State Governments have reduced the employment size from 20 to 10 for the purpose of applicability of this Act.

j) **Industrial Disputes Act 1947:** The Act lays down the machinery and procedure for resolution of Industrial disputes, in what situations a strike or lock-out becomes illegal and what are the requirements for laying off or retrenching the employees or closing down the establishment.

k) **Industrial Employment (Standing Orders) Act 1946:** It is applicable to all establishments employing 100 or more workmen (employment size reduced by some of the States and Central Government to 50). The Act provides for laying down rules governing the conditions of employment by the PPHC on matters provided in the Act and get the same certified by the designated Authority.

l) **Trade Unions Act 1926:** The Act lays down the procedure for registration of trade unions of workmen and PPHCs. The Trade Unions registered under the Act have been given certain immunities from civil and criminal liabilities.

m) **Child Labour (Prohibition & Regulation) Act 1986:** The Act prohibits employment of children below 14 years of age in certain occupations and processes and provides for regulation of employment of children in all other occupations and processes. Employment of Child Labour is prohibited in Building and Construction Industry.

n) **Inter-State Migrant workmen's (Regulation of Employment & Conditions of Service) Act 1979:** The Act is applicable to an establishment which employs 5 or more inter-state migrant workmen through an intermediary (who has recruited workmen in one state for employment in the establishment situated in another state). The Inter- State migrant workmen, in an establishment to which this Act becomes applicable, are required to be provided certain facilities such as housing, medical aid, traveling expenses from home upto the establishment and back, etc.

o) **The Building and Other Construction workers (Regulation of Employment and Conditions of Service) Act 1996 and the Cess Act of 1996:** All the establishments who carry on any building or other construction work and employs 10 or more workers are covered under this Act. All such establishments are required to pay cess at the rate not exceeding 2% of the cost of construction as may be modified by the Government. The PPHC of the establishment is required to provide safety measures at the Building or construction work and other welfare measures, such as Canteens, First-Aid facilities, Ambulance, Housing accommodations for workers near the work place etc. The PPHC to whom the Act applies has to obtain a registration certificate from the Registering Officer appointed by the Government.

p) **Factories Act 1948:** The Act lays down the procedure for approval at plans before setting up

a factory, health and safety provisions, welfare provisions, working hours, annual earned leave and rendering information regarding accidents or dangerous occurrences to designated authorities. It is applicable to premises employing 10 persons or more with aid of power or 20 or more persons without the aid of power engaged in manufacturing process.

1.57) LIMITATION OF SYSTEM INTEGRATOR (SI)'S LIABILITY

Limitation of System Integrator (SI)'s Liability towards the PPHC:

i. Neither Party shall be liable to the other Party for any indirect or consequential loss or damage (including loss of revenue and profits) arising out of or relating to the Contract. The limitation of liability upto 100% of project value.

ii. Except in case of gross negligence or willful misconduct on the part of System Integrator (SI) or on the part of any person or company acting on behalf of System Integrator (SI) in carrying out the Services,

iii. System Integrator (SI), with respect to damage caused by System Integrator (SI) to PPHC's property, shall not be liable to PPHC:

a. for any indirect or consequential loss or damage; and

b. for any direct loss or damage that exceeds (A) the total payments payable under the Contract to System Integrator (SI) hereunder, or (B) the proceeds System Integrator (SI) may be entitled to receive from any insurance maintained by System Integrator (SI) to cover such a liability upto project value.

iv. This limitation of liability shall not affect System Integrator (SI) liability, if any, for damage to Third Parties caused by System Integrator (SI) or any person or company acting on behalf of System Integrator (SI) in carrying out the Services.

1.58) ADDITIONAL CONDITIONS:

1) OEM certified Compliance Sheet for the products mentioned in the BOQ must be submitted alongwith tender.
2) The contractor shall not be entitled for any payment on account of work done till the agency signs the agreement and the same is accepted by the competent authority. The payment will be released on Pro-Rata basis.
3) Unless otherwise specified in the schedule of quantities, the rates for respective items shall be all inclusive and apply to the following: -
4) All lifts & all heights, floors including terrace, leads and depths.
5) All taxes, duties, transportation charges up to site, octroi, insurance charges, tools and plants, labour, tools and plants and other inputs involved in the execution of the item.
6) Any of the conditions and specifications mentioned in the tender documents.

7) Performance test of the entire installation(s) before the work is finally accepted.
8) All incidental charges for cartage, storage and safe custody of materials brought to site.
9) No escalation over the quoted rates shall be admissible during execution of work.
10) The contractor shall be responsible for the watch and ward / guard of the material/services, safety of all fittings arranged/ installed by him against pilferage till the work is handed over to the Department. No extra payment shall be made on this account.
11) The work should be planned in a systematic manner so that trench/ chase cuttings in theroads is minimized. The contractor shall ensure proper co-ordination of various authorities viz. Municipal Corporation, Deptt. Of sanitary & water supply/horticulture & electricity etc.
12) Contractor shall take all precautionary measures to avoid any damage to Govt. infrastructure. All necessary arrangement shall be made at his own cost.
13) The contractor shall procure the required materials in advance so that there is sufficient time for testing of the materials and approval of the same before use in the work.
14) All material shall only be brought at site as per program finalized with the Engineer-in- Charge. Any pre-delivery of the material not required for immediate consumption shall not be accepted and thus not paid for.
15) The Contractor will have to submit 'As built drawings' and Operating/Maintenance Manual (in duplicate) after completion of work. Final payment of the work will be released after submission of these documents.
16) In case the firm does not adhere to the OEM's maintenance schedule attending of complaints as mentioned above, the same shall be got done and attended to at the risk and cost of firm and amount the expenditure incurred will be recovered from the firm from his pending dues/ security deposit.
17) Works shall be executed strictly in accordance with the specifications mentioned in this DNIT and as per Punjab Public Works Department Specifications wherever applicable and to the entire satisfaction of the Engineer-in-charge.
18) The Executive Engineer reserves the option to take away any item of work or any part thereof at any time during the currency of contract and to reallot it to any other agency with due notice to the Contractor without liability of any kind of payment of any compensation.
19) Any material left at site on month after completion of work shall become the property of the PPHC and no payment shall be made to the contractor for the material.
20) The contractor has to make own arrangement for all items/all materials required directly or indirectly for completion of work at his own cost.
21) Surplus earth from excavation of foundations shall be placed by the contractor at the places approved by the Engineer-in-charge.
22) The contractor will make his own arrangements for water required for works. If he does not arrange water at his own level and use water from some Govt. sources, recovery as applicable shall be made.

23) The arrangement of Electric supply required for execution of work shall be made by the contractor at own cost.
24) No compensation shall be payable for any damaged caused to materials, unfinished or finished items or work due to rain, floods or any other act of God. The contractor shall ensure due insurance for such risks.
25) The contractor will arrange his own tools and plants required for work.
26) The jurisdiction of Civil Court for matters under dispute shall be SAS Nagar, Punjab, India.
27) The Contracting Agency will supply its Employee Provident Fund Code number before execution of Contract Agreement with PPHC and will submit to PPHC clearance certificate from the Regional Provident Fund Commissioner Authorities from time to time. Security of the Contracting agency will be released only after the Contracting agency willsupply NOC from the Regional Provident Fund Commissioner Authorities.
28) Any item not existing in this DNIT but actually executed at site as per CSR item then it shall be paid at the same rate at which the work has been allotted to the agency but in any case not above the CSR.
29) Rate of NS Item shall be submitted by the Contractor for approval on the basis of prevailing market rates at the time of execution.
30) The agency will follow the COVID-19 guidelines issued by the Govt. applicable at the time of execution of work and as amended from time to time.
31) The electrical will shall be carried out as per Punjab PWD specifications.
32) Although the bidders are supposed to submit MAF of hardware & Software from the OEMs, the SI may also provide any brand of hardware & Software out of the technically complied brands offered by other bidders, with the consent of PPHC in the favour of project.
33) The PPHC may ask the SI to arrange MPLS connectivity from any of the technically qualified ISPs.
34) Quantity of any item can be increased or decreased/ withdrawn at any stage.

SECTION-II
BILL OF QUANTITIES:

Bidder shall enclose the unpriced Bill of quantity with makes and models offered in the Technical Bid for Supply, Installation, Testing & Commissioning & Operations & Comprehensive Maintenance (SITC & O&M) of following equipment.

2.1) IMPLEMENTATION PHASE:

S. NO.	ITEM	UNIT	QTY
A	**Field Locations:**		
1.	SITC of Outdoor ANPR 2MP Box Camera with all accessories and external IR illuminator	Each	216
2.	SITC of Outdoor ANPR 2MP Bullet/ Box Camera with all accessories and external IR illuminator (For SVD)	Each	16
3.	SITC of Outdoor RLVD 2MP Bullet/ Box Camera with all accessories	Each	63
4.	SITC of RADAR/ Video Based Speed Violation Detection System for both sides i.e. To and Fro for 3 or 4 lane road including Traffic Management software with licences suitable for achieving desired accuracy during day/ night and in all weather conditions, complete with following items:- - RADAR-as required - PLB/ HDPE/ Flexible pipes from RADAR/ cameras to Main Outdoor Cabinet box (At junctions) - Armoured Cat-6 cables, Armoured Optical Fibre Cable, Power cabling from RADAR/ cameras to Main Outdoor Cabinet box, underground/ on surface (At junctions), - Termination boxes, - Managed L2 Network switches (Industrial grade), Passive items i.e. LIUs, Pigtails, Patch cords on Poles. - Any other item, if required to made the system operational to achieve the desired functionality. **Note:** ITMS Server, ANPR Cameras, Poles, Outdoor Cabinet boxes, Network switches & LIUs, LPUs in Main Outdoor Cabinet boxes, UPS & earthing excluded. It is mandatory to provide ANPR with Global Shutter for SVD system. (Attach item wise unpriced BOQ for all locations in Technical Bid)	Locations	2
5.	SITC Outdoor IP 4MP Surveillance Bullet/ Box Camera with all accessories	Each	104
6.	SITC of IP HD 2MP PTZ Camera with all accessories	Each	22
7.	SITC of Manageable Layer-2 Industrial grade Switch 8 Port PoE with surge protection	Each	23
8.	SITC of Manageable Layer-2 Industrial grade Switch 24 Port PoE with surge protection	Each	18
9.	SITC of Industrial grade Local Processing Unit (LPU)	Each	72
10.	SITC of 1KVA Online UPS for Outdoor Junction (120 minutes battery backup)	Each	41
11.	SITC of 2KVA Online UPS for Outdoor Junction (120 minutes battery backup)	Each	2
12.	SITC of Outdoor Cabinet Box with Accessories	Each	41
13.	SITC of Termination Box with Accessories	Each	91
14.	SITC of 6.3Mtr Octagonal Pole with foundation (with 6mtr Single Arm)	Each	49
15.	SITC of 6.3Mtr Octagonal Pole with foundation (with 8mtr Single Arm)	Each	35
16.	SITC of 6.3Mtr Octagonal Pole with foundation (with 8mtr Double Arm)	Each	7
17.	SITC of 4 mtr long Pole with foundation	Each	72
18.	SITC of 8 mtr long Pole with foundation	Each	22
19.	SITC of 3 Core 1.5mm Armoured copper cables	Mtrs	12000
20.	SITC of 2 Core 10 sqmm Armoured aluminum cables	Mtrs	6000
21.	SITC of 6 Core Optical fibre cables (OFC)	Mtrs	3000
22.	SITC of Cat 6 Armoured cables	Mtrs	11000

23.	SITC of PLB HDPE duct in trenchless laying	Mtrs	7500
24.	SITC of Electricity Meter with outdoor enclosure	Each	22
25.	SITC of G.I Rod earth System	Each	162
26.	SITC of Passive Components for field locations i.e. LIUs, Pigtail, OFC/Cat 6 patch cords etc.	Lots	23
B	**Data Centre and Integrated Command & Control Centre:**		
1.	SITC of IR IP Indoor dome cameras	Each	5
2.	SITC of Firewall for network protection in N+1 configuration	Each	2
3.	SITC of Rack mount Servers for Video Management in N+1 configuration with OS, MS office & any application	Each	6
4.	SITC of Video Analytics server	Each	1
5.	SITC of Video Management Software	Lot	1
6.	SITC of Video Analytics software	Lot	1
7.	SITC of Traffic Management software with e-Challan module	Lot	1
8.	SITC of External enterprise grade storage for recording data with usable space for 35 days (Minimum 1000TB)	Lot	1
9.	SITC of PTZ 3-axis Joystick Controller with LED display. Same make of PTZ camera	Each	5
10.	SITC of 42 U Network Racks with KVM	Each	3
11.	SITC of Operator Workstations with dual monitors	Each	10
12.	SITC of 24 Port Managed L3 Network Switch	Each	2
13.	SITC of 24 Port aggregation L2 Managed Network switch	Each	2
14.	SITC of Passive Components for Command & Control Centre/ Server Room i.e. IO's, face plates, LIUs, Pigtail, HDMI Cables OFC/ Cat 6 patch cords etc. as required to make the system operational	Lot	1
15.	SITC of 55" LED Video display	Each	1
16.	SITC of 55" LED based Video Wall with Controller in 5x2 grid	Set	1
17.	SITC of Multi-function Laser Printer	Each	1
18.	SITC of 2x20 KVA UPS with battery backup of 01 Hr	Each	1
19.	SITC of 62 KVA DG set in soundproof canopy	Each	1
20.	SITC of 2x5 TR Precision Airconditioning for Data Centre in redundant mode	Each	1
21.	SITC of 2.0 Ton rating split AC	Each	2
22.	Timer for 2 nos. Split ACs	Each	1
23.	SITC of Chemical Earthing system	Each	4
24.	Providing & Fixing of antistatic Raised flooring with suitable pedestal assembly.	Sft	400
25.	SITC of Biometric Access Control System (For 6 nos. doors) complete with controller, readers and cards etc.	System	1
26.	SITC of Anti-Virus end point software suite for servers, workstations, Local processing units and any other devices (Valid upto 3 years after Go-live)	Lot	1
27.	Establishment of Enterprise & Network Management System (EMS & NMS) for monitoring and measurement of the SLA parameters identified for the project	Job	1
	NOTE:- All components will be covered under free Comprehensive AMC for 12 months, by deploying minimum 01no. Project Manager and 04 nos. field Engineers including T&P and sky lift as required. Any additional manpower, if required to maintain SLA will be provided by the SI at his own cost**.**		

2.2) POST IMPLEMENTATION:

1	Annual comprehensive Maintenance for 2 years for all equipment mentioned above (Table 2.1), after expiry of free AMC of One year (Including deployment of minimum manpower, required tools and plants and sky lift as required).	Job	01
2	MPLS connectivity at Edge locations with required bandwidth & redundancy connected with Command & Control Centre, for 3 years from	Locations	20

	the date of Go-Live		
3	MPLS connectivity with required bandwidth at Command & Control Centre, for 3 years from the date of Go-Live	Location	01

2.3) DECLARATION OF BILL OF QUANTITIES

a) The bill of quantities shall be read in conjunction with the instruction to bidders, general and special conditions of contract, minimum technical specifications and other details given in DNIT in conjunction with the independent assessment done by bidder.

b) Delivery of equipment should be done as per Low Level Design / High Level Design submitted by bidder & approved by PPHC.

c) The quantities given in the bill of quantities are given to provide a common basis for bidding. The basis of payment will be the actual quantities of work ordered and carried out, as measured by the contractor and verified by the engineer in charge.

d) The rates and prices tendered in the priced bill of quantities shall include all constructional activity, tools & plant, labour supervision, materials, maintenance, insurance, profit, taxes and duties, together with the all general risks, liabilities and obligations set out in the contract.

e) Nothing extra will be paid due to losses/damages caused by Rains/Floods, War, epidemic, strike of the departmental officials or any other act of God or any other cause whatsoever. The same shall be got covered under appropriate insurance / coverage by bidder till making the system junction wise go-live.

f) The item for which no rate or price has been entered in will not be paid for by the PPHC when executed and shall be deemed to be covered by the other rates and prices in the Bill of Quantities. Nothing extra shall be paid on account of such items.

g) Bidder shall mention rates and values of all Taxes, levies.

SECTION-III
DETAILED SPECIFICATIONS:

The below specifications are minimum specifications and bidder is free to offer higher specification products. Compliance status of all parameters of specifications should be attached along with the Make & Models in the Technical bids

3.1) Outdoor ANPR 2MP Box Camera with all accessories

S. No.	Features	Min Technical Specifications and Standards
1	Image Sensor	1/3" progressive scan RGB CMOS
2	Operating Frequency	50 Hz
3	Day/ Night Operation	Yes with IR Cut Filter
4	Minimum Illumination	Color: 0.20 lux, B/W: 0.05 lux, 0 lux with IR illumination on IR - 50 Mtr External
5	Low light Capability	The camera shall be able to provide usable Color video in low light conditions
6	Lens	12~40mm motorized lens, P-IRIS lens
7	Type of Shutter	Rolling/ Global
8	Shutter Speed	1/12000 s or better to 1/5 s or better
9	Image Resolution	1920 x 1080, 1280 x 720, 800 x 450, 480 x 270, 320 x 240
10	Compression	H.265,H.264, Motion JPEG or better
11	Image Compression	H.265 or better .The camera support scene adaptive bitrate control, in order to lowering bandwidth and storage requirements. The camera shall support automatic dynamic GOP for optimal bitrate utilisation. The camera shall support automatic dynamic ROI to reduce bitrate in unprioritized regions
12	Frame Rate and Bit Rate	25 FPS at all resolutions with Controllable Bit Rate/ Bandwidth and Frame Rate
13	Video Streams	Minimum 3 Streams which can be individually control and configure.
14	Electronic Exposure & Control	Automatic/ Manual
15	Wide Dynamic Range	120 dB or better
16	Image settings	defogging, electronic image stabilization, barrel distortion correction, scene profiles
17	Connectors	1 Input & 1 Output for Alarm Interface
18	Audio	External microphone input, line output, Two way Audio
19	Event Triggers	Intelligent video, Edge event, External Input, Motion Detection, Network, Time scheduled, Manual Trigger, Alarm Input Trigger
20	Event Actions	File upload: sFTP/HTTPS, NAS and email Notification: email, SMTP, PTZ function, Edge Storage/ NAS Storage, Pre & Post Alarm Recording, Actions configurable by web interface, External Output activation
21	Edge Storage	Support minimum 128GB and 128GB SD card to be supplied along with camera.
22	Other feature	Support Focus assistant, Remote back focus
23	Storage	The Cameras shall have the feature to directly record the videos/ images onto NAS/SAN without any Software or integration
24	Protocols	IPv4, IPv6, HTTP, HTTPS, SSL/TLS, QoS,FTP, SMTP, SNMP, DNS, DynDNS, NTP, RTSP, RTP, TCP, UDP, IGMP, RTCP, ICMP, DHCP

25	Security	Password protection, IP address filtering, HTTPS encryption, AES-256, IEEE 802.1X, Digest authentication, User access log and unauthorized firmware installation, Camera should not support GB/T 28181 and there should not be any option to activate /de-activate this protocol through web browser or camera firmware, MAC address should be of same OEM. The camera should be equipped with Trusted Platform Module (TPM) /FIPS 140-2 certified/IEC-62443-4-1-2018 or the camera should have a cryptographically verifiable hardware module where a collection of certificates verify devise identification.
26	Firmware Upgrade	The firmware upgrade shall be done though web interface, The firmware shall be available free of cost
27	Logs	The camera shall provide a log file, containing information about the 250 latest connections and access attempts since the unit's latest restart
28	Other	The manufacturer must provide Long term support (LTS) firmware.
29	Interface	RJ 45, 100 Base TX
30	Power requirements for camera	PoE/PoE+ and AC/DC with suitable power adaptor as per OEM.
31	Operating Temperature	-5 °C to 50 °C
32	Operating Humidity	Humidity 10–90% RH
33	Certification	BIS, UL, CE, FCC, IEC/EN/UL 62368-1
34	Application Programmers Interface	The interface shall be available for integration with 3rd party analytics.
35	Housing	IP67/ IP 68 and IK10- rated including glass. Housing and IR should be from the same OEM.
36	Onvif Profile	S, G & T

3.2) Outdoor ANPR 2MP Bullet/ Box Camera with all accessories (For SVD)

S. No.	Features	Min Technical Specifications and Standards
1	Image Sensor	1/3" progressive scan RGB CMOS
2	Operating Frequency	50 Hz
3	Day/ Night Operation	Yes with IR Cut Filter
4	Minimum Illumination	Color: 0.20 lux, B/W: 0.05 lux, 0 lux with IR illumination on IR - 50 Mtr External
5	Low light Capability	The camera shall be able to provide usable Color video in low light conditions
6	Lens	12~40mm motorized lens, P-IRIS lens
7	Type of Shutter	Global
8	Shutter Speed	1/12000 s or better to 1/5 s or better
9	Image Resolution	1920 x 1080, 1280 x 720, 800 x 450, 480 x 270, 320 x 240
10	Compression	H.265,H.264, Motion JPEG or better
11	Image Compression	H.265 or better .The camera support scene adaptive bitrate control, in order to lowering bandwidth and storage requirements. The camera shall support automatic dynamic GOP for optimal bitrate utilisation. The camera shall support automatic dynamic ROI to reduce bitrate in unprioritized regions
12	Frame Rate and Bit Rate	25 FPS at all resolutions with Controllable Bit Rate/ Bandwidth and Frame Rate
13	Video Streams	Minimum 3 Streams which can be individually control and configure.
14	Electronic Exposure & Control	Automatic/ Manual
15	Wide Dynamic Range	120 dB or better
16	Image settings	defogging, electronic image stabilization, barrel distortion correction, scene profiles

17	Connectors	1 Input & 1 Output for Alarm Interface
18	Audio	External microphone input, line output, Two way Audio
19	Event Triggers	Intelligent video, Edge event, External Input, Motion Detection, Network, Time scheduled, Manual Trigger, Alarm Input Trigger
20	Event Actions	File upload: sFTP/HTTPS, NAS and email Notification: email, SMTP, PTZ function, Edge Storage/ NAS Storage, Pre & Post Alarm Recording, Actions configurable by web interface, External Output activation
21	Edge Storage	Support minimum 128GB and 128GB SD card to be supplied along with camera.
22	Other feature	Support Focus assistant, Remote back focus
23	Storage	The Cameras shall have the feature to directly record the videos/ images onto NAS/SAN without any Software or integration
24	Protocols	IPv4, IPv6, HTTP, HTTPS, SSL/TLS, QoS,FTP, SMTP, SNMP, DNS, DynDNS, NTP, RTSP, RTP, TCP, UDP, IGMP, RTCP, ICMP, DHCP
25	Security	Password protection, IP address filtering, HTTPS encryption, AES-256, IEEE 802.1X, Digest authentication, User access log and unauthorized firmware installation, Camera should not support GB/T 28181 and there should not be any option to activate /de-activate this protocol through web browser or camera firmware, MAC address should be of same OEM. The camera should be equipped with Trusted Platform Module (TPM) /FIPS 140-2 certified/IEC-62443-4-1-2018 or the camera should have a cryptographically verifiable hardware module where a collection of certificates verify devise identification.
26	Firmware Upgrade	The firmware upgrade shall be done though web interface, The firmware shall be available free of cost
27	Logs	The camera shall provide a log file, containing information about the 250 latest connections and access attempts since the unit's latest restart
28	Other	The manufacturer must provide Long term support (LTS) firmware.
29	Interface	RJ 45, 100 Base TX
30	Power requirements for camera	PoE/PoE+ and AC/DC with suitable power adaptor as per OEM.
31	Operating Temperature	-5 °C to 50 °C
32	Operating Humidity	Humidity 10–90% RH
33	Certification	BIS, UL, CE, FCC, IEC/EN/UL 62368-1
34	Application Programmers Interface	The interface shall be available for integration with 3rd party analytics.
35	Housing	IP67/ IP 68 and IK10- rated including glass. Housing, if required, and IR should be from the same OEM.
36	Onvif Profile	S & G

3.3) Outdoor RLVD 2MP Bullet/ Box Camera with all accessories

S. No.	Features	Min Technical Specifications and Standards
1	Image Sensor	Progressive scan RGB CMOS 1/2.8" or better
2	Minimum Illumination	Color: 0.20 lux B/W: 0.05 lux, 0 lux with IR illumination on
3	IR Illumination	Internal/ External Optimized IR with 40 meter range, 850nm, with adjustable angle and intensity or better
4	Lens	2.7-13.5 mm or better, Megapixel Lens with remote zoom and focus, P-IRIS lens,
5	Type of Shutter	Rolling/ Global
6	Shutter Speed	1/12000 s or better to 1/5 s or better
7	Image Resolution	1920x1080 or Better
8	Compression	H.265,H.264, Motion JPEG or better
9	Frame Rate and Bit Rate	25/30 FPS at all resolutions with Controllable Bit Rate/ Bandwidth

		and Frame Rate
10	GOP/ GOV	Ability to change the GOP/GOV Length to optimize the bandwidth and storage
11	Image reproduction	Compression, Color, Brightness, Sharpness, Contrast, Local contrast, White balance, Exposure control
12	Image Compression	The camera support scene adaptive bitrate control, in order to lowering bandwidth and storage requirements. The camera shall support automatic dynamic GOP for optimal bitrate utilisation. The camera shall support automatic dynamic ROI to reduce bitrate in unprioritized regions.
13	Video Streams	Minimum 3 Streams which can be individually control and configure.
14	Analytics	Inbuilt Motion Detection, Line crossing, Loitering, active tampering alarm
15	Aspect Ratio	16:9 or 9:16 without change in image quality or change in Image
16	Wide Dynamic Range	120 dB or Better
17	Privacy Masks	Required
18	Event Triggers	Analytics, Network disconnect, Alarm input, App event, Time schedule
19	Event Actions	File upload: sFTP/HTTPS, NAS and email Notification: email, SMTP, PTZ function, Edge Storage/ NAS Storage, Pre & Post Alarm Recording, Actions configurable by web interface, External Output activation
20	Edge Storage	Support minimum 128GB and 128GB SD card to be supplied along with camera
21	Storage	The Cameras shall have the feature to directly record the videos/ images onto Nas without any Software
22	Protocols	IPv4, IPv6, HTTP, HTTPS, SSL/TLS, QoS, FTP, SMTP, SNMP, DNS, DynDNS, NTP, RTSP, RTP, TCP, UDP, IGMP, RTCP, ICMP, DHCP
23	Text Overlay	Date & time, and a customer-specific text, camera name, graphical image etc.
24	Security	Password protection, IP address filtering, HTTPS encryption, AES-256, IEEE 802.1X, Digest authentication, User access log and unauthorized firmware installation, Camera should not support GB/T 28181 and there should not be any option to activate /de-activate this protocol through web browser or camera firmware, MAC address should be of same OEM. The camera should be equipped with Trusted Platform Module (TPM) /FIPS 140-2 certified/IEC-62443-4-1-2018 or the camera should have a cryptographically verifiable hardware module where a collection of certificates verify devise identification.
25	Audio streaming	Audio in, simplex, External microphone input or line input
26	Audio compression	AAC, G.711, G.726
27	Interface	RJ 45, 100 Base TX
28	Power requirements	PoE/PoE+ and AC/DC with suitable power adaptor as per OEM.
29	Operating Temperature	-5 °C to 50 °C
30	Operating Humidity	Humidity 10–90 RH
31	Protection	IP67/ IP68 and IK10- rated including glass
32	Certification	BIS, UL, CE, FCC, IEC/EN/UL 62368-1
33	Logs	The camera shall provide a log file, containing information about the 200 latest connections and access attempts since the unit's latest restart. The file shall include information about the connecting IP addresses and the time of connecting.
34	Other	The manufacturer must provide Long term support (LTS) firmware.
35	Onvif S & G & T	Required

3.4) IP HD 2MP PTZ Camera with all accessories

S. No.	Features	Min Technical Specifications and Standards
1	Image Sensor	1/3” Progressive Scan CMOS or better
2	Operating Frequency	50 Hz
3	Day/ Night Operation	Automatic with IR Cut Filter
4	Minimum Illumination	Color: 0.09 lux B/W: 0.02 lux, 0 lux with IR illumination on
5	high-speed pan-tilt functionality	360° endless pan range and a Tilt: -10 to +90°
6	Optical Zoom	Zoom: 40x optical, 12x digital
7	Lens	Varifocal, Minimum zoom 4.3-129 mm or better
8	Pan & Tilt	Pan: 360° endless, 0.05°–200°/s Tilt: -10 to +90°, 0.05°–140°/s
9	Image Resolution	1920 x 1080 or better
10	Compression	H.265,H.264, Motion JPEG or better
11	Frame Rate and Bit Rate	25/30 FPS at all resolutions with Controllable Bit Rate/ Bandwidth and Frame Rate. In CBR Priority to be defined for Video quality or frame rate and the bandwidth upper limit shall not exceed the defined limit
12	GOP/ GOV	Ability to change the GOP/GOV Length to optimize the bandwidth and storage
13	Video Streams	Minimum 2 Streams @ 1080P (1920x1080), 720P(1280× 720) minimum 25 fps which can be individually controlled and configured
14	Type of Shutter	Electronic/ Global
15	Shutter Speed	1/12000 s or better to 1/5 s or better
16	Electronic Exposure & Control	Automatic/ Manual
17	Wide Dynamic Range	120 dB or Better
18	BLC,HLC	Required
19	Electronic Image Stabilization	Required
20	Privacy Masks	Minimum 10 configurable zones or better
21	Preset Positions	Minimum 256 or better
22	Guard Tour	Minimum 6, max 15 minute duration each
23	Defog	Required
24	On-screen directional indicator	Required
25	Compression	The camera support scene adaptive bitrate control, in order to lowering bandwidth and storage requirements. The camera shall support automatic dynamic GOP for optimal bitrate utilisation. The camera shall support automatic dynamic ROI to reduce bitrate in unprioritized regions.
26	Event Triggers	Intelligent video, Edge event, External Input, Motion Detection, Network, Time scheduled, 3rd Party Analytics, Manual Trigger, Alarm Input Trigger, use Schedule.
27	Event Actions	File upload: sFTP/HTTPS, NAS and email Notification: email, SMTP, PTZ function, Edge Storage/ NAS Storage, Pre & Post Alarm Recording, Actions configurable by web interface, External Output activation.
28	Edge Storage	Support minimum 128GB and 128GB SD card to be supplied along with camera.
29	Storage	The Cameras shall have the feature to directly record the videos/ images onto Nas without any Software
30	Protocols	IPv4, IPv6, HTTP, HTTPS, SSL/TLS, QoS, FTP, SMTP, SNMP, DNS, DynDNS, NTP, RTSP, RTP, TCP, UDP, IGMP, RTCP, ICMP, DHCP

31	Text Overlay	Date & time, and a customer-specific text, camera name, graphical image etc.
32	Security	Password protection, IP address filtering, HTTPS encryption, AES-256, IEEE 802.1X, Digest authentication, User access log and unauthorized firmware installation, Camera should not support GB/T 28181 and there should not be any option to activate /de-activate this protocol through web browser or camera firmware, MAC address should be of same OEM. The camera should be equipped with Trusted Platform Module (TPM) /FIPS 140-2 certified/ PKI/ IEC-62443-4-1-2018 or the camera should have a cryptographically verifiable hardware module where a collection of certificates verify devise identification.
33	Logs	The camera shall provide a log file, containing information about the 250 latest connections and access attempts since the unit's latest restart
34	Other	The manufacturer must provide Long term support (LTS) firmware.
35	Interface	RJ 45, 100 Base TX
36	Enclosure	IK10 housing including glass, IP67- Metal casing (aluminium), repaintable skin cover, PVC Free
37	Mount	Wall / Pole Mount
38	Power requirements	PoE/PoE+ (802.3at) Class 4 and 12/24/48V AC/DC with suitable power adaptor as per OEM.
39	Inbuilt Analytics	Line crossing, Loitering, Auto tracking, Motion Detection
40	Operating Temperature	-5 °C to 50°C
41	IR illumination (Internal/External)	Optimized IR with power-efficient, long life 850 nm IR LEDs Range of reach 250 m or more depending on the scene
42	Operating Humidity	10–90% RH or better
43	Certification	BIS, UL, CE, FCC, IEC/EN/UL 62368-1
44	Onvif profiles	G, S & T

3.5) Outdoor IP 4MP Surveillance Bullet/ Box Camera with all accessories

S. No.	Features	Min Technical Specifications and Standards
1	Image Sensor	Progressive scan RGB CMOS 1/3" or better
2	Minimum Illumination	Color: 0.20 lux, B/W: 0.05 lux, 0 lux with IR illumination on
3	IR Illumination	Internal/ External Optimized IR with 40 meter range, 850nm, with adjustable angle and intensity or better
4	Lens	2.7~13.5 mm or better, IR corrected, Megapixel Lens with remote zoom and focus
5	Type of Shutter	Rolling/ Global
6	Shutter Speed	1/12000 s or better to 1/5 s or better
7	Image Resolution	2560x1440 or Better
8	Compression	H.265,H.264, Motion JPEG or better
9	Frame Rate and Bit Rate	25/30 FPS at all resolutions with Controllable Bit Rate/ Bandwidth and Frame Rate
10	GOP/ GOV	Ability to change the GOP/GOV Length to optimize the bandwidth and storage
11	Image reproduction	Compression, Color, Brightness, Sharpness, Contrast, Local contrast, White balance, Exposure control
12	Image Compression	The camera support scene adaptive bitrate control, in order to lowering bandwidth and storage requirements. The camera shall support automatic dynamic GOP for optimal bitrate utilisation. The camera shall support automatic dynamic ROI to reduce bitrate in unprioritized regions.

13	Video Streams	Minimum 3 Streams @ H.265, 2560x1440, 1080P (1920x1080), 720P (1280× 720), minimum 25/30 fps which can be individually control and configure.
14	Analytics	Inbuilt Motion Detection, Line crossing, Loitering, active tampering alarm
15	Aspect Ratio	16:9 or 9:16 without change in image quality or change in Image
16	Electronic Exposure & Control	Automatic/ Manual
17	Wide Dynamic Range	120 dB or Better
18	Privacy Masks	Required
19	Event Triggers	Intelligent video, Edge event, External Input, Motion Detection, Network, Time scheduled, Manual Trigger, Alarm Input Trigger
20	Event Actions	File upload: sFTP/HTTPS, NAS and email Notification: email, SMTP, PTZ trigger function, Edge Storage/ NAS Storage, Pre & Post Alarm Recording, Actions configurable by web interface, External Output activation
21	Edge Storage	Support minimum 128GB and 128GB SD card to be supplied along with camera.
22	Storage	The Cameras shall have the feature to directly record the videos/ images onto Nas without any Software
23	Protocols	IPv4, IPv6, HTTP, HTTPS, SSL/TLS, QoS, FTP, SMTP, SNMP, DNS, DynDNS, NTP, RTSP, RTP, TCP, UDP, IGMP, RTCP, ICMP, DHCP
24	Text Overlay	Date & time, and a customer-specific text, camera name, graphical image etc.
25	Security	Password protection, IP address filtering, HTTPS encryption, AES-256, IEEE 802.1X, Digest authentication, User access log and unauthorized firmware installation, Camera should not support GB/T 28181 and there should not be any option to activate /de-activate this protocol through web browser or camera firmware, MAC address should be of same OEM. The camera should be equipped with Trusted Platform Module (TPM) /FIPS 140-2 certified/IEC-62443-4-1-2018 or the camera should have a cryptographically verifiable hardware module where a collection of certificates verify devise identification.
26	Audio streaming	Audio in, simplex, External microphone input or line input
27	Audio compression	AAC, G.711, G.726
28	Interface	RJ 45, 100 Base TX
29	Power requirements	PoE/PoE+ and AC/DC with suitable power adaptor as per OEM.
30	Operating temperature	-5 °C to 50 °C
31	Operating Humidity	Humidity 10–90% RH
32	Protection	IP66/ IP67 and IK10- rated including glass
33	Certification	BIS, UL, CE, FCC, IEC/EN/UL 62368-1
34	Logs	The camera shall provide a log file, containing information about the 200 latest connections and access attempts since the unit's latest restart. The file shall include information about the connecting IP addresses and the time of connecting.
35	Other	The manufacturer must provide Long term support (LTS) firmware.
36	Onvif profiles	S, G & T

3.6) IP 2MP Indoor Dome Camera

S. No.	Features	Min Technical Specifications and Standards
1	Image Sensor	1/3" progressive scan RGB CMOS
2	Operating Frequency	50 Hz
3	Day/ Night Operation	Yes with Built-in IR Cut Filter

4	Minimum Illumination	Color: 0.20 lux, B/W: 0.05 lux, 0 lux with IR illumination on
5	IR Illumination	Built in Or External Optimized IR with 15 meter range
6	Lens	2.8/4-10 mm
7	Field of View	Horizontal field of view: 100°
8	Type of Shutter	Rolling/ Global
9	Shutter Speed	1/10000 s to 2 s
10	Image Resolution	1920x1080
11	Compression	H.265,H.264, Motion JPEG or better
12	Frame Rate and Bit Rate	Upto 25 FPS at all resolutions with Controllable Bit Rate/ Bandwidth and Frame Rate
13	GOP/ GOV	Ability to change the GOP/GOV Length to optimize the bandwidth and storage
14	Compression	The camera shall support for its H.265 implementation support scene adaptive bitrate control, in order to lowering bandwidth and storage requirements. The camera shall support automatic dynamic GOP for optimal bitrate utilisation. The camera shall support automatic dynamic ROI to reduce bitrate in unprioritized regions.
15	Motion Detection	Yes built in with multiple configurable areas in the video stream
16	Intelligent Analytics	Tampering alarm, line crossing
17	Aspect Ratio	16:9 or 9:16 without change in image quality or change in Image
18	Electronic Exposure & Control	Automatic/ Manual
19	Wide Dynamic Range	120 dB or Better
20	Privacy Masks	Required
21	Event Triggers	A. The camera shall be able to send and received trigger directly from any other camera without interface of VMS. B. Intelligent Video, Edge Storage Events, Software Alarms, Camera Tampering
22	Event Actions	File upload: sFTP/HTTPS, NAS and email Notification: email, SMTP, PTZ function, Edge Storage/ NAS Storage, Pre & Post Alarm Recording, Actions configurable by web interface, External Output activation
23	Edge Storage	Support minimum 128GB and 128GB SD card to be supplied along with camera.
24	Storage	The Cameras shall have the feature to directly record the videos/ images onto Nas without any Software
25	Protocols	IPv4, IPv6, HTTP, HTTPS, SSL/TLS, QoS,FTP, SMTP, SNMP, DNS, DynDNS, NTP, RTSP, RTP, TCP, UDP, IGMP, RTCP, ICMP, DHCP
26	Text Overlay	Date & time, and a customer-specific text, camera name, graphical image etc.
27	Security	Password protection, IP address filtering, HTTPS encryption, AES-256, IEEE 802.1X, Digest authentication, User access log and unauthorized firmware installation, Camera should not support GB/T 28181 and there should not be any option to activate /de-activate this protocol through web browser or camera firmware, MAC address should be of same OEM. The camera should be equipped with Trusted Platform Module (TPM) /FIPS 140-2 certified/IEC-62443-4-1-2018 or the camera should have a cryptographically verifiable hardware module where a collection of certificates verify devise identification.
28	Firmware Upgrade	The firmware upgrade shall be done though web interface, The firmware shall be available free of cost
29	Logs	The camera shall provide minimum 200 logs of latest connections, access attempts, users connected, changes in the cameras etc.
30	Other	The manufacturer must provide Long term support (LTS) firmware.

31	Interface	RJ 45, 100 Base TX
32	Enclosure	IP66 or better
33	Power requirements	PoE/PoE+/ AC/DC with suitable power adaptor as per OEM.
34	Operating Temperature	-5°C to 50 °C
35	Operating Humidity	Humidity 10–90% RH
36	Certification	BIS, UL, CE, FCC, IEC/EN/UL 62368-1
37	Mount	Wall/ Ceiling/ Surface
38	Onvif S,G,T	Required

3.7) Industrial grade Field Layer-2, 8 port POE Switch with Ethernet surge protection

S. No.	Parameter	Minimum Specifications
1.	General Performance	8 x 10/100/1000 Base-T PoE+ Ports auto negotiation Plus Loaded with SFP Trans receiver 2 x SFP Ports fully populated 2 x 1G SFP modules. (Each of RJ45 ports shall offer IEEE S02.3at PoE+ 30w per port) Switching fabric 20 Gbps or more
2.	Layer 2	802. 1Q VLAN on all ports with support for minimum 256 active VLANs or higher. Should support Improved resiliency with the support ITU-T standard or equivalent for ring topology which should provide 50ms ring convergence
3.	PoE	Switch should have minimum 240 W PoE power available or extra power injector should be provided in the junction box
4.	Layer 2	Spanning Tree Protocol as per IEEE 802.1d, 802.1s and802.1w
5.	Mounting	DIN mountable with accessories
6.	Protocols	Switch should support non-blocking throughput.
7.	L2	Switch should support classification and scheduling as per IEEE 802.1P
8.	QoS	Switch should support strict priority queuing or Policing or equivalent to guarantee that the highest-priority packets are serviced ahead of all other traffic.
9.	Certification	RoHS Compliant, IEEE 802.3af, 802.3at, NTP
10.	Environmental	Operating Temperature 0 deg C to (+) 70deg C with fan less design
		Relative Humidity of 5% or 95% Non-condensing
11.	Standard	FCC, EN61000-4-2 (ESD), EN61000-4 (RS), EN61000- 4-4 (EFT), EN 61000-4- 5 (Surge), EN61000-4-6 (CS), EN61000-4-8 IEEE 802.1D-2004 for Spanning Tree Protocol IEEE 802.1p for Class of Service ITU-T G.8032v1/v2 Ethernet Ring Protection Switching(ERPS) or Equivalent Ring Protection IEEE 802.3x for flow control IEEE802.3af / 802.3at for Power-over-Ethernet IEEE 802.1s for Multiple Spanning Tree Protocol IEEE 802.1w for Rapid Spanning Tree Protocol IEEE 802.1Q for VLAN Tagging IEEE 802.1X for authentication IEEE 802.1AB for LLDP (Link Layer Discovery Protocol)
12.	Security	Telnet/SSH, ARP, AAA radius server authentication TACACS+, SNTP, SNMPv3 based authentication and access security, RMON,LLDP, HTTP IPv6
13.	Surge Protector	Switch should support surge protection and shall be rugged in nature suitable for outdoor environment and highly effective for voltage surge and transient protection for RJ45 Ethernet-compatible equipment.
14.	Type	Plug-in type and shall be compatible with any RJ45

		10/100/1000 Tx Ethernet port without any operating power requirement.
15.	Compatibility	PoE Compatible: IEEE 802.3af and 802.3at compliant for PoE and PoE Plus.
16.	Test & Standards	EN61000-4-5/ IEC and EN55024/32/ IEC surge standards
17.	Standards	FCC, CE, ROHS Compliance

3.8) Industrial grade Field Layer-2, 24 port POE Switch with Ethernet surge protection

Sr. No	FEATURES	MINIMUM SPECIFICATION
1	Model	10BASE-T/100BASE-TX/1000BASE-T: 24 POE
		4 x SFP ports
2	Switch Category	Gigabit Managed Switches
3	Performance	Switching Capacity: 56Gbps
		Min. Packet Forwarding Rate: 41.7 Mbps
		MAC Address Table: 8K and 1.5 Mbits Packet Buffer.
		Static MAC Entries: 50 entries
		Power Supply: Internal AC Power Supply,
		Power Saving (Green)
		Power Budget: 240 Watts
4	Layer 2 Features	Spanning Tree (STP) 802.1d
		RSTP 802.1w , 802.1s MSTP
		Loopback Detection
		Link Aggregation LACP 802.3ad: Minimum 8 groups per device and 8 ports per group
		Port Mirroring : One-to-One / Many-to-One
		Jumbo Frame: 9K Bytes
		Dual image & Dual configuration or Signed image to prevent OS image tampering
		Multicast Filtering
5	Layer 2 Multicast Features	Internet Group Management Protocol (IGMP) Snooping : v1 /v2
		IGMP Snooping Groups : 256
6	Virtual LAN (VLAN) Features	Tagged VLAN 802.1Q, 256 Static VLANs
		Asymmetric VLAN or equivalent
7	Quality of Service (QoS) Features	No. of Queues: 8 Queues per port
		Traffic Prioritization 802.1p
		Strict Priority
		Weighted Round Robin (WRR) or Shaped Round Robin (SRR) or graceful congestion mgt
		QoS based on 802.1p Priority
		QoS based on Differentiated Services Code Point (DSCP)
		Port-based (Ingress/Egress, min granularity 10/100/1000 is 64 Kbps)
8	Access Control List (ACL) Features	Access Control List based on:
		• 802.1p priority
		• VLAN
		• MAC address• Ether type
		• IP address
		• DSCP
		• Protocol type
		• TCP/UDP port number
		• IPv6 Traffic Class
		Max. ACL Profiles / Rules: 50 / 768
9	AAA	802.1X Authentication
		• Supports local/RADIUS database
		• Supports port-based access control
		Mac 128 entries when using local database
		Support MD5 authentication or SLL or SSH for link security

10	Security Features	DoS attack prevention or equivalent
		Port Security: Max. 32 MAC Addresses per Port
		Broadcast/Multicast/Unicast Storm Control
		DHCP Server Screening
		ARP Spoofing Prevention
		Traffic segmentation
11	Environmental	Operating Temperature (-) 0 deg C to (+) 70deg C with fan less design
		Relative Humidity of 5% or 95% Non-condensing
12	Surge Protector	Switch should support surge protection and shall be rugged in nature suitable for outdoor environment and highly effective for voltage surge and transient protection for RJ45 Ethernet-compatible equipment.
13	Test & Standards	EN61000-4-5/ IEC and EN55024/32/ IEC surge standards
14	Standards	FCC/ CE/ IEC/ EN, ROHS Compliance

3.9) Industrial Grade Local Processing Unit (LPU)

S. No.	Parameter	Updated Minimum Specification
1	Processor	8th Gen. Core i5 Processor up to 3.0Ghz 6 cores
2	Chipset	Intel H310
3	Graphics	Intel UHD Graphics 630
4	Memory	8GB DDR4
5	Ethernet	2x1G RJ4
6	USB	2xUSB3.0
7	USB	2xUSB 2.0
8	Video Port	1xHDMI/ DVI-D port and 1xVGA/DP port
9	Serial Port	2 x RS-232/422/485 ports
10	Audio	1 x Mic-in and 1 x Line-out
11	Storage	1x internal SATA port for 2.5” HDD/ SSD
13	Storage	512GB 2.5" SATA SSD
14	Operating Temperature	0~60Deg C with 0.7 m/s airflow
15	Humidity	5~90%, non-condensing
16	EMC Certification	CE/FCC Class A, BIS

3.10) Online UPS 1 KVA with 120 mins battery back-up

S. NO	PARAMETER	MINIMUM SPECIFICATIONS
1.	**Capacity in KVA**	1KVA
2.	**UPS SystemDescription**	1 KVA True Online Pure Sine with 120 minutes backup on given site load on SMF batteries, interconnection cables andwire from UPS to battery bank.(SI is free to provide UPS of higher rating & higher battery bank, if required)
	Input (Rectifier)	
3.	Primary source voltage	230 VAC 1 Phase
4.	Input Voltage Range	160 - 300 V AC (depends on %age of load)
5.	Input Frequency Range	50-60Hz ± 5%
6.	Input power factor	> 0.99
	Output (Inverter)	
7.	DC volts	36/72 V as per OEM VDC Design

8.	Output voltage	230 VAC 1 Phase
9.	Output frequency	50 Hz (+/- 0.5/0.1 Hz) /60 Hz (+/- 0.5-3 Hz)
10.	Output Power Factor	0.8 or higher
11.	Overall Efficiency	> 85%
12.	Transfer Time (ACMode to battery mode)	Zero
13.	Panel	LCD/LED Panel
14.	Battery Management	Should be inbuilt within the UPS System
15.	Back-up	Min 120 Minutes on full load (min VAH of Batteries required is 2700VAH), Bidder to submit the OEM battery sizing sheet for backup calculation
16.	Status on LED/ LCD display	Normal Mode/Load on Battery/Load on Bypass/System Fault
17.	Audible Alarm	Mains Failure/Low battery /UPS Warning/ Overload/Fault & Bypass Mode etc.
18.	Ambient temperature	0 to 40 Deg. C
19.	Standards	ISO 9001:2015,ISO 140001, BIS Standard & ROHS/PEP compliance product

3.11) Online UPS 2 KVA with 120 mins battery back-up

S. NO	PARAMETER	MINIMUM SPECIFICATIONS
1.	**Capacity in KVA**	2KVA
2.	**UPS SystemDescription**	2 KVA True Online Pure Sine with 120 minutes backup on given site load on SMF batteries, interconnection cables and wire from UPS to battery bank.(SI is free to provide UPS of higher rating & higher battery bank, if required)
	Input (Rectifier)	
3.	Primary source voltage	230 VAC 1 Phase
4.	Input Voltage Range	160 - 300 V AC (depends on %age of load)
5.	Input Frequency Range	50-60Hz ± 5%
6.	Input power factor	> 0.99
	Output (Inverter)	
7.	DC volts	72 V /96 V as per OEM VDC Design
8.	Output voltage	230 VAC 1 Phase
9.	Output frequency	50 Hz (+/- 0.5/0.1 Hz) /60 Hz (+/- 0.5-3 Hz)
10.	Output Power Factor	0.8 or higher
11.	Overall Efficiency	> 85%
12.	Transfer Time (AC Mode to battery mode)	Zero
13.	Panel	LCD/LED Panel
14.	Battery Management	Should be inbuilt within the UPS System
15.	Back-up	Min 120 minutes on full load (min VAH of Batteries required is 5400VAH), Bidder to submit the OEM battery sizing sheet for each backup calculation
16.	Status on LED/ LCD display	Normal Mode/ Load on Battery/ Load on Bypass/ System Fault
17.	Audible Alarm	Mains Failure/Low battery / UPS Warning/ Overload/ Fault & Bypass Mode etc.
18.	Ambient temperature	0 to 40 Deg. C
19.	Standards	ISO 9001:2015, ISO 140001, BIS Standard & ROHS/ PEP compliance product

3.12) Outdoor Cabinet with Accessories

S. No	Minimum Specifications
a.	Approx. size: As per requirement at site
b.	Should be IP 55 rated floor mount and vandal resistant design. All the apertures and entry /exit slots should have a design for blockage of rodent and pest entry.
c.	Should be constructed based on metal profile with a front sheet steel door with3 point locking system to ensure the security of the cabinet.
d.	Should have integrated DIN rail mounting bracket slots for loading equipment. Theshelves should be modular to adjust the height of slots as per equipment
e.	Base/ Plinth including screw-fastened/ PG Gland plates for cable entry.
f.	Cabinet should have minimum 2 fans with Front door having filtered Air inlet and atop tapered canopy for water shower protection and heat dissipation.
g.	Painting: 7 tank process with PP Powder Coating.
h.	All the in & out cables must have sealable cable entry systems
i.	Should be ROHS compliant.
j.	The foundation of the cabinet shall be 600 mm or more above the nearest road / ground level (whichever is higher) to avoid water flooding with suitable foundationdesign for cable entry directly into the cabinet.

3.13) Termination Box with Accessories

S. No	Minimum Specifications
a.	Approx. size: As per requirement at site
b.	Should be IP 55 rated floor mount and vandal resistant design. All the apertures and entry /exit slots should have a design for blockage of rodent and pest entry.
c.	Should be constructed based on metal profile with a front sheet steel door with3point locking system to ensure the security of the cabinet.
d.	Should have integrated DIN rail mounting bracket slots for loading equipment. Theshelves should be modular to adjust the height of slots as per equipment
e.	To be fixed on Pole with help of suitable Steel clamps.
f.	Cabinet should have minimum 2 fans with Front door having filtered Air inlet and atop tapered canopy for water shower protection and heat dissipation.
g.	Painting:7 tank process with PP Powder Coating
h.	All the in & out cables must have sealable cable entry systems
i.	Should be ROHS compliant.

3.14) 6.3Mtr Octagonal Pole with foundation (with 6mtr/8mtr Single/Double Arm)

S No	Minimum Specification
1.	Load combination as per IS 800 : 2013
2.	Indian Standard IS 875 (Part 3):2015
3.	PLG 07
4.	Design Life of the structure should be minimum 15 years.
5.	Basic Wind Speed Load shall be 120 kmph.
6.	Gust Effect Factor 3.5
7.	Assumptions of equipment weight: Wight of equipment mentioned in 'Detail of Components'.
8.	Thickness of the pole and cantilever arm shall be suitable for the structural stability as per the OEM Design.
9.	Pole shaft: A36 or Equivalent
10.	Base plate: A572-50 or Equivalent
11.	Anchor bolts: EN8 Grade
12.	Structural shapes for Channel, angle, pipes as per IS:808 / IS:1161 / IS:2329 (YS: 240 Mpa or Equivalent)
13.	Connection Bolts: Grade 5.6
14.	Galvanizing of the octagonal structure need to be in compliance with A123/153. Average Galvanization thickness should be 80 microns or more.
15.	Each Pole should have Factory tested and test report traceability: direct, one-to-one traceability to pole shafts and base plates batch to other structural components. The test reports and inspection records are to be submitted when requested
16.	Octagonal CCTV Pole height must provide ground clearance of 6.0 meters in all the cases of installation of cantilever arms. Pole Height May vary in design as per junction requirement but extended ARM length should be min 6 & 8 Mtrs or more, as required
17.	Pole Structure should have design foundation with appropriate Grade of concrete (M25) and should be conducted after soil test report 10T / SBC for foundation with J bolt Anchor Design
18.	Steel anchor J bolts to be embedded in the foundation. Base plate is welded at bottom of pole shaft and pole is fixed to foundation through base plate to anchor bolts connection.
19.	The quantity, length and diameter of foundation bolts shall be suitable for the structural stability as per the OEM Design.
20.	Executing agency should get their Galvanized Octagonal Pole Structures, required to accommodate the load of proposed equipment, designed by qualified structural engineers and certified for structural safety from NABL certified labs/ Govt. Labs/ Govt. certified Labs/ Licensed Structure Engineer, and submit all the necessary details and documents required to the get this approval before Installation.
21.	The agency will get the bearing capacity tested, before designing the foundation. The design calculation, approved by OEM of poles, shall also be supplied by the agency and same should be vetted by any Govt. Lab/ Engineering Institutions. Expenditure on this account, if any, shall be borne by the agency.

3.15) 4 Mtr/ 8 Mtr long poles

01. Galvanized Steel Octagonal Pole (Sheet thickness 3mm) designed to withstand wind speed 169km/ hour of suitable length (Raw material confirming to IS-5986 (FE-510), BSEN-10025S-355J or ASTM-572A (GR-50)) and thickness of galvanization (80microns or more) conforming to IS:2629/IS:2633/ IS:4759,BSEN-ISO-1461-1991ORASTM-A123/A123M-
02. The pole shall be in single piece hot dip galvanized and shall be tapered towards the top. The bottom section shall have openable slot with exterior surface door & shall have suitable locking

arrangement for housing three phase 4wire cable connection, bakelite sheet, MCB, loop in and out arrangement for incoming/ outgoing cables. There shall be suitable arrangement for earthing. Rigid Base plate of suitable size and thickness shall be welded inside and outside at the bottom of pole with provision for fixing 4 foundation bolts. The octagonal pole shall be bolted on a pre- cast foundation with a set of four foundation bolts for greater rigidity. The foundation shall be erected over cement concrete M20 with steel of given size as per design of original pole manufacturer to fixed up to a required planting depth below ground level as required:-

Sr. No.	Parameters	4 Mtr Long	8 Mtr Long
1	Top dia.(A/F)	70mm	70mm
2	Bottom dia. (A/F)	130mm	135mm
3	Foundation size below G/L	550mm x 550mm x 900mm	550mm x 550mm x1500mm
4	No. of Foundation Bolts	04 nos.	04 nos.
5	Size of Foundation Bolts	16mm dia	20mm dia
6	Length of Foundation Bolts	≥450mm	≥750mm
7	Base Plate Dimensions	200mm x 200mm x12mm Thick	240mmx240mm x16mm Thick

3.16) 6 Core OFC: Unitube, Gel filled cable

S. No.	Minimum Specifications
1.	6 Core, single mode, outdoor Unitube, Gel filled Cable
2.	Attenuation: @850nm≤ 3.0db/KM
3.	Tensile rating: 1200N
4.	Maximum Crush resistance: 2000N
5.	Outer sheath: HDPE
6.	Inner sheath: PE
7.	Operating Temperature: (-) 05 deg C to (+) 70 Deg C

3.17) Cat-6 Armoured Cable

Sr. No	Minimum Specifications
a.	Cable Shielding: HDPE Outer Jacket with Steel Tape/ Aluminium wire armouring for ruggedized purpose
b.	Cable: Should be Cat-6, UTP 23 AWG.
c.	Pairs: 04 pairs
d.	Bend Radius:10x Diameter (min.)
e.	Cables should be UL/CPR certified/ EIA/TIA 568.C.2
f.	OEM should have ISO 9001:2008 Certification for Quality Management System
g.	All cables at field locations/ signal points must be concealed (i.e. inside the pole or underground laying shall be done by SI).

3.18) PLB HDPE Duct

Sr. No.	Description	Parameter
1	Single Piece Length	upto 200-300 Meters
2	Application	Telecom
3	Thickness	3.5 mm
4	Color	Blue/ Orange
5	PN Grade	16

6	Packing Type	Coil
7	Pressure	PN 16
8	Diameter	40 mm OD / 33 mm ID
9	Packaging Type	Coil
10	Weight	380 gms/Meter
11	Temperature	60 DEG. C
12	Usage/Application	TELECOM DUCT
13	Material	HDPE
14	UV Resistant	Yes
15	Size Range	40 mm OD / 33 mm ID

3.19) Passive Items:-

Sr. No	Name of item	Minimum specifications or better
1	**LIU 12 port with Splice tray**	Material: Should be made powder coated Mild Steel ports :12 Shall have slide-out type drawer enclosure for Easy access to splicing tray, Easy access to back side of the connector and have labels for better identification. Shall have splicing trays for Fiber management and should have greater access during installation and rework. Should be loaded with Splice try and LC type Adapter Plate. All Fiber and copper cabling should be from same OEM and should have 25-year Performance warranty (Warranty to cover Bandwidth of the specified and installed cabling system)
2	**LIU 24 port with Splice tray :**	Material: Should be made powder coated Mild Steel ports :24 Shall have slide-out type drawer enclosure for Easy access to splicing tray, Easy access to back side of the connector and have labels for better identification. Shall have splicing trays for Fiber management and should have greater access during installation and rework. Should be loaded with Splice try and LC type Adapter Plate. All Fiber and copper cabling should be from same OEM and should have 25-year Performance warranty (Warranty to cover Bandwidth of the specified and installed cabling system)
3	**HDMI Lead 5 MTR. :**	HDMI Lead 5 Mtr. Latest version HDTV Cable supporting 3D content, Ethernet and Audio return Channel, Guaranteed 4K 1140P, 1080, 720p, 480p, Resolution, Supports refresh rate up to 240H2 and 48 Bit deep Colour, Support Transfer HDd Dolby 7.1 and DTS-HD Master audio
4	**Fiber pigtail:**	Fiber pigtail should be Single mode OS2 9/125µ, duplex LC type. Should be LSZH. All pigtails should be factory terminated and packed. Should be ROHS. Length 1meter/1.5 meter All Fiber and copper cabling should be from same OEM.
5	**Fiber Patch Cord LC-LC:**	Fiber Patch Cord should be Single mode OS2 9/125µ, Duplex. Should be LSZH. All patch cords should be factory terminated and packed. Should be ROHS. All Fiber and copper cabling should be from same OEM.
6	**Splicing of OFC Cable**	Testing & Termination of Fiber core with splicing and Optical connectivity test report for per core of OFC
7	**Fiber adaptor:**	Quick mounting clip integrated. Protection class of IP-20 rated with Material: fiber- (halogen-free)
9	**SFP Module**	Supply installation testing and commissioning of SFP 1000 base Single mode LC up to 5 Km support fiber module of same OEM for switch.
10	**Cat-6 Patch Cord**	Cat6 cable of desired length which Should be Compliant with Cat.6 requirements of ISO/IEC 11801, EN 50173 and EIA/TIA

		568.C.2, Should be stranded bare copper, Should be unshielded
11	**Face Plate:**	Face Plate single/ Duplex, Square plate, 86mmx86mm, Material :ABS Plastic
12	**Information Outlet (IO)**	Information outlet belongs to Category 6, EIA/TIA 568-C.2 Information outlets Should accept cables from 22-24 AWG copper, Should be Tool less will be added advantage, Information outlet should be of Halogen Free material, Should be Certified by independent labs like ETL/GHMT/3P or Govt. approved Lab. for compliance. Information Cat6 Information outlet which should be tool less support and having mating cycle of 750 or more

3.20) Firewall

1	The Firewall should be hardware based, reliable, purpose-built security appliance with hardened operating system that eliminates the security risks associated with general purpose operating systems.
2	The firewall OS family shall be tested and certified for EAL4 as per Indian Common Criteria Certification Scheme (IC3S) or equivalent organization.
3	Certifications: NDPP / NSS / ICSA Labs
4	Appliance should have at least 8 x GbE copper, 2 x SFP and 2 x SFP+ loaded ports. The system should have the 1XRJ45, 1 X COM RJ45 and 1 Micro USB Port for Management purpose.
5	Form factor should not exceed 2 RU.
6	Appliance should have minimum throughputs: Firewall : 27 Gbps; IPS /NGFW : 5 Gbps or above, VPN Throughput : 13Gbps or above, Concurrent sessions : 3 Million or above
7	The proposed solution should support identity awareness on the security gateway via Active Directory or RADIUS
8	Solution should be compatible with any networking equipment with SD-WAN support for Jitter/ Latency / Packet Loss.
9	High Availability (Active-Active & Active-Passive)
10	Two factor authentication for user creation
11	Deployment of minimum 2 or more ISP links. The system should support 2000 VPN Users.
12	802.3ad interface link aggregation
13	High-performance, next-gen IPS deep packet inspection engine with selective IPSpatterns for maximum performance and protection. The solution should be able to address aspects of the Advanced Persistent Threat (APT).
14	Second independent malware detection engine for dual-scanning.
15	Fully transparent proxy for anti-malware and web-filtering; support for HTTP and HTTPS scanning on as per user or network policy basis with customizable rules and exceptions. TLS 1.3 scanning capabilities. The system should support IPS, User Authentication, Advance Threat Protection, Sandboxing, Zero day protection, CASB, Traffic Shaping, Web filtering, Application Filtering
17	Enhanced application control with signatures and Layer 7 patterns for applications
18	Should provide risk level of applications on the network.
19	Solution should be capable to provide CLI and GUI. Solution should support Role based administration.
20	IPv4 and IPv6 supported.
21	The system should provide provision to prevent unauthorized access to private data by encrypting user identity data / log/report on appliance reporting. The local storage /SSD to be provided

3.21) VMS Application & Analytics Server

S. No.	Parameter	Minimum Specifications
1.	Processor	3rd Generation Intel Xeon Scalable processors (as per architecture)
2.	Memory slots	32 DDR4 DIMM slots, speed ~ 3200 MT/s scalable to 2TB
3.	Disks supported	2.5” SAS/SATA/NVMe (HDD/SSD)
4.	RAID Controller	Latest as per OEM
5.	I/O slots	Minimum - 3 x PCIe Gen4 slots all x16

6.	Network ports	2x 1G Ethernet / 2x 10G SFP+ / 2x 16G FC
7.	Certified for OS	Windows Server / Enterprise Linux
8.	Power Supply	Platinum rated redundant hot swappable power supply
9.	GPU Support	As per application requirements
10.	Power & temperature	Real-time power meter, graphing, thresholds, alerts & cappingwith historical power counters. Temperature monitoring & graphing
11.	Pre-failure alert	Should provide failure monitoring & alerts of actual component failure of fan, power supply, memory, CPU, RAID, NIC, HDD
12.	Configuration & Management	•Real-time out-of-band hardware performance monitoring & alerting •Agent-free monitoring, driver updates & configuration, power monitoring & capping, RAID management, monitoring of FC, HBA & CNA & system health
		• Out-of-band hardware & firmware inventory • Support for Redfish API for simple and secure managementof scalable platform hardware
13.	HTML5 support	HTML5 support for virtual console & virtual media without using Java or ActiveX plugins
14.	Server security	Should have a cyber resilient architecture for a hardened server design for protection, detection & recovery from cyber attacks
		Should protect against firmware which executes before the OS boots
		Should provide effective protection, reliable detection & rapid recovery using Signed firmware updates, Secure default passwords, Configuration and firmware drift detection, Persistent event logging including user activity, Secure alerting, Automatic BIOS recovery, System erase
		Configuration upgrades should be only with cryptographically signed firmware and software
		Should provide system lockdown feature to prevent change (or “drift”) in system firmware image(s) & prevent malicious modification of server firmware
15.	Intrusion alert	Intrusion alert in case chassis being opened
16	Storage	Minimum last 7 days recording should be available at Server. It should get synchronized with the main storage after restoration from failure, if occur. (The calculation of storage at full frame rate & full resolution with 24x7 recording should be attached with the technical bid)

3.22) Video Management Software

S. No.	Minimum Specifications
1.	Video Management Software: Video Management Software shall maintain coherent operations between all servers and workstations. It shall host Management Server/s, Media / Recorder Server/s and Database Server/s.
2.	Video Recorder Software: The Video Recorder Server shall be a dedicated server that shall store and process video with the help of Video Management System.

3.	Video Analytics: Video Analytics Software shall be installed in the Video Analytics Server to analyze live video in real-time & to detect, identify events of interest as per configuration. It shall automatically issue alerts to the appropriate personnel and initiate appropriate follow-up action according to predefined rules. It should be possible to configure video analytics on any of the cameras from the Video Management System desktop/ client Application. The video feeds shall be connected over the network to the Video Analytics Server. Video Analytics Server shall perform all event detection functions. Analytics shall include Detection of crowd count/ Overcrowding and Loitering etc. Traffic Analytics shall also include, if the same are not available in ITMS Software:- Wrong direction movement of vehicles – All RLVD (Evidence)/ ANPR cameras Triple riding on two wheelers - All RLVD (Evidence)/ ANPR cameras Wrong parking - All RLVD (Evidence)/ ANPR cameras
4.	The VMS architecture should comprise of centralized & decentralized architecture. The VMS should have system components such as Management Server to manage the system, Recorder or Media Server to stream and store the video feeds from the cameras and Database Server to store metadata information.
5.	The VMS should support single site or multi-site deployment scenarios. The VMS should have capability to aggregate videos from multiple sites to the central site for recording or monitoring. The VMS should also have the capability to aggregate the system alerts such as video analytics, user created alerts and system health alerts to the central site.
6.	The VMS should support single or multiple recorder server deployments. The proposed VMS should support unlimited IP and Analog cameras by augmenting the computing and storage hardware. The proposed VMS should support unlimited number of VMS clients.
7.	The VMS architecture should support automatic assignment of the available cameras on the network to the available Recorder Servers based on the recorder server's resources such as number of cores, available RAM and resource utilization.
8.	The VMS should be computing hardware agnostic and should work on commercially off the Shelf (COTS) servers and storage solutions.
9.	The VMS should support virtual computing environment and should support all the industry leading virtualization platforms available for Windows or Linux or Unix environment
10.	The VMS should support 64-bit architecture OS and hardware environments.
11.	VMS shall store the system's configuration in a relational database, either on the management server computer or on the network.
12.	The VMS should support redundancy at each level to avoid single point of failure. The redundancy should be built into the platform and should offer failover support for Management Servers, Media / Recording Servers, Database Servers and Storage Medium.
13.	VMS should support Failover against temporary disconnection of DBMS Service, without any loss of camera video. As soon as the DBMS service resumes, all data should automatically be synchronized to the Database.
14.	Each media/ recorder server should have its own storage configuration. It should be possible to select storage location/s out of all available storage locations for each media server. The media server should support load balancing and fail-safe operation by distributing the video data on all selected storage locations. So, in case of failure of any storage location, the entire video data is not lost.
15.	The surveillance system shall provide a scalable and reliable platform to enable customized, network-based surveillance applications.
16.	The VMS should support mix of storage technologies such as SAN / NAS storage. Each media server should support such storage locations simultaneously.
17.	In case of the failure of the Recording Server, the VMS should automatically assign the cameras on the failed recording server to other operational/redundant recording servers on the network. Manual intervention of any kind should not be required in such a case. When the failed server becomes active again, the cameras should be automatically allocated to the recorder server again without manual intervention.
18.	The Server should allow recording of camera feeds on network storage. In case the network storage fails, the recording server should start recording on the local storage. The local

	recording should get synchronized with the network storage as and when it is available again.
19.	The system shall provide for integration with other software applications through an open and published Application Programming Interface (API). Such applications shall include, but not be limited to, access control, video analytics, incident management system and other alarm and sensor inputs. It shall be possible to integrate VMS into the Command & Control system.
20.	VMS shall be open to any video wall system integration.
21.	VMS should have Open Interface to send Analytics event alerts and other Maintenance Alerts (e.g. Camera disconnection, Storage Full, DBMS disconnection, etc.) over HTTP / HTTPS protocol to any external application running in a different machine in the same LAN. This is required for integration with command & control software or any other 3rd party incident management system.
22.	This shall allow operations managers and system integrator to build customized video surveillance networks that meet the city requirements. VMS shall be a scalable and flexible video management system which should support unlimited cameras by adding licenses and augmenting the computing and storage infrastructure. The VMS should not have any cap on the number of client workstations.
23.	VMS Server Management systems should support network time protocol (NTP) on server, which automatically sets the server time and date.
24.	The Video Surveillance System should support high availability (HA) architecture.
25.	VMS should support H.265, H.264 and MJPEG or latest stream for both live view and Recording independently. Compression rate should be manageable.
26.	The VMS Software should be ONVIF S, G, T Profile Compliant. The Declaration of ONVIF S, G, T Profile shall be available on ONVIF Official Website www.onvif.org in the Conformant Devices.
27.	The VMS should be able to stream standard H.265, H.264 and MJPEG or latest camera video streams to any external software on demand basis.
28.	VMS video feeds shall be either directly encrypted or carried through encrypted tunnelusing VPN or HTTPS end to end encryption technology from cameras to video recording servers.
29.	The VMS should intelligently synchronize the edge recording on the camera with the central recording in case the camera loses the network connection.
30.	Archive retention period should be configurable on per camera basis. The system should allow both retention based as well as First In First Out (FIFO) based deletion policy.
31.	The VMS should support multiple directory access protocols such as Microsoft Active Directory or Lightweight Directory Access Protocol (LDAP).
32.	VMS should integrate with multiple third-party mapping platforms in multiple layers
33.	Central dashboard layout should be customizable as per operator's requirement and various users should be able to save their customized views for future use.
34.	Dashboard should support the data presentation in various forms including bar graphs, Pie charts, line charts and should support different filters like Date/Time, Locations, violations and should be able to compare the similar data with previous week/month/year basis
35.	VMS should allow managing clusters of cameras. Quantity of cameras per cluster should be unlimited. VMS should allow assigning each camera to one or several clusters simultaneously.
36.	The Client Viewer should support real-time simultaneous view of 1, 2x2, 3x3, 4x4, 1+5, 1+7, 1+11, 8x8 or customized multi-screens video display and a simple click should allow enlarging any of the multi-screen displays into a full screen display. On clicking again on the enlarged display, multi-screen display should reappear.
37.	VMS should select the appropriate video stream from camera for display depending on the display resolution to optimize the network bandwidth.
38.	The Client Viewer should support the use of standard PTZ controller or 3-axis USB joysticks for control of pan, tilt, zoom and auxiliary camera functions.

39.	VMS should provide options for export format type (AVI/MP4/MJPEG), timestamp, frame rate, digital zoom export, and AVI CODEC. Video clip may be exported to desktop/CD/DVD or a specific file path. All audio associated with the video being exported should automatically be included in the AVI export. VMS should watermark every frame of the Video files with watermarks to authenticate the source & sanctity of the exported video and create an alert for any watermark tampered video after export. While exporting video segments to external media (CD/DVD) or to any folder in workstations, the VMS should allow encryption of the video files. The user should be able to create customized watermark text.
40.	The VMS desktop client should show vital system parameters for components such as Database Server, Media Servers, Local Workstation and Storage System (all available storages). The client should show the parameters such as CPU Core Usage, RAM Utilization and Storage Utilization. AES 56 encryption should be provided.
41.	The VMS should have reports such as camera uptime availability, camera recording percentage, recording status, critical events, incident video, etc.
42.	The System health status like Server failure, Camera Disconnection, Storage Full Indication, etc. should always be displayed within the client workstation GUI all the time.
43.	VMS should maintain a continuous log of Server Status Messages, Camera Connectivity, Storage Status, Recording ON/OFF, User Activity Logs, etc. which should be accessed from the workstations using different filters.
44.	The system should give full audit trail of the user activities in the system.
45.	The VMS should allow the user to bookmark any recorded video for ready reference at any later point of time.
46.	The system should allow the user to tag critical Event clips so that they do not get removed from the storage based on FIFO/Retention period settings. The system should also store such flagged data in a virtual partition in storage or a separate storage stack.
47.	The VMS should allow multi-monitor support for the client workstation.
48.	The VMS should allow the captured image / video to be tamper proof for all legal and regulatory perspective using latest encryption, watermark and other features.
49.	The VMS provider should provide an SDK (Software development kit) for customization on the software for future requirement
50.	The video management system should be able to produce graphical & tabular dashboard of configurations, events, performance and operations. The user shall be able to customize the reports and dashboards as per requirements from time to time. The reports shall be exportable to open file formats like MS - Excel, MS – Word etc
51.	Data viewing and report generation at central dashboard should support multi-layer hierarchy like State -> Districts -> Zones -> Police stations -> Junctions -> cameras
52.	The Offered VMS Application should have an Application Security Audit Certificate from CERT-IN Empaneled Auditors for the VMS as per the OWASP Guidelines to Mitigate Cyber Security Attacks. Documentary proof to be submitted before the GO-Live of the project. **Undertaking from OEM of VMS Software for the same must be submitted alongwith bid.**
53.	The Intellectual Property Rights of offered VMS, ITMS/ANPR/RLVD/SVDS Application OEM must not reside in Country sharing Land Border with India. OEM shall provide a declaration about the intellectual property rights.

3.23) ITMS Server Software

S. No.	Minimum Specifications
1.	The System should automatically detect a vehicle in the camera view using video detection and activate license plate recognition. The System should automatically detect the license plate in the captured video feed in real-time. The system should perform Optical Character Recognition (OCR) of the license plate characters. The System should store JPEG image of vehicle and license plate and enter the license plate number into database management system like MSSQL, MySQL, and PostgreSQL etc. along with date timestamp and site location details. System should be able to detect and recognize the alpha numeric license plate in standard fonts and formats for all classes of vehicles such as cars, HCV, and LCV and other vehicles. The system should be robust to variation in License Plates in terms of font, size, contrast and colour and should work with average 90% (Minimum) accuracy for HSRP and Non HSRP.
2.	The system should detect the colour of all vehicles on best effort basis, in the camera view and label them as per the predefined list of configured system colours. The system should store the colour information available of each vehicle along with the license plate information for each transaction in the database. The system should have options to search historical records for post event analysis by the vehicle colour and with vehicle colour with license plate and date time combinations.
3.	White List & Black List. The system should have option to input certain license plates according to the hot listed categories like “Wanted”, “Suspicious”, “Stolen”, etc. by authorized personnel. The system should be able to generate automatic alarms to alert the control room personnel for further action, in the event of detection of any vehicle falling in the hot listed categories
4.	On successful recognition of the number plate, system should be able generate automatic alarm to alert the control room for vehicles which have been marked as “Wanted”, “Suspicious”, “Stolen”, “Expired". (System should have provision/ expansion option to add more categories for future need). The Instantaneous and automatic generation of alarms. In case of identity of vehicle in any category which is define by user.
5.	The system should enable easy and quick retrieval of snapshots, video and other data for post incident analysis and investigations. The system should be able to generate suitable MIS reports that will provide meaningful data to concerned authorities and facilitate optimum utilization of resources. These reports should include: Report of vehicle flow at each of the installed locations for Last Day, Last Week and Last Month. Report of vehicles in the detected categories at each of the installed locations for Last Day, Last Week and Last Month. Report of Vehicle Status change in different Vehicle Categories. The system should have Search option to tune the reports based on license plate number, date and time, site location as per the need of the authorities. The system should have option to save custom reports for subsequent use. The system should have option to export report being viewed to common format for use outside of the ANPRS or exporting into other systems. The system should provide advanced and smart searching facility of License plates from the database. There should be an option of searching number plates almost matching with the specific number entered (up to 1 and 2-character distance).

6.	The system should have option to input certain license plates according to category like "Wanted", "Suspicious", “Stolen", and “Expired" etc. by Authorized personnel. The system should have an option to add new category by authorized personnel. The system should have option to update vehicle status in specific category by authorized personnel. Eg. on retrieval of stolen vehicle, system entry should be changed from "Stolen" to "Retrieved". System should have option to specify maximum time to retain vehicle records in specific categories.
7.	The Central Management Module should run on the ANPRS Central Server in control booth. It should be possible to view records and edit hotlists from the Central Server
8.	Besides recording the snaps & video clips of every license plate extracted, it is also required that a centralized video management software is also supplied to achieve the below: - Continuous recording of every lane video irrespective of presence of vehicle. Such recording schedules can be continuous, event based, schedule based, trigger based etc. Archive Search using dates, time, event etc. High Availability/Redundancy of Recording & Database. Health monitoring module - To allow for continuous monitoring of the operational status and event-triggered alarms from servers, cameras and other devices. The health monitoring module should provide a real-time overview of alarm status or technical problems while allowing for immediate visual verification and troubleshooting. Virtual Matrix – To allow viewing of live video in different layouts on operator screen.
9.	The system should instantaneously (within 4 seconds) receive the processed ANPR of the violating vehicle
10.	ANPR shall be based on non-intrusive type of enforcement for detecting various traffic violations.
11.	The system should store a continuous unaltered video clip of the vehicle approaching and leaving the violation location in the database
12.	The system should work in both day and night conditions with the defined minimum accuracy levels.
13.	The offered software should have cleared at least one integration with NIC platforms with security integrity testing with Vaahan or Saarthi database. The solution shall alsointegrate with hand held e-Challan & PoS devices. Proof of this integration to beattached in technical bid
14	The Offered ITMS Application should have an Application Security Audit Certificate from CERT-IN Empaneled Auditors for the ITMS as per the OWASP Guidelines to Mitigate Cyber Security Attacks. Documentary proof to be submitted before the GO-Live of the project. **Undertaking from OEM of ITMS Software for the same must be submitted alongwith bid.**

a. The sub-systems should work with minimum intervention and maximum accuracy levels in all weather conditions
b. The system should be able to provide Unique Case number for each of the Violation Detected.
c. The system should be capable of providing a search and filter option to find specific cases at any point of time.
d. **Type of traffic Violations —** The following type of traffic violations but not limited to, should be automatically detected using the appropriate non-intrusive technology in each case as applicable. All the traffic violations should be seamlessly integrated withthe ANPR system and maintain the status of each violation till closure in the database.
   - Stop Line Violation
   - Wrong direction movement
   - Red Light Violation & Detection (RLVD) with ANPR
   - Speed violation detection with ANPR
   - No helmet detection.
   - Triple riding.

e. **Red Light Violation Detection (RLVD) —** The non-intrusive RLVD Sub- system should be capable of capturing multiple Infraction Vehicle Data (IVD) simultaneously on each arm of the junction at any point of time.

f. **Zebra Crossing / Stop Line Violation at Junctions —** The non-intrusive system should be capable of detecting the violations and capturing multiple IVD simultaneously on each arm of the junction at any point of time.

g. **Wrong Direction Vehicle Movement —** The non-intrusive system should be installed at critical junctions to capture the wrong direction vehicle movement. The system should identify and capture multiple IVD. The E-Challan standard procedure should betriggered.

h. **Speed Violations -** The nonintrusive system shall be capable of measuring speed of vehicles and capture over speed vehicles. The Speed measurement should support multiple methods for calculation of speed –Instantaneous Speed Measurement methods. Speed accuracy of +/- 2% should be achieved. SVD software should be ARAI certified. (for both RADAR based and Video based SVD system)

i. The system should have provision to be seamlessly integrate with e-challan sub- system which is capable of providing a legally binding court evidence following a proven and a robust procedure.

j. The system shall have an operations monitoring dashboard, located at the ICCC.

k. The ANPR sub-system is also used for identifying hot-listed vehicles. The system should have option to input certain license plates according to the hot listed categories like “Wanted”, “Suspicious”, “Stolen”, etc. by authorized personnel.

l. The following are the minimum details of the infracting vehicle to be captured:
    - Location Name and ID along with Latitude and Longitude
    - Date & Time of the instance
    - Vehicle Number plate (Captured & Processed)
    - Colour of the vehicle
    - Instantaneous Speed of the vehicle

m. **Court Evidence Standard Procedure –** The system should provide the video & image evidence with number plate data, of infracting vehicles to be submitted in the court.

n. **Court Evidence Standard Procedure –** The system should provide the video & image evidence with number plate data, of infracting vehicles to be submitted in the court.

o. The tamper-proof video extract shall be provided as supporting evidence (for submission in a court of law) to each infracting vehicle and the video length shall be t-5 to t+5 seconds where t being the instant at which the infraction occurred. The time period of before and after infraction video shall be customizable to less than or more than 5 secs. The video output should be in color during day and in any industry standard format such as MJPEG, MP4, AVI etc., with visually readable license plate number.

p. The system shall have options to search historical records for post event analysis by the vehicle color or the vehicle color with license plate and date time combinations

<u>E-Challan Challan Generation Software Module:</u>

Architecture:

- The e-Challan system should have a robust client-server architecture. The system should support hand held terminals and workstations to generate e-Challans.
- The system should be computing hardware agnostic and should work on Commercially Off The Shelf (COTS) servers and storage solutions.
- The system should support virtual computing environment and should support all the

industry leading virtualization platforms available for Windows, Linux or Unix environment.

- The system should support 64 bit architecture OS and hardware environments.
- The system should support open source DBMS, e.g., PostgreSQL, MySQL besides MS SQL and Oracle. The VMS should be capable of supporting SQL or No SQL databases.
- The system should support redundancy at each level such as application server and database server to avoid single point of failure.
- The system should have full audit trail which should register all the user and system generated events.
- The system should support role based access control for the operator logins.
- The e-Challan generation system should be pre-integrated with the red light violation,speed violation, no helmet detection and triple ride detection system. The Challan generating system should generate and print Challans for all the red light violation, speed violation, no helmet detection and triple ride detection events that are duly validated and checked by the operator.
- The system should support secure data communication through the clients (workstation and hand held equipment).
- The system should have the capability of integrating hand held equipment from multiple OEMs. The integration should happen on the standard APIs available from the hand held equipment OEM.
- The system should support integration with the vehicle databases such as VAHAN, SARATHI, etc.
- The system should support web based and client based access to the system. In other words, the system should support thick and thin clients.
- The system should support integration with the SMS gateway to communicate with the vehicle owners regarding the challan information. The SMS charges will be reimbursed/ paid by PPHC/ Police department.
- The e-Challan software should be supplied with a desktop or web client based payment collection application. The system should have the functionality to accept payments by cash/debit/credit card using 3rd party application/gateway. However once the payment is made, the operator should be able to input the payment information in the software against challan ID so that further reports are maintained in the system.
- The Challan generation system should have a provision to be integrated with the third party website such as police website, so that the owner of the vehicle can view the challan details along with Evidence snaps by logging in to the website.

- The system should support integration of the digital signatures of the competent authorities for signing the e-Challans.

Functional Requirements:-

- The system should generate e-Challans based on the various violations detected in the Intelligent Traffic Management System. The system should allow the operator to validate each auto-generated e-Challan before printing.
- The system should allow generating an e-Challan based on the easy selection of violations under the relevant Motor Vehicle acts. The system should have the capability to configure such Motor Vehicle Act provisions and associated fines / penalties in the system for easy selection.
- The e-Challan system should have the hand held device based spot e-Challan machines for issuing e-Challans on the road by the traffic police officers. The hand held machines and the central e-Challan systems should be integrated so that challan information is duly updated as a common database/repository.
- The hand held equipment should be connected with the central server through WiFi, 3G/4G/5G GPRS connectivity.
- The hand held equipment should allow the operator to login to the device to access the functionalities. The device should not show any records / system information without log-in.
- The hand held equipment should work in online as well as offline mode. In case there is no connection with the server, the hand held equipment should store the events in the local database. The local events should be synchronized with the central server database when the network connectivity is restored.
- In case of spot challan of a violation using handled device, the traffic police officer should get historical information of such violator such as past violations, earlier challan paid/unpaid etc.
- The operator should have the option to mark the challan as PAID when the violator deposits the money against the challan. At the end of the day, the system should generate a report for the challans paid and the total amount collected.
- It should be possible to store the e-Challans based on the functional and legal requirements in terms of the number of days. The system should not delete the e-Challans which are under legal or court procedure.
- The system should have postal record maintenance facility to keep track of dispatched, returned, refused, etc. e-Challans sent to the violating individuals.

Reports:-

- The e-Challan generation system should be able to generate e-Challans in local language and English.
- The Operator should have the option to filter the violations based on the following criteria for generation of Challans:
  - Number of violations by the same vehicle multiple times
  - For a particular category (e.g., 4-wheelers) of vehicles
  - For a particular Thana / police jurisdiction
  - Paid/Unpaid Challans
  - Ageing Analysis of Pending Challans
- The system should generate end-of-the-day report of the e-Challans generated. The report should contain the number of e-Challans generated, number of e-Challans paid, pending e-Challans, etc. The system should have the capability to send such reports via email to the designated persons.
- The system should have robust search functionality to search the violations by violation types, date and time duration, police jurisdiction / thana, operator, hand held device, location, vehicle number, etc.
- The system should generate statistical reports in terms of bar charts based on various categories.

3.24) 3D Speed Detection RADAR (In case RADAR is used for SVD)

S. No	Minimum Specification	
1	Application	Speed Violation Detection
2	Technology	3D or better
3	Vehicle Detection range (Max)	Range 180m
4	Detection Range (Min)	5m
5	Speed Accuracy	+/-2% (km/h) for the speed upto 200 KMPH
6	Speed Interval	-68.3m/sec to -0.1m/sec and 0.1m/sec to 68.3m/sec (opt. +/-88.8m/sec)
7	Refresh Time	50 m/s
8	Tracking	track upto 10 vehicles simultaneously capture atleast 10 over speeding vehicles per second
9	Form Factor	Enclosure Rugged, Watertight Casing (metal /Polycarbonate (UV stabilized)
10	IP protection	IP66
11	Operating Temp	Operation Temperature (-)5 deg C to + 60 deg C
12	INPUT Power	12 to 32 VDC; 3.7 Watts
13	Serial	Communication Interface RS485, CAN Bus, Relay Contacts(option)
14	Simulation	Compatible with Simulator of any make
15	Calibration	Certificate from Govt. accredited certifying bodies

3.25) Video Based Speed Detection (In case Cameras are used for SVD)

S. No	Minimum Specification	
1	Application	Speed Violation Detection
2	Technology	Video Based
3	Speed Accuracy	+/-2% (km/h) for the speed upto 200 KMPH
4	Certification	SVD software should be ARAI certified.
5	ANPR Camera	As per Specifications (Item no. 3.2)

3.26) External Enterprise grade Storage

S. No.	Parameter	Minimum Specifications
1.	Storage Purpose	This storage will be used to store all camera's video feeds at 1080p resolution 25 fps or higher with H.264, H.265 or latest video compression standards for 35 days storage of Video data & flagged data.
2.	Storage Capacity	As per application architecture requirements
3.	Storage Type	Block Storage System with iSCSI or Fiber Channel protocols
4.	Controller	Dual redundant Active-Active Controllers. Each Controller shall have minimum dual core 2.2 GHz CPU.
5.	Cache	24 GB memory spread across both controllers. SSDs will not be considered as Cache. Offered Storage shall also have optional support for Flash cache using SSD / Flash drives.
6.	RAID	RAID levels supported: 0, 1, 5, 6, 10
7.	Scalability	Proposed Solution should be scalable to 50% increment capacity in the same Storage Array without upgrading thecontrollers.
8.	Host Connectivity per Controller	As per application architecture requirements
9.	Throughput	Proposed Storage System should ensure a minimum throughput of 2000 Mbps Video Management Workload. Bidder to provide the workload management architecture in support of the device offered along with a validation document.
10.	Storage Management	Web based management- HTML5, Single-array intuitive element manager, Use common management tasks from convenient locations, Manage storage profiles, network connections, alerts etc.
11.	Power Supply	Dual Redundant Power Supply. Replacement without downtime
12	Additional	Autonomous Drive Regeneration (ADR) or equivalent

3.27) 3 Axis Joystick Controller

S. No.	Parameters	Minimum Requirements
1	Functionality	Functionality Hall-effect joystick with three axis:X/Y: for pan and tilt, Z: knob for zoom Deflection by square delimiter, (± 15o) for XY, (± 25o) for Z 6 hotkeys with customizable function
2	Connectors	USB Type B/ USB Type A
3	Power	Via USB interface
4	Casing	Polycarbonate ABS
5	Operating conditions	0 °C to +60 °C Humidity 10–85% RH (non-condensing)
6	Storage conditions	-10 °C to +60 °C
7	Approvals	EN/IEC/UL 62368–1 & CE
8	System requirements	Windows 10, Windows 7, Windows XP, Windows Vista, Windows 12 (Server), Windows 8 (Server), USB 2.0port
9	Included accessories	1 x USB cable 200 cm (78.7 in) for connection to PC

3.28) 42U Rack with KVM

S. NO.	MINIMUM SPECIFICATION
1.	Description: 42U OEM Server rack to mount servers, KVM Switch, Ethernet Switch etc.
2.	Size: 42U with width & depth of 800mm x 1200mm
3.	KVM Switch: KVM Switch Shall be put in SI Scope
4.	Monitor, Keyboard, Mouse: Keyboard, Mouse, Folding 15" LCD/LED Monitor per rack to be connected through KVM
5.	Appearance: HFSR (Hex door in front and split hex doors behind)
6.	Power: Vertical Power Distribution Unit with 12 x 5/15 sockets Round Pin, 230 Volts AC, 32 Amp with Plug.
7.	Accessories: Adjustable mounting depth Casters, leveling feet included. Cable pass-through in top & bottom of rack Vertical cable management bar One stationary shelves One sliding shelves
8.	Cooling Fans: Fan tray with 4 fans for Colling the servers
9.	OEM should have ISO 9001:2008 Certification for Quality Management System

3.29) Workstations with dual monitor:

S. NO	PARAMETER	SPECIFICATION
1	Processor	Intel i7
2	Operating System	Windows 10 Pro, 64 bit or latest
3	Chipset	Business Series
4	Ram	32 GB or better
5	HDD / SSD	256GB SSD and 1TB SATA HDD
6	Keyboard & Mouse	Keyboard & Mouse (Same make as Workstation)
7	Monitor	21" FHD Monitor or higher (Dual)
8	I/O	4 USB 3.1, RJ45, VGA Port, HDMI, 1 Audio input Jack/Microphone & Headphone
9	Energy Efficiency and other compliance	Energy Efficiency and other Compliance: EPEAT/ROHS and EPR compliant
10	Security	Workstation should be supplied with proper antivirus and its support for the contract period free of cost (As per specifications mentioned below).

3.30) 24 Port Managed L3 Core Switch for Command Centre

Sr. No.	Specifications	
1		Should have 24x GE/ 10GE SFP+ and 2x 40GE / 100GE QSFP+port Transceiver must be included in BOQ
		Should have 880 Gbps of Switching capacity or more
		Should have 654 Mbps of Packets Forwarding or more
		Should have 32K or more MAC address table
		Should have 4K or more VLAN support
		Should have dual Dual hot swappable power supply

	General Requirements	Switch should support simple management access i.e. without the need for local management clients (HTTPS preferred)
		Switch should support SNMP for polling of system statistics, SNMP traps, SNMP MIB download from GUI
		Should log all authentication events Locally and to Syslog Server
		Should support backup of the full system configuration via the GUI
		Should support a local user database
		Should have built-in tcp dump-like tool and log collecting functionality
		Should support multiple configuration files with 2 bootable partitionsfor better availability and easy upgrade / fall back.
		Should offer hardware warranty
		Should support auto-ranging power supply with input voltages between100 and 240V AC
2	***Layer 2 Requirement***	Should support jumbo frames
		Should support link auto-negotiation
		Should support Spanning Tree Protocol, STP Root Guard, BPDU Guard
		Should support Edge Port / Port Fast
		Should support IEEE 802.1p Mapping to priority queue and VLAN tagging
		Should support IEEE 802.3ad Link Aggregation with LACP
		Should support load balancing algorithms with Link Aggregation
		Should support MCLAG (Multi Chassis Link Aggregation)/VSS Support.
3	***Layer 3 Requirement***	Should support Static Routing
		Should support Dynamic Routing Protocols: OSPFv2, RIPv2, VRRP, BGP
		Should support Multicast Protocols: PIM-SSM, ECMP
		Should support Bidirectional Forwarding Detection (BFD),DHCP Relay, DHCP server, IP conflict detection and notification
		Should support Unicast Reverse Path Forwarding/ RPF, IPv6 route filtering
		Should support Filtering route maps based on routing protocol, Port Mirroring, Admin Authentication Via RFC 2865 RADIUS
		Should support IEEE 802.1X Authentication Port-based, IEEE 802.1XAuthentication MAC-based, IEEE 802.1X Guest VLAN, IEEE 802.1X MAC Access Bypass (MAB) or GUEST VLAN, IEEE 802.1X Dynamic VLAN Assignment
		Should support Radius CoA (Change of Authority),Radius Accounting, MAC-IP Bindings Flow
		Should support ACL , ACL Schedule
		Should support IEEE 802.1ab Link Layer Discovery Protocol (LLDP),IEEE 802.1ab LLDP-MED.
		Should support DHCP-Snooping, Dynamic ARP Inspection, Sticky MAC and MAC Limit
		Should support IEEE 802.1X auth, IEEE 802.1X EAP pass-through, Flow Export (Net Flow/IPFIX/sflow or equivalent)
		Should support Per-port and per-VLAN MAC learning limit
		Should support Assign VLANs via Radius attributes (RFC 4675)
		Should support Auto Discovery of Multiple Switches from central management system

4	***Management requirements***	Switch management software must support atleast 50 switch license from day one.
		Should support 802.1x MAC-based authentication
		Should support MAC Authentication Bypass (MAB) or GUEST VLAN Support.
		Should support load balancing algorithms with Link Aggregation
		Should support full line rate without traffic oversubscription
		Should support Time-Domain Reflect cometry (TDR) or equivalent Support, this will be used for testing cable systems and able to detect andpinpoint issues
		Should support telnet/SSH
		Should support SNTP
		Should support firmware download via TFTP/FTP/GUI
5	***Authentication Requirements***	Should support captive portal for authentication
		Should support LLDP, LLDP-MED
		Should support MAC, IP and Protocol based VLAN assignment (802.1v)
		Should support 802.1x port-based authentication, guest VLAN assignment, MAC-based authentication
		Should support MAC Authentication Bypass (MAB) or Guest VLAN
6	***Security***	Should support Policy-Based Routing through switch/ NMS
		Should support Black listing and While Listing of MAC address from central management system
		Should support Storm Control
		Should support Loop Guard
		Should support IGMP snooping
		Should support DHCP snooping and DHCP relay
		Should support Port mirroring
		Should support s Flow
		Should support ACL, Schedule ACL
		Should support security checks
		Should support port MAC limit
		Should support static MAC
		Should support Dynamic ARP Inspection
		Should support IP-MAC binding
7	***QoS***	Should support 8 queues per port
		Should support packet classification, packet marking, packet queuing
		Should support 802.1p, TOS/DSCP for priority queueing
		Should Support Weighted Round Robin (WRR) or Shaped Round Robin (SRR) or equivalent
		Should support policer
		Should support QoS per VLAN
		Support for Switch must be quoted for 5 years
		OEM Should be International certification for the, Information Security Management System: ISO 27001, ISO 9001 Quality Management System

3.31) 24 Port Managed L2 Aggregation Switch for Command Centre

Sr. No.		Specifications
1	***General Requirements***	Should have 24x GE RJ45 and 4x 10GE SFP+ for Uplink connectivity Transceiver must be included in BOQ
		Should have 128 Gbps of Switching capacity or more
		Should have 95 Mbps of Packets Forwarding or more

		Should have 16K or more MAC address table
		Should have 4K or more VLAN support
		Switch should support simple management access i.e. without the need for local management clients (HTTPS preferred)
		Switch should support SNMP for polling of system statistics, SNMP traps, SNMP MIB download from GUI
		Should log all authentication events Locally and to Syslog Server
		Should support backup of the full system configuration via the GUI
		Should support a local user database
		Should have built-in tcp dump-like tool and log collectingfunctionality
		Should support multiple configuration files with 2 bootable partitions for better availability and easy upgrade / fall back.
		Should offer hardware warranty
		Should support auto-ranging power supply with input voltages between 100 and 240V AC
2	***Layer 2 Requirement***	Should support jumbo frames
		Should support link auto-negotiation
		Should support Spanning Tree Protocol, STP Root Guard, BPDU Guard
		Should support Edge Port / Port Fast
		Should support IEEE 802.1p Mapping to priority queue and VLAN tagging
		Should support IEEE 802.3ad Link Aggregation with LACP
		Should support load balancing algorithms with LinkAggregation
		Should support MCLAG (Multi Chassis Link Aggregation) or LACP (Link aggregation Control Protocol)
3	***Management requirements***	Should support Auto Discovery of Multiple Switches from central management system
		Should support 802.1x MAC-based authentication
		Should support MAC Authentication Bypass (MAB) or GUEST VLAN Support.
		Should support load balancing algorithms with Link Aggregation
		Should support virtual Stacking or equivalent
		Should support full line rate without traffic oversubscription
		Should support 802.1ag Connectivity Fault Management (CFM) or equivalent
		Should support telnet/SSH
		Should support SNTP
		Should support firmware download via TFTP/FTP/GUI
4	***Authentication Requirements***	Should support captive portal for authentication
		Should support LLDP, LLDP-MED
		Should support MAC, IP and Protocol based VLAN assignment (802.1v)
		Should support 802.1x port-based authentication, guest VLAN assignment, authentication fail VLAN for unauthenticated users, MAC-based authentication
		Should support MAC Authentication Bypass (MAB) or GUEST VLAN Support.
5	***Security***	Should support Policy-Based Routing from central management system
		Should support Policy Control of Users and Devices from central management system
		Should support Black listing and While Listing of MAC address from central management system
		Should support Storm Control
		Should support Loop Guard

		Should support IGMP snooping
		Should support DHCP snooping and DHCP relay
		Should support Port mirroring
		Should support s Flow
		Should support ACL, ACL classifier, ACL drop action, ACL policer action, ACL counter action, ACL mirror action, ACL redirect action
		Should support security checks
		Should support port MAC limit
		Should support static MAC
		Should support Dynamic ARP Inspection
		Should support Sticky Mac
6	***QoS***	Should support 8 queues per port
		Should support packet classification, packet marking, packet queuing
		Should support 802.1p, TOS/DSCP for priority queueing
		Weighted Round Robin (WRR) or Shaped Round Robin (SRR) or equivalent
		Should support policer
		Should support QoS per VLAN
		Support for Switch and Controller must be quoted for 5years
		OEM should be International certification for the, Information Security Management System: ISO 27001, ISO 9001 Quality Management System..

3.32) 55” LED Video Display for individual monitoring

S. No	Minimum specifications
1	Diagonal Size: 55".
2	Light Source: LED
3	Resolution: 1920x1080
4	Brightness: 450 nits
5	Contrast ratio: 1000:1
6	Display Colours: 8 bit - 16.7M
7	Input Terminals: 1xDVI-D over HDMI, HDMI 2.0 (1), DP 1.2 (1)
8	Output Terminals: 1 RS-232, 1 RJ-45 In/Out is same
9	Power Supply: AC 100 - 240 V, 50Hz.
10	All accessories including cables and wall mount kit

3.33) LED based Video Wall, in 5x2 Grid:

S. No	Minimum specifications
1	55” LCD based display panels in a 5x2 & 2x2 configuration complete with wall mount.
2	The light source shall be LED with 16 at least million colours
3	1920 x 1080 native resolution.
4	Response time 8ms
5	Contrast Ratio – 1400:1
6	Viewing angle 160 deg or more
7	Brightness : 500 CD/m2
8	Combined bezel of side to side monitors should be less than or equal to 3.5 mm
9	Signal Interface 2 x HDMI in/ 1x Display port1x RS232C Dsub-9 / RJ 45 1x Digital DVI/ HDMI out / Display port out 1x USB
10	Input resolution compatibility : 1080P and higher
11	List OSD format for nomenclature
12	Accessories from OEM makeIR remote Wall mount brackets
13	Operating ambient temperature : 0 deg to 40 deg Operating relative humidity : 20% to 80%

3.34) Video Wall controller

.	Controller and displays must be from same OEM and the Controller should be able to make complete video wall display as one logical screen with grid management tools from remote workstations. It should be possible to display any or all the inputs on the video wall in any desired configuration.
Architecture	Should be based on Server architecture with at least Intel Xeon Octa core 2.1 GHz Processor (3.0GHz max. Turbo frequency)
Operating System	Windows 10 or higher -64 bit
RAM	8GB or higher
Power Supply	Redundant Power Supply
Outputs	To connect video wall of 5x2 configuration with resolutions of 1080P minimum
Inputs	5 DVI/HDMI Inputs & Dual LAN
Chassis	19" rack mount
Noise Level	50 dbA or less
Controller certification	Controller should be have BIS certification
Controller Software	
Scaling and display	Software to enable the user to display multiple sources in any size and anywhere on the display wall.
Auto Source Detection	Software should support for auto source detection
Layout Management	Should support for Video, RGB, DVI, ,Internet Explorer, Desktop Application and Remote Desktop Monitoring Layouts
Scenarios	Software should able to Save and Load desktop layouts from Local or remote machines
Layout Scheduler	All the Layouts can be scheduled as per user convenience
	Software should support auto launch of Layouts according to specified time or event by user
Launch Application	Software should be able to support
Live Preview	Software should able to provide live preview of video wall
Scalable GUI	Scalable GUI to any size of Video wall screen
Scheduler	User can schedule the layout on specific date & time, weekday, weekend , start & end date
Source positioning	User can position the source input on Video wall with single click

3.35) Multi-function Laser Printer

S. No	Minimum specifications
1.	Print technology: Laser
2.	Printer Output – Black & White only
3.	Functions : Multi-function (Print, copy & scan)
4.	Print speed: Up to 15 PPM
5.	Scan resolution, optical: Up to 1200 dpi
6.	Copy resolution (black text): Up to 600 x 600 dpi
7.	Scanner type: Flat bed
8.	Scan file format: JPEG, TIFF (compressed and uncompressed), PDF, GIF, and BMP
9.	Print resolution – Up to 600 x 600 DPI
10.	Display: 2 inch LCD
11.	Connectivity – USB
12.	Paper handling input, standard: 10-sheet priority feed slot, 150-sheet input tray

3.36) 2x20 KVA UPS with battery backup of 1 Hrs

Sr. No.	Parameter/ requirement	Minimum specifications/requirements
1	Capacity in KVA	2x20 KVA in PRS mode

2	UPS SystemDescription	2 units of 20 KVA DSP Controlled True Online Pure Sinewave with 01 Hour backup on SMF batteries, battery Rack, interconnection cables and wire from UPS to battery bank. UPS should be with IGBT Based Rectifier and IGBT based Inverter and with LCD/LED touch screen display for monitoring its status. Both the UPS should have separate battery bank with min 32000VAH and UPS should operate in PRS mode with 50% load sharing basis in such a way that if one UPS fails then load should transfer to second UPS without any interruption to load.
	Input (Rectifier)	
3	Primary source voltage	400 VAC 3 Phase, 5-wire
4	Input Voltage Range	400V (+/-20%)Three phase 320VAC -480VAC
5	Input Frequency Range	45-55 Hz
6	Input power factor	> 0.99
	Output (Inverter)	
7	Output voltage	380, 400, 415V AC (3Ph+N+PE) (Adjustable)
8	Output frequency	50Hz/60Hz ± 5%
9	Output Power Factor	0.9 or higher
10	Overall Efficiency	upto 93%
11	Voltage Harmonic Distortion	< 2% for linear load
12	Transfer Time (AC Mode to battery mode)	Zero
13	Display Panel	LCD / LED Touch screen Display
14	Battery Management	Should be inbuilt within the UPS System
15	Battery Bank & Back-up time	12V SMF batteries bank with Min 1 Hour on rated load (min VAH of Batteries required is 32000VAH with each UPS), Bidder to submit the OEM battery sizing sheet for backup calculation, At least 2 batteries Flexibility should be available in Battery bank.
16	Status on LED/ LCD display	Normal Mode/ Load on Battery/ Load on Bypass/ System Fault
18	Bypass	UPS Should have built in Manual and automatic Bypass
19	Other Features	UPS should have EPO feature (Emergency Power off) & cold start feature, input phase sequence protection as well as correction with dust filters.
20	Audible Alarm	Mains Failure/ Low battery/ UPS Warning/ Overload/ Fault & Bypass Mode etc.
21	Ambient temperature	0 to 40 Deg.C
22	Standards Certifications	ISO 9001:2015,ISO 14001, IEC EN 62040-1, -2 & -3 Standard Compliance product .
	Network Management	
23	Smart RS232 & USB	Supports Windows 2000/2003/ 2008/7/8/10/ Linux and MAC
24	SNMP	Inbuilt for web monitoring
25	Warranty Terms	Two Years on UPS & Batteries with onsite service Support

3.37) DG Set 62 KVA with silent canopy

Silent DG- 62 KVAwith AMF Control Panel	62 KVA Auto Starting Silent DG Set mounted on a common base frame with AVM (Anti- Vibration) pads, residential silencer with exhaust piping, complete conforming to ISO 8528 specifications and CPCB certified for emissions. Engine: Radiator cooled, multi cylinder, 1500 RPM diesel engine, with electronic/manual governor and electrical starting arrangement complete with battery. Alternator: Self-exciting, self-regulating type alternator rated at 0.8 PF or better, 415 Volts, 3 Phase, 4 wires, 50 cycles/sec, 1500 RPM, Windings of 100% Copper, class H insulation, Protection as per IP 23. AMF Panel should be fitted inside the enclosure, with the following

	meters/ Indicators/ Essentials: - Incoming and outgoing voltage - Current in all phases - Frequency - KVA and power factor - Time indication for hours/minutes of operation - Fuel Level in fuel tank, low fuel indication - Emergency Stop button - Auto/Manual/Test selector switch - MCCB/ Circuit breaker for short-circuit and overload protection - Control Fuses, Earth Terminal - Any other switch, instrument, relay etc. essential for automatic functioning of DG set with AMF panel The DG set shall be provided with acoustic enclosure / canopy to reduce the sound level and to house the entire DG set (Engine & Alternator set) assembly outside (open-air). The enclosure must be weather resistant powder coated, with insulation designed to meet latest MOEF/CPCB norms for DG sets, capable to withstand Mohali climate. The enclosure must have ventilation system, doors for easy access for maintenance, secure locking arrangement. It should be sufficient and suitable for containing fuel for minimum 12 hours continuous operation, Complete with level indicator, fuel inlet and outlet, air vent, drain plug, inlet arrangement for direct filling and set of fuel hoses for inlet and return. Fuel should be provided and maintained by SI and the cost of fuel will be reimbursed by Punjab Police upon submission of bills. Make: Sudhir/ Kirlosker/ Greaves This item is inclusive of Cables, Earthing, Platform and any other item required to make the same operational

3.38) 2x5TR Precision Air Conditioning for Data Centre:

Sr. No.	Minimum Specifications
1.	The AC Units should have high sensible heat ratios, to match high sensible loads of Data Centre. A Microprocessor Controlled Precision package AC system with R-410a refrigerant shall be suitable to take thermal and air qty. inputs from the server and adjust its operation accordingly to achieve highest levels of performance and efficiency.
2.	The Indoor unit shall comprise of Variable capacity Digital Scroll Compressor in independent or Tandem circuit, corrosion resistant EC fans, Evaporator DX Cooling Coil with hydrophilic coating, Microprocessor controllers, Expansion valves, Driers, G4 Filter, Suction and Discharge piping, Internal power and Control wiring, Crankcase heaters, Infrared Humidifier, Heaters, HP/LP Cutouts, Power and Control contactors, water leak detectors and Other Electrical accessories.
3.	Fixed capacity compressor technology like multiple scrolls or Fixed Capacity Tandem scrolls are strictly not allowed. Systems should mandatorily be having variable capacity compressors and performance as per above specifications.
4.	Each Precision AC unit should be provided with in-built electrical panel. Necessary 415 Volts +/- 10%, 3 Phase, 4 Wire (With Neutral), 50 Hz +/- 5% Power shall be provided by Customer at each unit's electrical panel. Balance distribution of Power is in the Scope of Bidder. All Electrical cabling should be of armored Copper.
5.	The Control System is microprocessor based, 32 bit RISC. It can be programmed to control the function of every device within the unit via I/O.
6.	Unit return air temperature : 32 Deg C (Unit Shall capable of taking up to 38 Deg C)
7.	Ambient air design temperature : 45 Deg C
8.	Cooling Capacity : 5.0 Ton
9.	Air Flow Direction : Bottom discharge-top return
10	Air inlet Temp & RH : Set point ± 1°C (DB) & Set point ± 5% (Return Air)
11	Air Quantity : Modulating as per server (110 cfm/kW (min)

12	Units should be able to provide 1 SHR at part load
13	Units should be able to work for fixed supply air logic instead of return air control logic. Units should be also connected to cold aisle remote sensors for taking the temperature feedback from the top of the racks at multiple places. This would help to regulate the fan for required airflow in the cold aisles
14	The Units shall be designed for 68-69 DBA at 1.5 meter from the unit outlet quiet operation with all moving parts mounted on anti-vibration mounting and carefully balanced to ensure minimum vibration The Units shall be designed for 68-69 DBA at 1.5 meter from the unit outlet quiet operation with all moving parts mounted on anti- vibration mounting and carefully balanced to ensure minimum vibration.
15	If required, the unit shall be tested at site for performance rating before acceptance. Performance test shall be a heat load test using heater supplied by the Precision unit supplier.
16	It's mandatory to submit software selection output of the proposed unit.

3.39) Split Air Conditioner - 2.0 Ton rating

1.	Type : Split units
2.	Capacity: minimum 2.0 Ton
3.	Cooling Capacity: minimum 24000 BTU / Hr
4.	Noise Level: < 50 dB
5.	Operation: Remote Control
6.	Power: 230VAC, 50Hz
7.	Type of Refrigerant : R 32 (CFC free) of better
8.	Inclusive of Cables, Refrigerant pipes, drain pipes, MS Stand or any other item if required to make the same operational.

3.40) TIMER FOR 2 NOS. SPLIT ACs:

1	Type	Analog/ Digital
2	Input Supply	230 Volts AC +/- 15 %, 50Hz.
3	Output Supply	230 Volts AC
4	Output Contacts	02 nos.
5	Rated Current	30Amp. (Separately for 2 ACs)
6	Change over time delay	4/8 Hr (site selectable switch)
7	Protections	Over voltage, Under Voltage with indication
8	Mounting	Wall mount

3.41) Biometric Access Control System

1	IP-enabled 2- door intelligent controller with a 32-bit CPU architecture, TCP/ IP protocol support, flash memory for firmware and large local card holder database with CE and FCC approvals	06 Doors
2	A Reader which is capable of doing 3 levels of authentication Biometric IP Based Finger Print RFID Reader/ Controller, Mullion Mount for both entry & exit Smart/ Proximity Card Pin (keypad)	12 Readers
3	Smart/Proximity Card	50 Nos
4	600 lbs Electromagnetic locks	06 Doors
5	Emergency Door Release	06 Nos
6	Access Control System software package with real time access event monitoring, control and response, door Modes, advanced reporting facility, Anti pass-back processing and other comprehensive security solution features	01 No.
7	Finger Print Reader enrolment kit	01 No.
8	Proximity card enrolment Kit	01 No.
9	Accessories, cabling etc.	As required

3.42) Chemical Earthing:

Bidder shall be responsible for the erection of Chemical earthing of Sheet metal/ iron clad Switches and metallic cases etc., with 3 mtr long copper pipe (50mm dia) including erectionof the same, 3 metre

below ground level with necessary Ground Resistance Improvement Powder (Backfill Compound), watering tank etc. and making connections to the distribution with 4mm dia copper wire (including boring & refilling).Resistance of the earthing should notbe more than one ohm

3.43) Raised Flooring:

a) False Floor: The system should be of nominal 600mm x 600mm (35mm thick) modular panels with heavy-duty pedestals. The total system weight shall not exceed 40 kg/sq.m. The system shall be able to withstand a UDL of 2000 kg/m^2 (1117Kg/Sq.mt) and a point load of 450 kg/in^2.

b) Panel: Panels shall be made of steel with an infill of lightweight, non-combustible having fire resistance, thermal insulation, and flame resistance properties at 60 min cementitious material. The bottom of the panel shall be embossed in hemispherical shape togive strength and flexural rigidity. The infill material shall not become loose when the panelsare cut. Panel should be able to withstand an ultimate load of at least 2.5 (3.00) times the designed point load. The entire panel shall be coated with epoxy powder coating of 75-100 microns on the exposed surface. The panel top & bottom sheet shall be 0.9/0.7mm (0.7/0.6mm) resp. Flatness should be 0.2mm. The panels shall have a top covering of anti- static high pressure laminate with resistance of $2x10^5$ — $5x10^8$ ohms. The HPL thickness shall not be less than 1.5mm. (1.6mm).

c) Pedestal: Pedestal base shall be square of 100mm x 100mm x 2mm bearing area and assemble to a tube of 22mm dia with 2mm wall thickness. The pedestal head should notbe less than 3mm thick. The pedestals should be able to accept an axial load of 22,500N and an ultimate load not less than 45,000N. Pedestal head shall be galvanized die-cut steelwelded to a counter locking leveling nut. The pedestal head shall be provided with PVC cap for acoustic dampening. Height of pedestal shall be approximate 300 mm for floor /150mm for steps.

d) Stringer: Stringer shall be 20mm x 30mm x 1.5mm thick (25mm x 25mm x 1mm) galvanized steel hollow pipe capable of supporting a 205kg point load at the center of the stringer. Stringers shall be individual and rigidly fastened to the pedestals with 6.35mm (1/4")fasteners.

Note: The False Floor System shall be tested as per CISCA standards for Point Load, Uniform Load, Ultimate Load and Pedestal Axial Load. Test reports from an accredited laboratory to be furnished along with the quotation.

3.44) Antivirus

Sr. No.	Client OS Protection Specifications
1	Client- Server Architecture with Centralized Administration/Console from Antivirus Server onto Client machines. Both Client Protection module and existing Server Protection module should belong to single OEM for true integration along with a singular datalake.
2	Protection against various types of Viruses
3	Protection against Malware including worms, Trojans, spyware, Bots, adware, rootkits etc.
4	Advanced machine learning capability for improved static detection of malware, thereby stopping large classes of malware. Should have AMSI based protection.
5	Website control: URL lookup and blocking of known malicious websites. Blocked sites should include those that may run JavaScript to perform cryptomining and sites that harvest

	user authentication credentials and other sensitive data.
6	Peripheral Control: Ability to block devices/ control access privileges for access to peripherals and removable media viz. USB/Pen drives, CD/DVD etc.
7	Inbuilt protection against Ransomware and Anti-Exploit capabilities.
8	Application control to block certain legitimate applications from running on work computers such as games or instant messaging programs.
9	Capability to exclude certain files/folders/paths from monitoring procedures.
10	Solution should have provision to share threat and app telemetry with network security appliance to take automated security decisions.
11	Solution should include a stable AI/ML/DL based module
12	Solution should be able to block zero-day attacks
13	Solution should be capable of making a detailed analysis of root cause of attack.
14	Should have provisions to mitigate active adversaries
15	Should have HIPS by analysing behaviour in runtime
16	Should support cache server capability to avoid unnecessary bandwidth consumption for updates.
17	Solution should support running of scheduled queries in EDR. EDR should be able to allow administrator to remotely investigate and take action on endpoint.
18	Management console login feature should support multi factor authentication.
19	Capability to prevent a user from tampering the endpoint.
20	Free upgrades and updates from the internet for a period of **three years** from installation date.
21	Should support Windows 10 and 11 client operating system having 8 GB or above RAM and Corei3 or above processor
22	Should have scheduled scanning and updates of signature
23	OEM should be Leader in latest Gartner Magic Quadrant for Endpoint Protection Platforms., 2021
24	Should be capable of Powerful Investigative Capabilities (EDR) including :
25	Investigation and IOC Sweeping (server-side metadata sweep)
25.1	Patient Zero ID / Root Cause Analysis and IOA Behaviour Hunting/Detection
25.2	API's for query / automation and Unknown file guidance
25.3	Variant Protection to detects mutations of malicious samples by recognizing known fragments of malware code
25.4	Runtime Machine Learning scores real-time behaviour against a cloud model to detect previously unknown threats
25.5	IOA Behavioral Analysis detects behavior that matches known indicators of attack (IOA), including ransomware encryption behaviors, script launching
25.6	In-memory runtime analysis malicious script detection, malicious code injection, runtime un-pack detection
25.7	Isolation, Quarantine, Process kill, Execution block and Damage rollback
25.8	Provides context-aware endpoint investigation and response (EDR), recording and detailed reporting of system-level activities to allow threat analysts to rapidly assess the nature and extent of an attack. Custom detection, intelligence, and controls
25.9	Record detailed system-level activities and perform multi-level search across endpoints using rich-search criteria such as OpenI OC, Yara, and suspicious objects.
25.10	Detect and analyse advanced threat indicators such as file less attacks.
25.11	Root cause analysis for simple or full "kill chain
25.12	Search by multiple parameters by Open IOC rule for disk scans and Yara rules for memory scans
25.13	Detect and intelligently prioritize advanced threats automatically
25.14	Understand complex attacks at a glance with the MITRE-based detection framework
25.15	XDR integration between endpoint and firewall logs
26	Isolate at-risk endpoints to run an investigation and resolve security issues. Restore the connection promptly when all issues have been resolved.

3.45) Enterprise & Network Management System:

1	**GENERAL FEATURES:**
i	The tool should provide complete cross domain visibility of IT infrastructure issues

ii	Integrate events from all network devices.
iii	Automated discovery of the infrastructure CIs
iv	The tool should classify events based on service levels
v	Offer relevant tools, run books/playbooks, graphs in context of a selected event
vi	The tool should have correlation functionality
vi	Tool should provide superior view of infrastructure health across system, networks, IT/Network infrastructure and end user into a consolidated, central console.
vii	Tool should allow customizable dashboard with widgets. The tool should different metrics to be monitored.
viii	Tool should provide support for maintenance windows and scheduled downtimes.
ix	End-to-end visibility of infrastructure, alerts and SLAs that are impacted.
x	Tool should be able to show priority of an event.
xi	Tool should allow to browse performance metrics by selecting CIs
	The login screen should have option for CAPTCHA for secure user login.
xii	Ability to launch in context to performance graphs or reports.
xiii	The tool should have threshold capability and indicate the current infrastructure resource utilization
xiv	The system needs to provide a single interface to view events, tickets and assist in root cause analysis
	Should perform required correlation to reduce the number of messages presented to the operators and to assist in the root cause.
2	**Auto-Discovery**
I	Proposed NMS should have the auto-discovery feature.
ii	Should be able to collect information from routers, switches, network devices, storage, servers, and firewalls.
iii	Discovery should be automatic and continuous to detect real time changes in the IT infrastructure.
iv	Discovery should work without requiring agent installation.
v	Should use Industry standard protocols such as WMI, ICMP, SNMP V1/V2/V3, API based to perform discovery without requiring the installation of an agent.
vi	Discovery systems should have the ability to modify out-of-box discovery scripts, create customized discovery scripts.
vii	Discovery system should support bulk device discovery via xml file.
3	**Consolidated & Customizable Dashboards & Reports**
i	Should provide drill down reports for performance, availability, inventory, operation, virtualization
ii	Should provide drill down based reports such as server resource consumption, performance, alerts etc. and reports scheduling from same page
iii	Reports should be accessible via web browser.
iv	Reports can be scheduled to publish automatically or they can be produced on demand
v	The system should provide drill down function and reports
vi	Reports can be published in HTML, PDF, Excel formats.
vii	Should be possible to send reports via email from the GUI
viii	Automated report generation and publishing
ix	Reports should be able to collect and collate specific information regarding the resource utilization and its performance
X	Tool should be able to deliver comprehensive, long term, and customizable reporting.
xi	Tool should support long term data retention up to 60 Months
xii	Tool should provide a multiple predefined reports that can be launched in the context of fault, performance, services etc.
xiii	Tool should provide reports from both Network devices, Servers, Services from the same console.
xiv	The tool should support customizable reporting
xv	Tool should allow to configure downtime for Devices
4	**ITSM – Helpdesk/Ticketing Tool & SLA Mgmt.**
i	Solution should provide Hourly, Daily, Weekly, Monthly and yearly CI wise SLA Details.
ii	Should have feature for manual and auto ticketing
iii	Report for SLAs
iv	Should have seamless integration with NMS

v	Ability to define CI in CMDB
vi	Should support incident mgmt., problem mgmt., change mgmt. and SLA mgmt.
vii	Should allow to define SLA based on different working scenarios for 24*7 and non-24*7 basis (working hours based/exclusion for holidays if required)
viii	Should allow to create Escalation Matrix for Escalations mgmt.
ix	Incident mgmt. should have the ability to define priority to tickets
x	Allows to create different hierarchy of users like technicians, team lead, change manager, CAB team for the project
xi	Should allow users to attach documents/images to the tickets.
xii	Should also allow to attach RCA documents and text to the resolved tickets.
xiii	Email notifications for status of the tickets, escalations & updates related to actions performed on the tickets
xiv	Ability to define different SLAs for CI
xv	Ability to provide different reports for open tickets, closed tickets, resolved tickets
xvi	Should allow tickets to be re-opened
5	**Network Topology & Automation**
i	Should be able to generate a graphical representation of the network and identify which devices are up, down or having issues
ii	Manage network by comparing devices configuration to defined/standard configuration
iii	Manage SNMPv3 configurations and communicate over SNMPv3.
iv	Store different configuration backups of the devices.
	The proposed tool should have inbuilt automation like ansible
v	Should have role-based access for the users and support integration with common authentication systems like Active Directory/ LDAP/Key-cloak etc.
vi	Automate routine network configuration tasks for updates, community string changes and configuration upload and download, compare configuration, bulk configurations and configuration backup.
vii	Scalability – The network management solution should be highly scalable capable.
	The proposed system should support horizontal and vertical scalability without any architectural changes in the system
6	**Network Fault Management**
I	The solution should allow for discovery to be run on a continuous basis which tracks dynamic changes near real time; in order to keep the topology always up to date. This discovery should run at a low overhead, incrementally discovering devices and interfaces.
ii	The NMS must immediately determine the impact of a component failure
iii	Polling intervals should be configurable on a need basis through GUI to ensure that key systems are monitored as frequently as necessary.
iv	The topology of the entire Network should be available with a Network state poller with aggressive/customizable polling intervals.
V	The NMS application should provide a Unified Fault, Availability and Performance function from a single station only to reduce network and device loads with unified fault & performance polling.
vi	The NMS performance system must provide predefined and highly customizable reporting across the network.
vii	The Network performance operator console should provide operators with seamless transitions from fault data to performance reports and back. For example- select a node in NMS fault mgmt. system and cross launch it for historical and near real time data.
viii	NMS should be cloud native, should assist in Root Cause Analysis
	The NMS should support deployment on physical server, virtual machines and Microservices based system
ix	NMS should have Global Management capability, where it can work in a distributed environment, horizontal & vertical scalability
X	NMS should have support for SNMPv3.
xi	It should be able to identify multiple occurrences of a specific fault on a device within a specified time frame to enable detection of chronic problems.
xii	The system should support a multiple of discovery protocols like SNMP, CDP (Cisco), LLDP (link-level discovery protocol).
	The system should support Netconf, Streaming telemetry, MQTT protocols for monitoring
xiii	The proposed Licensing should not be based to ports in a device.

7	**Network Performance Management**
I	Should establish the status of network devices and interfaces with unified status calculation and visualization of network fault & performance data.
ii	Network Performance reporting tool must provide the following capabilities:
	Data collection and thresholding of network device ports (any that support MIB2 including virtual interfaces): Bytes In, Bytes Out, Discards, Errors, Network Delay
iii	Should be able to schedule key reports for automated delivery a) Distribute reports by email in HTML, Excel or pdf formats. b) System should be scalable to support unlimited number of device c) Ability to store data for up to 60 months
8	**Monitoring Features – Servers & Virtualization**
I	Should offer operations management of the IT environment to manage distributed, heterogeneous systems - Windows, LINUX from a single management station.
ii	Centralized view for Agent based and agentless monitoring managed from one central console.
iii	Should provide a centralized point of control and management of the IP/SNMP based IT infrastructure components
iv	Should support Virtual platforms – VMware, KVM, Hyper-V etc.
v	The system should support agent based for managing the nodes if required to have the capability of storing events / data locally if communication to the centralized server is not possible due to some problem. This capability will help to avoid losing critical events.
vi	Should be configurable to suppress events for key systems/devices that are down for routine maintenance or planned outage.
vii	Should provide a console/ web browser interface that can be accessed from anywhere using industry standard web browsers.
viii	Each operator should be provided with user roles that should include operational views enabling operators to quickly determine impact on devices
ix	Should have Role based access for the users
x	The system should integrate with Helpdesk / Service desk tool for automated incident logging and also notify alerts or events via e-mail or SMS.
xi	The system should have management policies to monitor and manage WMI, SNMP and support automatic ticket creation and tickets should be available in NMS GUI
i	OEM of NMS software application should have CMMI Level 3 certification. OEM shall facilitate the validation of CMMI L3 Certificate from CMMI Institute.
ii	OEM of proposed NMS software application shall have ISO 9001:2015, ISO 27034-1, ISO 27001:2013 and ISO 45001:2018
iii	NMS software application should have qualified Security verification certificate for OWASP Top 10 vulnerability guidelines by Government agency from the list of CERT-IN empanelled Information Security Auditing organization for testing and issuing the certificate /clearance.

Executive Engineer (E)
PPHC, S.A.S. Nagar

SECTION-IV

ANNEXURES TO BE SUBMITTED WITH TECHNICAL BID:

4.1) CHECKLIST FOR THE DOCUMENTS TO BE INCLUDED IN THE PRE-QUALIFICATION & TECHNICAL BID

Sr. No.	DOCUMENTS TO BE SUBMITTED	SUBMITTED (Y /N)	DOCUMENTARY PROOF (PAGE NO.)
1.	Proof of payment of EMD, tender fee, and processing fee		
2.	Copy of Certificate of Incorporation/ Certificate of Registration of the Company		
3.	Consortium agreement, if applicable		
4.	Power of attorney/ board resolution to the authorized Signatory of the Bid (If applicable)		
5.	Copy of PAN card		
6.	Copy of GST		
7.	EPF Number		
8.	ESI Number		
9.	4.1) Check List		
10.	4.2) Annex-A: Bid Covering Letter		
11.	4.3) Annex-B: Manufacturer's Authorization Form ((From OEM of various items except CCTV Cameras)		
12.	4.3) Annex-B(i): Manufacturer's Authorization Form (From OEM of CCTV Cameras)		
13.	4.4) Annex-C: Detail of quoted items		
14.	4.5) Annex-D: Undertaking on Office Premises		
15.	4.6) Annex-E: Sole/ Lead Bidder & Consortium Member profile.		
16.	4.7) Annex-F: Technical Compliance sheet		
17.	4.8) Annex-G: Self Declaration For Rule 144 GFR		
18.	4.9) Annex-H: Non blacklisting undertaking		
19.	4.10) Annex-I: Financial Details		
20.	4.11) Annex-J: Experience details		
21.	4.12) Annex-K: Curriculum Vitae for proposed staff		
22.	4.13) Annex-L: Undertaking for SLA		
23.	4.14) Annex-M: Warranty Certificate		
24.	4.15) Annex-N: Undertaking for Data Retrieval (By Bidder)		

4.2) ANNEX-A: BID COVER LETTER

(To be submitted on Company's Letterhead and signed by Board Director / Power ofAttorney holder)

To,
Punjab Police Housing Corporation Ltd.
Sec 61, SAS Nagar, Punjab

Ref: Tender Notice No.<<…..>>dated<<…..>>

Dear Sir,

With reference and in response to your Request for Proposals for the work of _____ _____

We hereby submit our Prequalification bid, Technical bid, and Financial bid proposal, forthe same.

We hereby declare that:

a. We hereby acknowledge and unconditionally accept that PPHC can at its absolute discretion to apply whatever criteria it deems appropriate, not just limiting to those criteriaset out in the DNIT and related documents.
b. We have submitted EMD of INR_____in the form of RTGS/NEFT/IMPS to account no…… vide transaction no……. /BG no_____ from (Bank Name, Branch).
c. We have submitted DNIT document fee of INR_____in the form of RTGS/NEFT/IMPS to account no…… vide transaction no……. from (Bank Name, Branch).
d. We hereby declare that all information and details furnished by us in the Bid as per the requirements of and are true and correct. All documents accompanying such application are true copies of their respective originals.
e. We agree to abide by our offer for a period of 90 days from the date of submission prescribed by PPHC and that we shall remain bound by a communication of acceptancewithin that time.

f. We have carefully read and understood the terms and conditions of the contract applicable to the work. We do hereby undertake to provision as per these terms and conditions or higher specifications with qualified proof.

g. To supply the equipment / system software and commence services as stipulated in the schedule of delivery forming a part of the attached technical bid.

h. We affirm that the prices quoted are inclusive of delivery, installation, and commissioning charges and all taxes. Cost mentioned in financial bid covers the complete cost of project covering the scope of work mentioned in.

i. We enclose herewith the complete Technical Bid as required by you. This includes:

- Bid particulars
- Bid letter
- Proposed system architecture, detailed technical solution, equipment and servicesoffered
- Proposed Project Plan and Implementation Schedule
- Schedule of delivery
- Warranty for the period of 3 years for all the equipment's / system software supplied
- Through this including supply of consumable items.
- Manufacturer's authorization form(s)

j. In the event of acceptance of our bid, we do hereby undertake:

- We shall give benefit of any price reduction found by the time of placing the supply order.
- We agree to accept partial order if it is placed.
- To provide all the services as stipulated in the or asked by PPHC as the project demands
- We hereby undertake that all the components/parts/assembly are original and no duplicate parts are used in the entire manufacturing process.
- To undertake the services for entire contract period from the date of signing of the contract as mentioned in the.
- We affirm that the prices quoted are covering all taxes and duties, entire scope of work for duration of contract and no other expenses shall be paid by PPHC whatsoever.

k. We do hereby undertake, that, until a formal agreement contract is prepared and executed, this bid, together with your written acceptance thereof and notification of award of contract shall constitute a binding contract between us.

l. We understand accept that the PPHC may cancel the bidding process at any time and that PPHC is not bound to accept any bid that it may receive without incurring any liability towards the System Integrator (SI).

m. We fully understand and agree to comply that on verification, if any of the information provided in our bid is found to be misleading the selection process, we are liable to be dismissed from the selection process or termination of the contract during the project, if selected to do so In case of any clarifications please contact XXXXXXXXXX by email at XXX@XXX,XXX……..

Thanking you,

Yours sincerely,

(Signature of the Bidder Board Director / Power of Attorney Holder)Name
Designation Business AddressSeal
Date:
Place

4.3) ANNEX-B: MANUFACTURER'S AUTHORIZATION FORM

(This form has to be provided by the OEMs of the products proposed (Except OEM of CCTV Cameras). Separate MAF's to be provided from OEM if same item has different parts from different OEM)

Tender Ref. Date:

To:
Managing Director
Punjab Police Housing Corporation,
Sec 61, S.A.S. Nagar, Punjab

Dear Sir,

Subject: OEM Authorization for Supply, Installation, Testing, Commissioning, Operations & Maintenance of City Surveillance & Traffic Management System with Command & Control Centre in S.A.S Nagar City, Punjab

We_____ (Name of the OEM) who are established and reputable manufacturers of _____ (Name of Goods) having factories or product development centers at the locations _____ or as per list attached, do hereby authorize. _____ (Name and address of the Bidder) to bid, negotiate and conclude the contract with you against RFP No. _____Dated _____for the above goods manufactured or developed by us.

We hereby extend, our warranty for the hardware goods supplied by the bidder and or maintenance or support services for software products against this invitation for bid by _____ (Name of the Bidder) as per requirements and for the duration of contract as specified in this RFP.

We confirm the following:

a) that our offered product will not be end of life for minimum of three (03) years from the date of commissioning and the support for such offered product/s will be available for minimum of five (05) years from the date of bidding.
b) that we have customer support centers and spares available in India.
c) that the MAC address of the products are registered in our name only.
d) that, we have not been blacklisted/ terminated in any manner whatsoever by any of the State/UT and/or central government in India.
e) that Our brand has not been debarred by ONVIF and we are active member of ONVIF.
f) That we own the IPR in our name.- (Applicable for OEMs of Softwares only).

Yours faithfully,

For and on behalf of M/s (Name of the manufacturer)

Signature Name:_____

Designation Address:_____

Date:_____

Directorate Seal:_____

Note: This letter of authority shall be on the letterhead of the manufacturer and shall be signed by a person competent and having the power of attorney to bind the manufacturer. The Bidder in its Bid shall include it.

ANNEX-B(i): MANUFACTURER'S AUTHORIZATION FORM

(To be signed by the OEM of CCTV Cameras)

Tender Ref. Date:

To:
Managing Director
Punjab Police Housing Corporation,
Sec 61, S.A.S. Nagar, Punjab

Dear Sir,

Subject: OEM Authorization for Supply, Installation, Testing, Commissioning, Operations & Maintenance of City Surveillance & Traffic Management System with Command & Control Centre in S.A.S Nagar City, Punjab

We_____ (Name of the OEM) who are established and reputable manufacturers of _____ (Name of Goods) having factories or product development centers at the locations _____ or as per list attached, do hereby authorize. _____ (Name and address of the Bidder) to bid, negotiate and conclude the contract with you against RFP No. _____Dated _____for the above goods manufactured or developed by us.

We hereby extend, our warranty for the hardware goods supplied by the bidder and or maintenance or support services for software products against this invitation for bid by _____ (Name of the Bidder) as per requirements and for the duration of contract as specified in this RFP.

We confirm the following:

g) Our offered product will not be end of life for minimum of three (03) years from the date of commissioning and the support for such offered product/s will be available for minimum of five (05) years (if contract extended for 2 more years) from the date of bidding.
h) We have customer support centers and spares available in India.
i) The MAC address of the products is registered in our company's name only.
j) We have not been blacklisted/ terminated in any manner whatsoever by any of the State/UT and/or central government in India.
k) Our brand has not been debarred by ONVIF and we are active member of ONVIF
l) We own the IPR in our name.
m) Our company is in CCTV manufacturing for 10 years. (Proof attached).
n) Our company is registered in India for Last 7 years. (Proof attached).
o) Proof for supply of CCTV cameras in India for SITC of City Surveillance/ ITMS project for a Central Govt./ State Govt. department, as required in the RFP, attached herewith.
p) The components / parts / assembly of CCTV cameras are not complying with GB28181, GB/T 28181-2011 standards.
q) MTBF of offered CCTV Cameras is minimum 100000 hours.
r) The offered cameras are compliant with RoHS certification.
s) The offered Models of cameras are not any country specific. These are global models.
t) The offered Cameras have not Hisilicon chipset/Processor.
u) Chipset of the network cameras is not from the countries sharing land border with India.
v) We will provide 24x7 technical support in India. Detail of support centres attached herewith.
w) Our company has ISO 27001:2013 certification, for Information Security Management System.

Yours faithfully,

For and on behalf of M/s (Name of the manufacturer)

Signature Name:_____

Designation Address:_____

Date:_____

Directorate Seal:_____

Note: This letter of authority shall be on the letterhead of the manufacturer and shall be signed by a person competent and having the power of attorney to bind the manufacturer. The Bidder in its Bid shall include it.

4.4) ANNEX-C: DETAIL OF QUOTED ITEMS:

The Bidder shall complete the below given table with details of all OEMs as proposed for this assignment.

Sr. No.	Name of Item	Name of OEM and brand/ Make	Model no.
1			
2			
3			
5			
6			
7			
8			
9			
10			
11			
12			

4.5) ANNEX.-D: UNDERTAKING ON OFFICE PREMISES

(To be declared in the bidder's letterhead)

This is to certify that we << COMPANY NAME >>

- Has an office in SAS Nagar / Chandigarh UT / Panchkula. Relevant address proof and supporting documents are enclosed.

or

- We undertake to open an office in SAS Nagar / Chandigarh UT / Panchkula within 10 days after issuance of Letter of Acceptance.

Company Secretary / Authorized Signatory Name of Signatory:

Bidder Name:

Date

Place

4.6) ANNEX.-E: SOLE/ LEAD BIDDER & CONSORTIUM MEMBER PROFILE

Sr. No	Details	
1	Name of the Firm	
2	Registered Office address Telephone Number e-mail	
3	Correspondence/ contact address	
4	Details of Board members / Directors (Name, designation, address etc.)Telephone Numbere-mail	
5	Details of Contact person (Name, designation, address etc.)Telephone Numbere-mail	
6	Is the firm a registered company? If yes, submit documentary proof.Year and Place of the establishment of the Company	
7	Is the firm registered with sales tax department? If yes, submit valid sales tax registration certificate with latest filed return.	
8	Is the firm registered under Labour Laws Contract Act? If yes, submit valid registration certificate.	
9	Attach the organizational chart showing the structure of the organization including the names of the directors and the position ofthe officers. Total number of employees	
10	Number of years of experience in similar field	
11	Are you registered with any Government / Department / Public Sector Undertaking (if yes, give details)	
12	Details of regional offices	
13	Do you have a local representation /office in Punjab? If so, please give the address and the details of staff, infrastructure etc in the office and no. of years of operation of the local office	
14	Please give details of Key Technical and Administrative staff who will be involved in this project, their role in the project, their Qualifications & experience and the certification attained from network product vendor. (Documentary proof to be submitted)	
15	Were you ever required to suspend a project for a period of more than three months continuously after you started? If so, give thenames of project and reasons for the same.	
17	In how many projects you were imposed penalties for delay?Please give details.	
18	The Sole/ Lead Bidder shall disclose details pertaining to all contingent liabilities, claims, disputes, matters in appeal & in court and any pending litigation against the Sole / Lead Bidder. If nil, an undertaking from the Sole / Lead Bidder/ prime member of the consortium mentioning the same	
19	Whether your organization has Bank's certificate of solvency. If yes,submit documentary proof.	
20	Have you ever been denied tendering facilities by any Government/ Department/ Public sector Undertaking? (Give details)	

4.7) ANNEX-F: TECHNICAL QUALIFICATION COMPLIANCE SHEET

(To be declared on the OEM's letterhead)

Use separate sheet for each item

Name of Item:_____
Item no. in BOQ. _____
Make_____
Model No._____

S. NO.	DESCRIPTION	TECHNICAL PARAMETERS	COMPLIANCE (YES/NO)	REMARKS
1.				
2.				
3.				
4.				
5.				
6.				
7.				

For and on behalf of M/s (Name of the manufacturer)

Signature _____

Name:_____

Designation _____

Address:_____

Mob. No. _____

Date:_____

Directorate Seal:_____

4.8) ANNEX-G: FORMAT FOR SELF DECLARATION FOR RULE 144 OF GFR

(To be declared in the bidder's letter head)

Date:

To,
Managing Director,
Punjab Police Housing Corporation,
Sec 61, S.A.S. Nagar, Punjab

Subject: Self-declaration for Rule 144 of GFR reg. in response to the RFP for<<**>>

Ref:

Dear Sir,

With reference to above, I/We, declare that we comply with orders/ instructions regarding restrictions under rule 144(xi) of the general financial rule (GFR's) 2017 issued by the Ministry of Finance, government of India,

I/We hereby agree to provide copy of and/or produce original of all such documents as may be necessarily required to be submitted for evidence in this regard.

Place:
Date:

Authorized Signatory's Name and Designation:
Bidder's Company Seal:
Authorized Signatory's Signature:

4.9) ANNEX-H: UNDERTAKING FROM BIDDER ON NON-JUDICIAL STAMP PAPER DULY ATTESTED BY NOTARY PUBLIC.

(To be enclosed with Technical bid)

To

The Punjab Police Housing Corporation,
S.A.S. Nagar.

We_____ (Name of the bidder) declare that we have not been blacklisted/ debarred by any Central Government/ State Government/ Semi Government/ Corporation/ PSU or any other Government Institution/ Regional Authority in India at the time of bid submission.

Dated: this ……..

(Signature of Authorized Signatory)

Name :

Designation:

Phone/ Mobile/ Fax No.

Full Name

Full address

4.10) ANNEX-I: FORMAT TO SHARE FINANCIAL DETAILS

Financial Turnover of last 3 years

The financial turnover of the company has to be provided as per the following table:

ANNUAL TURNOVER DETAILS (CERTIFIED)			
S NO.	**FY- 2020-21**	**FY- 2021-22**	**FY- 2022-23**

NET WORTH			
S NO.	**FY- 2020-21**	**FY- 2021-22**	**FY- 2022-23**

a. Copy of audited financial statements or declaration from the appointed statutory auditor to be provided as proof of the financial turnover.

b. Net worth of the last three financial years should be positive. Copy of self-certified statutory auditor certificate to be submitted along with the bid.

4.11) ANNEX-J: FORMAT TO SHARE EXPERIENCE DETAILS

1. Project Summary Sheet:

SR. NO.	PROJECT NAME	PROJECT DESCRIPTION	PROJECT PERIOD	PROJECT COST

2. Self-certificate for Project execution experience (In Bidding Entity's Letter Head)

This is to certify that <Name of the Bidding entity> has been awarded with < Name of the Project > as detailed under:

Name of the Project

Client's Name, Contact no. and Complete Address	
Contract Value for the bidder (in INR)	
Current status of the project (Completed/Ongoing)	
Activities completed by bidding entity as on bid submission date *(N.B Only relevant activities as sought in the Criteria to be included)*	
Value of Work completed for which payment has been received from the client.	
Date of Start	
Date of Completion	

(Authorized Signatory)
Signature:
Name:
Designation:
Bidding entity's name
Address:
Seal and Date:
(NOTE: PO/Work order & Completion Certificate may also be attached)

4.12) ANNEX-K: CURRICULUM VITAE FOR PROPOSED STAFF

1. **Proposed Position** [only one candidate shall be nominated for each position]:
2. **Name of Firm** [Insert name of firm proposing the staff]:
3. **Name of Staff** [Insert full name]:
4. Date of Birth:
5. **Nationality**:
6. **Education** [Indicate college/university and other specialized education of staff member, giving names of institutions, degrees obtained, and dates of obtainment]:
7. Membership of Professional Associations:

   Other Training [Indicate significant training since degrees under 5 - Education were obtained]:
8. **Countries of Work Experience**: [List countries where staff has worked in the last ten years]:
9. **Languages** [For each language indicate proficiency: good, fair, or poor in speaking, reading, and writing]:
10. **Employment Record** [Starting with present position, list in reverse order every employment held by staff member since graduation, giving for each employment (see format here below): dates of employment, name of employing organization, positions held.]:

From [Year]: To [Year]:

Positions held:

Detailed Tasks Assigned	**Work Undertaken that Best Illustrates Capability to Handle the Tasks Assigned**
[List all tasks to be performed under this assignment]	[Among the assignments in which the staff has been involved, indicate the following information for those assignments that best illustrate staff capability to handle thetasks listed under point 11.] Name of assignment or project: Year:Location: Client: Main project features: Positions held: Activities performed

Certification:

I, the undersigned, certify that to the best of my knowledge and belief, this CV correctly describes myself, my qualifications, and my experience. I understand that any willful misstatement described herein may lead to my disqualification or dismissal, if engaged.

Date: Signature of staff member or authorizedrepresentative of the staff

Full name of authorized representative

4.13) ANNEX-L: UNDERTAKING OF SERVICE LEVEL COMPLIANCE

Tender Ref:

Date:

To,
Managing Director,
Punjab Police Housing Corporation,Sec 61, S.A.S. Nagar,
Punjab

Dear Sir,

Sub: Undertaking on Service Level Compliance

1. I/We as Implementing Agency do hereby undertake that we shall monitor, maintain, and comply with the service levels stated in the DNIT to provide quality service to PPHC.

2. However, if the proposed resources and IT components are found to be insufficient in meeting the and/ or the service level requirements given by PPHC, then we will augment the same without any additional cost to PPHC.

Yours faithfully,

(Signature of the Authorized signatory of theBidding Organization)
Name :
Designation :
Date :
Seal :
Business Address

4.14) ANNEX-M: WARRANTY CERTIFICATE

(On the bidder's Letterhead)

Tender Ref. Date: dd/mm/yyyy

To,
Managing Director
Punjab Police Housing Corporation,
Sec 61, S.A.S. Nagar, Punjab

Sir,

We warrant that the equipment(s) supplied under the contract would be newlymanufactured, free from all encumbrances, defects and faults in material or workmanship or manufacture, shall be of the highest grade and quality, shall be consistent with the established and generally accepted standards for materials of thetype ordered, shall be in full conformity with the specifications, drawings of samples, if any, and shall operate as designed. We shall be fully responsible for its efficient and effective operation. We also warrant that the services provided under the contract shall be as per the Service Level Agreement (SLA) with PPHC. This warranty shall survive inspection of and payment for, and acceptance of the Equipment and Services and shall expire only after 3 years after their successful installation and acceptance by PPHC.

We warrant that all services to be provided under the contract shall be as per our Service Level Agreement (SLA) with PPHC. This warranty on services provided shall remain valid for the entire duration of the services contract from the date of acceptance by PPHC.

The obligations under the warranty expressed above shall include all costs relating to labour, tools, spares, maintenance (preventive as well as unscheduled), and transport charges from site to manufacturer's works / service facilities and back for repair or modification or replacement at site of the equipment or any part of the equipment, which under normal care and proper use and maintenance proves defective in design, material or workmanship or fails to operate effectively and efficiently or conform to the specifications and for which notice is promptly given by PPHC to us (bidder). We shall provide on-site support for all the equipment and services supplied hereunder during the period of this warranty (3 years after acceptance for equipment and entire service period for services).

Authorized Signatory
Name :
Designation:
Seal:

4.15) ANNEX-N: UNDERTAKING FOR DATA RETRIVAL (Mandatory)

Ref. No. Date:

To,
Managing Director,
Punjab Police Housing Corporation,Sec 61, S.A.S. Nagar, Punjab

Subject: Undertaking for Data Retrieval in response to the Tender No. _____

Dear Sir,

I/ We << COMPANY NAME >> will retrieve the data from the damaged Hard disk as per clause 1.14 without any additional cost. If the Hard disks are required to carry away from the Control Centre to any laboratory for this purpose, I/ we will make standby arrangement at our own cost till restoration of the same.

It is undertaken that I/ we will maintain Data integrity and Privacy during data retrievalprocess.

Authorized Signatory
Name of Signatory:
Bidder Name:
Date
Place

SECTION-V

DOCUMENTS TO BE SUBMITTED WITH FINANCIAL BID:

5.1) TOTAL PRICE SUMMARY

TO BE UPLOADED AS BOQ SHEET (Xls format)

Punjab Police Housing Corporation			
SHORT E-TENDER NOTICE NO. 22/2023-24 DATED 01.01.2024			
Financial Bid			
Supply, Installation, Testing, Commissioning, Operations & Maintenance of City Surveillance & Traffic Management System with Integrated Command & Control Centre in S.A.S. Nagar,Punjab			
Name of the Bidder			
Address, Contact Details of the Bidder			
Sr. No	**ITEM**	**PRICE IN FIGURES (INCLUDING ALL TAXES)**	**TOTAL PRICE IN WORDS**
1	Total CAPEX price *(from table 5.2 of Section V) including free comprehensive warranty for 01 year*		
2	Total OPEX price and MPLS connectivity *(from table at 5.3 of Section V)*		
Total			

5.2) PRICE COMPONENT FOR CAPEX (INCLUDING COMPREHENSIVE WARRANTY FOR 01 YEAR FROM THE DATE OF GO-LIVE) INCLUDING MANPOWER:

(TO BE UPLOADED IN PDF FORMAT)

S. No	ITEM	UNIT	QTY	UNIT BASE PRICE (EXC. GST)	TOTAL PRICE (INR)	GST	TOTAL PRICE INCLUDING TAXES (INR)
			A	B	C (AxB)	D	C+D
A	**Field Locations:**						
1.	SITC of Outdoor ANPR 2MP Box Camera with all accessories and external IR illuminator	Each	216				
2.	SITC of Outdoor ANPR 2MP Bullet/ Box Camera with all accessories and external IR illuminator (For SVD)	Each	16				
3.	SITC of Outdoor RLVD 2MP Bullet/ Box Camera with all accessories	Each	63				
4.	SITC of RADAR/ Video Based Speed Violation Detection System for both sides i.e. To and Fro for 3 or 4 lane road including Traffic Management software with licences suitable for achieving desired accuracy during day/ night and in all weather conditions, complete with following items:- - RADAR-as required - PLB/ HDPE/ Flexible pipes from RADAR/ cameras to Main Outdoor Cabinet box (At junctions) - Armoured Cat-6 cables, Armoured Optical Fibre Cable, Power cabling from RADAR/ cameras to Main Outdoor Cabinet box, underground/ on surface (At junctions), - Termination boxes, - Managed L2 Network switches (Industrial grade), Passive items i.e. LIUs, Pigtails, Patch cords on Poles. - Any other item, if required to made the system operational to achieve the desired functionality. **Note:** ITMS Server, ANPR Cameras, Poles, Outdoor Cabinet boxes, Network switches & LIUs, LPUs in Main Outdoor Cabinet boxes, UPS & earthing excluded. It is mandatory to provide ANPR with Global Shutter for SVD system.	Locations	02				

	(Attach item wise unpriced BOQ for all locations in Technical Bid)						
5.	SITC Outdoor IP 4MP Surveillance Bullet/ Box Camera with all accessories	Each	104				
6.	SITC of IP HD 2MP PTZ Camera with all accessories	Each	22				
7.	SITC of Manageable Layer-2 Industrial grade Switch 8 Port PoE with surge protection	Each	23				
8.	SITC of Manageable Layer-2 Industrial grade Switch 24 Port PoE with surge protection	Each	18				
9.	SITC of Industrial grade Local Processing Unit (LPU)	Each	72				
10.	SITC of 1KVA Online UPS for Outdoor Junction (120 minutes battery backup)	Each	41				
11.	SITC of 2KVA Online UPS for Outdoor Junction (120 minutes battery backup)	Each	02				
12.	SITC of Outdoor Cabinet Box with Accessories	Each	41				
13.	SITC of Termination Box with Accessories	Each	91				
14.	SITC of 6.3Mtr Octagonal Pole with foundation (with 6mtr Single Arm)	Each	49				
15.	SITC of 6.3Mtr Octagonal Pole with foundation (with 8mtr Single Arm)	Each	35				
16.	SITC of 6.3Mtr Octagonal Pole with foundation (with 8mtr Double Arm)	Each	07				
17.	SITC of 4 mtr long Pole with foundation	Each	72				
18.	SITC of 8 mtr long Pole with foundation	Each	22				
19.	SITC of 3 Core 1.5mm Armoured copper cables	Mtrs	12000				
20.	SITC of 2 Core 10 sqmm Armoured aluminum cables	Mtrs	6000				
21.	SITC of 6 Core Optical fibre cables (OFC)	Mtrs	3000				
22.	SITC of Cat 6 Armoured cables	Mtrs	11000				
23.	SITC of PLB HDPE duct in trenchless laying	Mtrs	7500				
24.	SITC of Electricity Meter with outdoor enclosure	Each	22				
25.	SITC of G.I Rod earth System	Each	162				
26.	SITC of Passive Components for field locations i.e. LIUs, Pigtail, OFC/Cat 6 patch cords etc.	Lots	23				
B	**Data Centre and Integrated Command & Control Centre:**						
1.	SITC of IR IP Indoor dome cameras	Each	05				
2.	SITC of Firewall for network protection in N+1 configuration	Each	02				
3.	SITC of Rack mount Servers for Video Management in N+1 configuration with OS, MS office &	Each	06				

	any application						
4.	SITC of Video Analytics server	Each	01				
5.	SITC of Video Management Software	Lot	01				
6.	SITC of Video Analytics software	Lot	01				
7.	SITC of Traffic Management software with e-Challan module	Lot	01				
8.	SITC of External enterprise grade storage for recording data with usable space for 35 days (Minimum 1000TB)	Lot	01				
9.	SITC of PTZ 3-axis Joystick Controller with LED display. Same make of PTZ camera	Each	05				
10.	SITC of 42 U Network Racks with KVM	Each	03				
11.	SITC of Operator Workstations with dual monitors	Each	10				
12.	SITC of 24 Port Managed L3 Network Switch	Each	02				
13.	SITC of 24 Port aggregation L2 Managed Network switch	Each	02				
14.	SITC of Passive Components for Command & Control Centre/ Server Room i.e. IO's, face plates, LIUs, Pigtail, HDMI Cables OFC/ Cat 6 patch cords etc. as required to make the system operational	Lot	01				
15.	SITC of 55" LED Video display	Each	01				
16.	SITC of 55" LED based Video Wall with Controller in 5x2 grid	Set	01				
17.	SITC of Multi-function Laser Printer	Each	01				
18.	SITC of 2x20 KVA UPS with battery backup of 01 Hr	Each	01				
19.	SITC of 62 KVA DG set in soundproof canopy	Each	01				
20.	SITC of 2x5 TR Precision Airconditioning for Data Centre in redundant mode	Each	01				
21.	SITC of 2.0 Ton rating split AC	Each	02				
22.	Timer for 2 nos. Split ACs	Each	01				
23.	SITC of Chemical Earthing system	Each	04				
24.	Providing & Fixing of antistatic Raised flooring with suitable pedestal assembly.	Sft	400				
25.	SITC of Biometric Access Control System (For 6 nos. doors) complete with controller, readers and cards etc.	System	01				
26.	SITC of Anti-Virus end point software suite for servers, workstations, Local processing units and any other devices (Valid upto 3 years after Go-live)	Lot	01				
27.	Establishment of Enterprise & Network Management System	Job	01				

	(EMS & NMS) for monitoring and measurement of the SLA parameters identified for the project						
	NOTE:- All components will be covered under free Comprehensive AMC for 12 months, by deploying minimum 01no. Project Manager and 04 nos. field Engineers including T&P and sky lift as required. Any additional manpower, if required to maintain SLA will be provided by the SI at his own cost**.**						
	TOTAL:						

Price (in words) _____

NOTE:-

All components will be covered under free Comprehensive AMC for 12 months, by deploying minimum 01no. Project Manager and 04 nos. field Engineers including T&P and sky lift as required. Any additional manpower, if required to maintain SLA will be provided by the SI at his own cost**.**

5.3) PRICE COMPONENT FOR OPEX AND MPLS CONNECTIVITY, INCLUDING MANPOWER:

(TO BE UPLOADED IN PDF FORMAT)

S. NO.	ITEM	UNIT	QTY	UNIT BASE PRICE (EXC. GST)	TOTAL PRICE (INR)	GST	TOTAL PRICE INCLUDING TAXES (INR)
			A	B	C (AxB)	D	C+D
A	**Annual comprehensive Maintenance:**						
1	Annual comprehensive Maintenance for 2 years for all equipment mentioned above (Table 5.2), after expiry of free AMC of One year, including deployment of minimum manpower, required tools and plants and sky lift as required. Any additional manpower, if required to maintain SLA will be provided by the SI at his own cost**.**						
i	For 1st Year after expiry of free AMC of One Year	Job	01				
ii	For 2nd Year after expiry of free AMC of One Year	Job	01				
B	**MPLS Connectivity:**						
1	MPLS connectivity at Edge locations with required bandwidth & redundancy connected with Command & Control Centre						
i	For First year from the date of Go-Live.	Locations	20				
ii	For Second year from the date of Go-Live.	Locations	20				
iii	For Third year from the date of Go-Live.	Locations	20				
2	MPLS connectivity with required bandwidth at Command & Control Centre						
i	For First year from the date of Go-Live.	Location	01				
ii	For Second year from the date of Go-Live.	Location	01				
iii	For Third year from the date of Go-Live.	Location	01				
	Sub Total-2						

Price (in words) _____

TEMPLATES

FORMAT OF BID SECURITY (BANK GUARANTEE)

(By any scheduled commercial bank/Nationalized Bank)

WHEREAS,_____[name of Bidder] (hereinafter called "the Bidder") has submitted his Bid dated_____[date] for the work of Supply, Installation, Testing, Commissioning, Operations & Maintenance of City Surveillance & Traffic Management System with Integrated Command & Control Centre in S.A.S. Nagar, Punjab [name of Contract] (hereinafter called "the Bid") KNOW ALL PEOPLE by these presents that We _____ [name of bank] of_____[name of country] having our registered office at_____(hereinafter called "the Bank") are bound unto Punjab Police Housing Corporation Ltd., Phase 7, S.A.S. Nagar [name of Employer] (hereinafter called "the Employer") in the sum of Rs._____ (Rupees_____ only) for which payment well and truly to be made to the said Employerthe Bank binds itself, his successors and assigns by these presents.

SEALED with the Common Seal of the said Bank this_____day of _____ 20____.

THE CONDITIONS of this obligation are:

1) If after Bid opening the Bidder withdraws his bid during the period of Bid validity specified in the Form of Bid;
   or
2) If the Bidder having been notified of the acceptance of his bid by the Employer during the period of Bid validity:
   (a) fails or refuses to execute the Form of Agreement in accordance with theInstructions to Bidders, if required;
   or
   (b) fails or refuses to furnish the Performance Security, in accordance with the Instruction to Bidders;
   or
   (c) does not accept the correction of the Bid Price;

we undertake to pay to the Employer up to the above amount upon receipt of his first written demand, without the Employer having to substantiate his demand, provided that in his demand the Employer will note that the amount claimed by him is due to him owing to the occurrence of one or any of the three conditions, specifying the occurred condition or conditions.

This Guarantee will remain in force up to and including the date_____* days after the deadline for submission of Bids as such deadline is stated in the Instructions to Bidders or as it may be extended by the Employer, notice of which extension(s) to the Bank is hereby waived. Any demand in respect of this guarantee should reach the Bank notlater than the above date.

DATE_____ SIGNATURE OF THE BANK

WITNESS_____ SEAL _____

_____

[Signature, name, and address]___

* 45 days after the end of the validity period of the Bid.

Details of Beneficiary for Bank Guarantee:-

Beneficiary Bank Account Name :	MANAGING DIRECTOR PPHC PH 7 MOHALI PB
Beneficiary Bank Name & Branch :	HDFC Bank & Branch Phase-7, Mohali
Beneficiary Bank IFSC Code :	HDFC0001307
Beneficiary Bank Account Number :	50100189729533
Beneficiary PAN :	AAACP9909L
Beneficiary TAN :	PTLP11135F

FORMAT OF PERFORMANCE BANK GUARANTEE

(To be furnished by Scheduled Commercial/ Nationalized Bank)

(To be submitted by the successful bidder after LoA)

In favour of Managing Director
Punjab Police Housing Corporation,Sec 61, S.A.S. Nagar,
Punjab

WHEREAS_____[name and address of SI] (hereinafter called "the SI") has undertaken, in pursuance of Contract No. _____ dated_to execute Supply, Installation, Testing, Commissioning, Operations & Maintenance of City Surveillance & Traffic Management System with Integrated Command & Control Centre in S.A.S. Nagar, Punjab (hereinafter called "the Contract");

AND WHEREAS it has been stipulated by you in the said Contract that the SI shall furnish you with a Bank Guarantee by a recognized bank for the sum specified therein as security for compliance with his obligations in accordance with the Contract.

AND WHEREAS we have agreed to give the SI such a Bank Guarantee.

NOW THEREFORE we hereby affirm that we are the Guarantor and responsible to you, on behalf of the SI, up to a total of _____ [amount of guarantee] [in words], such sum being payable in the types and proportions of currencies in which the Contract Price is payable, and we undertake to pay you, upon your first written demand and without cavil or argument, any sum or sums within the limits of _____ [amount of guarantee] as aforesaid without your needing to prove or to show grounds or reasons for your demand for the sum specified therein.

We hereby waive the necessity of your demanding the said debt from the SI before presenting us with the demand.

We further agree that no change or addition to or other modification of the terms of the Contract or of the Works to be performed thereunder or of any of the Contract documents which may be made between you and the SI shall in any way release us from any liability under this guarantee, and we hereby waive notice of any such change, addition, or modification.

This guarantee shall be valid until …… (i.e.) 30 days from the date of expiry of the Defects Liability/ AMC Period.

Signature and seal of the guarantor _____ Name of Bank ___
_____ Address _
Date _____

Details of Beneficiary for Bank Guarantee:-

Beneficiary Bank Account Name :	MANAGING DIRECTOR PPHC PH 7 MOHALI PB
Beneficiary Bank Name & Branch :	HDFC Bank & Branch Phase-7, Mohali
Beneficiary Bank IFSC Code :	HDFC0001307
Beneficiary Bank Account Number :	50100189729533
Beneficiary PAN :	AAACP9909L
Beneficiary TAN :	PTLP11135F

FORMAT OF AGREEMENT:

This agreement, made the _____day of_____20____ between Executive Engineer (Elect.), PPHC, S.A.S. Nagar on behalf of Managing Director, PPHC, Sahibzada Ajit Singh Nagar, Punjab (hereinafter called "the Department/ PPHC") of the one part and _____ name and address of contractor] (hereinafter called "the System integrator") of the otherpart.

Whereas the Department/ PPHC is desirous that the System integrator execute _____ [name and identification number of Contract] (hereinafter called "the Work")and the Department/ PPHC has accepted the Bid by the System integrator for the executionand completion of such Works and the remedying of any defects therein, at a contract price of Rs.....

NOW THIS AGREEMENT WITNESSETH as follows:

1. In this Agreement, words and expression shall have the same meanings as are respectively assigned to them in the Conditions of Contract hereinafter referred to, and they shall be deemed to form and be read and construed as part of this Agreement.

2. In consideration of the payments to be made by the Department/ PPHC to the System integrator as hereinafter mentioned, the System integrator hereby covenants with the Department/ PPHC to execute and complete the Works and remedy any defects therein in conformity in all aspects with the provisions of the Contract.

3. The Department/ PPHC hereby covenants to pay the System integrator in consideration of the execution and completion of the Works and the remedying the defects wherein the Contract Price or such other sum as may become payable under the provisions of the Contract at the times and in the manner prescribed by the Contract.

4. The following documents shall be deemed to form and be read and construed as partof this Agreement, viz.:

a. Letter of Acceptance;
b. Notice to proceed with the works;
c. Contractor's Bid;
d. Contract Data;
e. Conditions of contract (including Special Conditions of Contract);
f. Specifications;
g. Bill of Quantities; and
h. Any other document listed in the Contract Data as forming part of the contract.

In witness whereof the parties thereto have caused this Agreement to be executed the dayand year first before written.

The Common Seal of _____

was hereunto affixed in the presence of:

Signed, Sealed and Delivered by the said _____
_____

in the presence of:

Binding Signature of Department/ PPHC _____

Binding Signature of System integrator _____

DETAIL OF PROPOSED COMPONENTS:

Sr No	Location	Location Type	6.3Mtr Pole with 6 mtr arm	6.3Mtr Pole with 8 mtr arm	6.3Mtr Pole with 8mtr double arm	4 mtr pole	8Mtr Pole	IR IP HD PTZ Camera	IR HD Bullet Camera	ANPR Box Camera	ANPR Box/ Bullet Camera (For SVD)	RLVD Camera
21.	New Sunny Enclave gate	ITMS	3	2	0	3	2	2	5	13	0	4
22.	Cheema Boiler Chowk	ITMS	4	1	-	3	1	1	5	11	0	3
23.	Radha Swami Chowk	ITMS	4	2	-	6	1	1	7	15	0	4
24.	T-point Ajit Smachar Sec 90/Phase 8B Dividing	ITMS	-	3	-	3	1	1	4	8	0	3
25.	Landra Location (2 Chowks)	ITMS	3	2	-	6	2	2	6	9	0	5
26.	Purab Apartments Crossing Sec 88/89	ITMS	-	4	-	7	1	1	7	12	0	4
27.	Sec. 105/106 Dividing	Entry/ Exit	-	-	1	-	-	-	2	6	0	-
28.	Gurudwara Singh Shaheedan Chowk	ITMS	6	1	-	5	2	2	9	15	0	4
29.	IISER Chowk	ITMS	4	2	-	5	1	1	6	14	0	3
30.	Airport Chowk (Bakarpur)	ITMS	8	-	-	6	2	2	8	16	0	4
31.	Dairi T-Point (Landran Banur Road)	Entry/ Exit	-	-	3	-	-	-	2	12	0	-
32.	PCA Stadium Crossing (Ph 9/10)	ITMS	4	-	-	2	1	1	5	8	0	4
33.	Phase 7 Crossing	ITMS	5	1	-	7	1	1	8	13	0	4
34.	Chawla Chowk Crossing	ITMS	2	2	-	4	1	1	5	10	0	4
35.	Phase 3/5 crossing	ITMS	-	4	-	4	2	2	6	12	0	4
36.	Madanpur chowk	ITMS	-	4	-	4	1	1	4	12	0	4
37.	Micro Tower, Phase 2/3A crossing (2 locations)	ITMS/ Entry/ Exit	1	4	1	4	2	2	7	19	0	6
38.	Max Hospital	ITMS/ Entry/ Exit	1	1	1	3	1	1	4	11	0	3
39.	Near TDI/ Gilco gate (Airport Road)	SVD/ Entry/ Exit	2	-	1	-	-	-	2	0	8	-
40.	Airport Chowk (Bakarpur) to Zirakpur Road	SVD/ Entry/ Exit	2	2	-	-	-	-	2	0	8	-
	TOTAL		**49**	**35**	**7**	**72**	**22**	**22**	**104**	**216**	**16**	**63**